

The Local Employment Effects of the Clean Water Act Construction Grants Program*

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Abstract

Large-scale environmental infrastructure investments are often criticized as costly, yet are often used as place-based policies to generate local employment during recessions. This paper estimates the local labor-market effects of the Clean Water Act (CWA) Construction Grants Program using a county-level staggered difference-in-differences design exploiting variation in the timing of first grant-assisted construction starts. Initiating construction raises construction employment by approximately twenty jobs per county-year over the first two post-treatment years, with accompanying increases in establishments and wages. Effects are substantially larger in counties with greater pre-existing labor-market slack and among counties entering the program during its earlier, more intensive years. Evidence of broader effects is less precise, with suggestive responses in non-construction employment and per-capita personal income. Overall, the results show that federally financed wastewater infrastructure generates measurable short-run labor demand during implementation, with the magnitude of the employment response depending importantly on local labor-market conditions.

Keywords: Clean Water Act, pollution abatement, public infrastructure, local employment

JEL Classification: H41, H54, R11, E62, Q58

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1 Introduction

“The waste treatment plant construction program is now the largest public works program in the United States, surpassing highway building. . . . While not quantifiable, the additional economic stimulus, the ‘multiplier effect,’ of a program that creates more than one million jobs, is just as real as its goal to make the Nation’s waters cleaner and safer.”

— U.S. EPA, *Clean Water: Construction Grants Program News*, Fall 1975, p. 9.

Water and wastewater systems are among the most capital-intensive components of local public infrastructure, and decades of underinvestment have left a large financing gap. Recent federal assessments put wastewater and stormwater needs at roughly \$630 billion over a 20-year horizon (U.S. EPA, 2024). At the same time, prior work suggests that the measured environmental benefits of large-scale water infrastructure programs may not always justify their substantial costs (Keiser & Shapiro, 2019; Keiser et al., 2019). Existing evaluations have focused primarily on environmental and public-health outcomes, providing less evidence on the short-run local economic consequences of these investments during implementation. This omission is notable because water infrastructure has repeatedly been used as a policy lever in stimulus and recovery efforts: the ARRA directed \$4 billion to water-quality infrastructure, and the IIJA provided an additional \$12.7 billion in federal support. Construction-intensive infrastructure spending may raise demand for local labor and income while projects are underway, yet evidence on these effects remains limited relative to other categories of public infrastructure, notably highways, and whether direct construction employment translates into broader local economic activity remains an open empirical question.

This paper studies these questions using the Clean Water Act (CWA) Construction Grants Program, the large federal program that financed municipal wastewater infrastructure primarily during the 1970s and 1980s. I assemble administrative records on grant-assisted projects and their construction dates and link them to county-level labor-market and economic outcomes. I estimate dynamic treatment effects using a staggered difference-in-differences design in which treatment is defined by the first grant-assisted construction start in each county. The design therefore exploits variation in when federally assisted construction begins rather than simply comparing counties that did and did

not receive federal assistance. States allocated projects through priority lists based principally on documented pollution-control needs and affected population, prohibiting the use of development objectives or projections of future population growth in establishing project priorities. Although treated and untreated counties differ in pre-program levels, pre-program growth rates do not systematically predict treatment or treatment timing, and event-study estimates show little evidence of differential trends before construction begins. The construction-employment results are also robust to restricting comparisons to eventually treated counties and to conditioning on pre-program county characteristics.

The clearest result is a direct expansion of local construction activity when grant-assisted construction begins. Construction employment increases by approximately 20 workers per county-year on average over the first two years of a county's initial construction episode. The response is concentrated in heavy construction, the subsector that includes water and sewer infrastructure work, while general building and specialized trade construction show little systematic response. Construction establishments and wages also increase. Translating the employment estimate into cumulative labor input yields approximately 39.8 job-years per treated county over the first two years, or about 0.41 job-years per \$100,000 of federal assistance in 2017 dollars.

The construction-employment response varies substantially across program eras and local economic conditions. Effects are strongest among counties entering the program during its early, more intensive years, when construction starts and federal expenditures were most heavily concentrated. Employment responses are also substantially larger in counties with greater pre-existing labor-market slack: counties with above-median unemployment exhibit clear construction-employment gains, whereas estimates in lower-unemployment counties are considerably smaller. These patterns are robust to alternative measures of local slack and do not appear to be explained simply by differences in grant size. The concentration of employment gains in higher-unemployment counties is particularly informative because additional labor demand is more likely to draw on underutilized workers when local labor markets have greater slack.

I also examine whether the direct construction response extends into the broader county economy. This exercise is less sharp because the timing and functional form of indirect effects are not

as clearly determined by the physical construction process. In levels, employment rises in several non-construction sectors and in total county employment, but most responses become small and imprecise when measured proportionally; retail provides the clearest evidence of an employment response outside construction. Personal income per capita provides additional suggestive evidence of broader effects, with the preferred specification implying approximately \$1.82 in cumulative local personal income per dollar of federal assistance over four years. I interpret this calculation as a reduced-form accounting benchmark rather than a fiscal multiplier. Overall, evidence of broader local effects is suggestive but considerably weaker than the direct construction-sector response.

This paper contributes to the literature on the local effects of public infrastructure investment by providing evidence from a major category of environmental infrastructure that has received comparatively little attention. The CWA Construction Grants Program provides a useful setting because federal funds financed fixed, place-specific capital projects, allowing the direct labor-market response to be linked closely to the timing of physical construction. The results show that federally assisted wastewater infrastructure investment generated measurable short-run employment gains in construction, with substantially larger responses in places with greater labor-market slack and during the program's earlier years. Evidence that these gains propagated through the broader local economy is more limited. The findings therefore distinguish a clear, localized labor-demand response during project implementation from the broader stimulus effects often attributed to public infrastructure spending, which are more difficult to identify and appear to depend more strongly on local economic conditions.

Related Literature This paper sits at the intersection of three strands of research. The first is a growing empirical macroeconomics and public finance literature that uses geographic cross-sectional variation in fiscal spending to estimate regional employment and income effects. The second is a body of work in urban and spatial economics studying place-based policies and localized labor demand shocks in spatial equilibrium. These literatures provide the conceptual foundation for this paper: the macro literature motivates the focus on local fiscal multipliers, while the urban and spatial literature speaks to how local economic structure shapes the incidence of federal spending. The third strand assesses the Clean Water Act (CWA) and wastewater infrastructure investment, which has

largely emphasized environmental quality and health benefits rather than short-run labor-market responses.

A large literature estimates the local incidence of fiscal spending using variation across regions in the timing and intensity of government spending. Seminal work by Nakamura and Steinsson (2014) exploits regional variation in military procurement to estimate an “open-economy relative multiplier” and finds sizable effects of government spending on local activity. Subsequent papers use a wide range of quasi-experimental sources of regional variation—including formula-driven transfers, public health expenditures, disaster relief, and other place-based programs—to estimate local employment and income effects (Chodorow-Reich, 2019; De Ridder et al., 2020; Dupor & Guerrero, 2021; Suárez Serrato & Wingender, 2016). A central lesson is that interpretation hinges on spatial adjustment and the mapping from spending to local outcomes: commuting and migration can reallocate employment across space, sectoral composition can shape where demand is absorbed, and local general equilibrium forces (e.g., wages and rents) can mediate measured effects. Empirical designs therefore emphasize plausibly exogenous variation in the distribution of spending and outcomes tightly linked to the proposed transmission mechanism.

Within this broader literature, evidence on *public investment*—and infrastructure investment in particular—is comparatively limited. Public investment differs from other government purchases because it is lumpy, project-based, and subject to implementation lags (Leeper et al., 2010; Ramey, 2020). Work on ARRA exploits cross-area variation in federal allocations, including “shovel-ready” infrastructure grants, to estimate local employment effects, but findings are mixed and can be sensitive to program category and measurement (Chodorow-Reich et al., 2012; Conley & Dupor, 2013; Dupor & Mehkari, 2016; Dupor et al., 2023; Wilson, 2012). Related macro evidence also suggests that investment multipliers may differ from consumption multipliers, though results are not uniform across settings (Ilzetzki et al., 2013; Koh, 2025). Closer to the methodological spirit of this paper, Ghomi and Pappa (2026) use European Investment Bank loans to public infrastructure projects across EU countries and implement inverse-probability-score weighted regressions to address selection. They report cumulative output multipliers of about 1.1 after one year and 3.3 after five years, with larger effects under favorable financial conditions.

A recurring limitation in this literature is that researchers often do not directly observe the project-level implementation channel through which public investment is delivered, making it difficult to identify the mechanisms linking spending to local labor-market outcomes (Glaeser & Poterba, 2021). This motivates attention to settings in which the link between spending and local labor demand is tight and measurable. Leduc and Wilson (2013) and Garin (2019) emphasize the value of focusing on construction-intensive highway infrastructure programs and tracking outcomes in sectors that should respond mechanically if spending increases local activity. Related work studies the long-run local effects of large federal investment programs and discrete place-based shocks, including the Tennessee Valley Authority, wartime plant construction, and other infrastructure-driven investments (Garin & Rothbaum, 2025; Kang et al., 2025; Kline & Moretti, 2014; Leff Yaffe, 2020). This paper follows that logic by using project-level construction timing under the CWA Construction Grants Program and by focusing on construction employment, establishments, and wages as the most direct margins of adjustment, before examining whether these sectoral effects propagate to total employment and income.

A related body of work in urban and spatial economics studies place-based policies and localized labor-demand shocks in spatial equilibrium. Garin (2019) is especially close in spirit, as it examines whether federally financed highway infrastructure investment operates as a spatially targeted tool for increasing local payrolls in the construction sector. More broadly, work on local employment multipliers studies whether an exogenous increase in tradable employment generates additional non-tradable jobs locally (Kline & Moretti, 2014; van Dijk, 2017). This paper complements that literature by studying a large federal infrastructure program whose initial demand impulse is concentrated in construction, allowing the data to speak to whether a localized construction shock is associated with broader county-level employment and income responses.

A separate literature evaluates the CWA and wastewater infrastructure investment primarily through environmental quality and welfare channels, including changes in water quality and valuation via housing markets (Keiser & Shapiro, 2019; Ren & West, 2023). More recent work also emphasizes health and other downstream benefits (Flynn & Marcus, 2023), and how local municipalities are affected by the environmental mandates set forth by the act (Flynn & Smith, 2022; Jerch, 2026). Relative to this literature, evidence on short-run local labor-market effects of CWA wastewater con-

struction is scarce. To the best of my knowledge, the only closely related employment study is Ai et al. (2025), which uses a difference-in-differences design around the 1977 amendments and attainment versus nonattainment status to estimate impacts on sectoral employment. This paper is complementary: it examines the short-run local economic consequences of federally financed wastewater construction during project implementation, with particular attention to the direct construction-sector labor-market response and the extent to which these effects extend to broader county employment and income.

The remainder of the paper proceeds as follows. Section 2 describes the CWA Construction Grants Program. Section 3 presents the data and empirical framework. Sections 4 and 5 report the effects on construction-sector labor market outcomes and examine treatment effect heterogeneity. Section 6 examines broader county-level effects beyond the construction sector. Section 7 concludes.

2 The Clean Water Act Construction Grants Program

The Federal Water Pollution Control Act Amendments of 1972 (P.L. 92-500)—commonly known as the Clean Water Act (CWA)—constituted a major federal intervention to reduce water pollution and improve water quality across the United States (Congressional Research Service, 2019b). A central statutory objective was to restore and maintain the integrity of the nation’s waters, requiring not only regulatory standards on pollutant discharges, but also large-scale public investment in wastewater treatment infrastructure.

This paper focuses on Title II of the 1972 amendments, which authorized the Construction Grants Program—a grant-financed program administered by the EPA that subsidized the construction and upgrading of publicly owned wastewater treatment works. From its inception in fiscal year 1973 through its phase-out in 1990, the program disbursed approximately \$52 billion in nominal terms (about \$145 billion in 2017 dollars), making it one of the largest public infrastructure investment programs in U.S. history (Congressional Research Service, 2019a). Federal grants covered a fixed share of eligible project costs—originally 75 percent, later reduced to 55 percent in 1981—with the remainder financed by state and local matching contributions.

Grant allocation proceeded in two stages. The EPA first distributed appropriations to states using a formula based on wastewater treatment needs and population. States then awarded grants to municipalities using priority lists reflecting the severity of nearby water pollution, affected population size, and conservation needs. According to the Handbook of Procedures of the Construction Grants Program for Municipal Wastewater Treatment Works (U.S. EPA, 1980), the CWA explicitly prohibited states from incorporating development objectives or projections of future population growth into these lists, and required states to certify a project's rank-order priority at each grant step and to hold a public hearing before revising the list, limiting states' scope to informally reorder projects toward localities that happened to be economically "ready" in a given year.¹ Flynn and Smith (2022) similarly find no evidence that grants were allocated according to recipient governments' finances or spending preferences. The empirical design exploits differences in the timing of construction starts across counties rather than the fact of receipt, and evaluates identifying assumptions directly using event-time dynamics.²

I focus on Step 3 and combined Step 2+3 grants,³ which include the on-site construction component and most directly translate federal funds into local labor demand.⁴ The construction start date used to define treatment timing is not itself a discretionary choice made at the state or local level: it is set only after the Step 3 grant is accepted, through a formal sequence of competitive bid advertisement, bid review, and contract award administered under EPA regulation (U.S. EPA, 1980).

3 Empirical Framework and Data

This section describes the data and empirical design used to estimate local labor-market responses to federally assisted wastewater infrastructure construction under the CWA Construction Grants Pro-

¹Additional details on the state priority-list system, program funding and reserves, and the point in the grant process at which a project's construction start date is actually determined is provided in Appendix Section B.

²Counties that ever receive grants differ from counties that never receive grants. I address these differences and assess robustness to alternative comparison groups in Section 4.1.

³Step 2+3 combination grants provided design and construction funding together, before design was complete, for smaller projects: municipalities with a population of 25,000 or less and an estimated building cost of \$8 million or less were eligible, with the grant based on estimated allowable costs from the facilities plan plus an allowance for design. This gave smaller communities assurance that construction funds were obligated while design was still underway, rather than requiring separate sequential Step 2 and Step 3 awards (U.S. EPA, 1980).

⁴A natural question is whether the local construction sector anticipated federal assistance when Step 1 and Step 2 grants were awarded. I address this in Section 4.1.

gram. The analysis links project-level grant records to county-level outcomes and exploits variation in the timing of construction starts across counties in a staggered difference-in-differences framework.

3.1 Data sources

The analysis combines administrative records on the CWA Construction Grants Program with county-level labor-market and economic data. First, I use administrative microdata on the universe of Clean Water Act (CWA) Construction Grants Program awards obtained from the EPA’s Grant Information Control System through a Freedom of Information Act request. These records report project award dates, construction start and completion dates, project characteristics (e.g., grant step and project type), and funding amounts.⁵ I merge the grant files to facility-level information from the Clean Watershed Needs Surveys (CWNS) using a unique facility identifier, and then aggregate to a county-year panel of grant receipt.⁶

The analysis focuses on construction-stage grants (Step 3 and combined Step 2+3 grants). For county i , the treatment year G_i is defined as the first year in which grant-assisted construction begins.⁷ I use construction start rather than grant award as the treatment date because it more closely corresponds to the onset of on-site construction activity and the associated demand for construction labor. The baseline treatment therefore captures a county’s first entry into grant-assisted construction.

County-level employment outcomes come from the Quarterly Census of Employment and Wages (QCEW), which provides annual average employment, establishments, and wage measures by county and industry. The primary construction-sector outcomes are employment, average weekly wages, and the number of establishments. I also use total county employment to examine whether employment responses extend beyond the construction sector. Because consistent county-by-industry

⁵As documented in Flynn and Smith (2022), the validity of the grant records is also assessed in Keiser and Shapiro (2019). I cross-check the extracted files against the datasets used in Flynn and Smith (2022), which rely on the same underlying system. Further details are provided in Appendix Section A.

⁶The Clean Watershed Needs Survey (CWNS) is conducted every two to four years and provides an assessment of the capital investments publicly owned wastewater treatment facilities require to meet Clean Water Act standards.

⁷When multiple projects begin in the same county, G_i is the earliest construction start year.

Table 1: Summary statistics of grants under the Clean Water Act Construction Grants Program

	Mean	SD	p5	Median	p95
Panel A. Counties across time					
<i>Counties receiving grants (N = 2,437)</i>					
Number of grants	4.89	7.01	1.00	3.00	16.00
Project cost (million)	62.05	232.36	0.91	10.91	268.71
Federal grant award (million)	45.26	171.25	0.65	7.96	195.85
Per capita project cost	619.36	704.43	59.26	436.06	1,684.97
Per capita federal grant award	454.56	524.23	44.81	322.30	1,259.25
Panel B. County-year observations					
<i>County-year observations with grants (N = 8,436)</i>					
Number of grants	1.41	1.07	1.00	1.00	3.00
Project cost (million)	17.93	57.58	0.37	4.65	72.13
Federal grant award (million)	13.07	42.34	0.26	3.32	52.70
Per capita project cost	178.92	307.80	5.25	85.93	656.24
Per capita federal grant award	131.32	229.88	3.72	61.34	485.22
Panel C. Individual projects/grants					
<i>Total projects/grants (N = 11,921)</i>					
Project cost (million)	12.68	36.83	0.33	3.91	47.81
Federal grant award (million)	9.25	26.64	0.24	2.81	35.17
Per capita project cost	126.63	257.64	1.73	47.54	509.42
Per capita federal grant award	92.93	192.47	1.23	34.30	378.76

Notes: This table reports summary statistics for federally assisted projects under the CWA Construction Grants Program. Dollar values are in real 2017 USD. Panel A reports statistics aggregated at the county level, Panel B at the county-year level, and Panel C at the individual project/grant level. *Project cost* refers to the total eligible project cost for federal assistance, including the federal grant award and the local cost share.

QCEW series begin in 1975, the baseline sample excludes counties whose first grant-assisted construction start occurs prior to 1975. I also use the BEA Regional Economic Accounts, which report county-level personal income and population starting in 1968, to construct personal income per capita. Appendix Table A.1 reports pre-program demographic and economic characteristics of ever- and never-treated counties.⁸

Turning to the grant data, Appendix Figure A.3 plots the distribution of first treatment years across counties. Treatment cohorts are largest in the mid-to-late 1970s and decline steadily through the 1980s, consistent with the program's front-loaded spending profile; 20.88% of counties never re-

⁸Appendix Table A.2 separately compares counties whose first construction start occurs before 1975 with those retained in the QCEW estimation sample. Although the groups differ in baseline levels, their pre-program growth rates are similar across the reported outcomes. Appendix Figure A.2 shows the geographic distribution of the excluded pre-1975 counties.

ceive a construction-stage grant. Appendix Figure A.1 shows the geographic distribution of grants. Panel (a) maps the first year of grant-assisted construction start by county. Treatment timing is staggered across the program period and geographically dispersed. Panel (b) maps total real federal awards per capita, showing substantial heterogeneity in grant intensity across treated counties.

Table 1 reports summary statistics at the project, county-year, and county levels. Project sizes are highly right-skewed: the median project carries \$3.91 million in eligible cost and a \$2.81 million federal award, while the 95th percentile reaches \$47.81 million and \$35.17 million, respectively (2017 dollars). At the county-year level (Panel B; $N = 8,436$) grant activity is typically episodic, with the median grant-receiving county-year involving a single project, a median project cost of \$4.65 million, and a median federal award of \$3.32 million. Aggregating to the county level over the full program period (Panel A; $N = 2,437$ ever-treated counties), the median treated county receives three grants totaling \$7.96 million, while the 95th percentile receives 16 grants totaling \$195.85 million. Together, these patterns document substantial variation in both the timing and scale of grant-assisted construction across counties.

3.2 Empirical framework

I estimate the effect of initiating grant-assisted construction using a staggered difference-in-differences design (Callaway & Sant’Anna, 2021). Standard errors are clustered at the county level. Let G_i denote the first year in which grant-assisted construction begins in county i (with $G_i = \infty$ for never-treated counties). The baseline estimand captures the response to a county *first* entering grant-assisted construction. To maintain a single-adoption setting, I censor a county beginning in the year of its second grant-assisted construction episode.⁹ I report dynamic event-study effects around G_i and average post-treatment effects over two- and four-year windows. These windows summarize near-term responses and span the period during which most first-episode projects remain active.¹⁰

⁹This restriction isolates the initial construction-start episode and avoids conflating subsequent construction waves with the first treatment. I discuss how the estimation sample is affected in Appendix Section A.1 and report analyses on an unrestricted sample in Appendix Section C.6.

¹⁰Among the 2,431 counties with a qualifying construction start, the share still under grant-assisted construction, restricting to the project(s) that define each county’s first construction start, is 100 percent at the construction-start year, 95.2 percent one year later, 69.0 percent at two years, 45.8 percent at three years, and 29.9 percent at four years. See Appendix Section B.3 for descriptive statistics of the time from construction start to completion.

Estimand and aggregation Let Y_{it} denote an outcome for county i in year t . For each cohort g and year $t \geq g$, the group-time average treatment effect on the treated is

$$ATT(g, t) = \mathbb{E}[Y_{it}(g) - Y_{it}(\infty) \mid G_i = g], \quad t \geq g, \quad (1)$$

where $Y_{it}(g)$ denotes the potential outcome at time t if county i first begins grant-assisted construction in year g , and $Y_{it}(\infty)$ denotes the potential outcome if it is never treated. Identification comes from comparing cohort- g counties to counties that are untreated at time t . In the baseline specification, the control set is

$$C_t = \{i : G_i > t\},$$

which includes both not-yet-treated counties (with finite $G_i > t$) and never-treated counties ($G_i = \infty$). The unconditional cohort-time comparison underlying the baseline specification is

$$\widehat{ATT}(g, t) = (\bar{Y}_{g,t} - \bar{Y}_{g,g-1}) - (\bar{Y}_{C_t,t} - \bar{Y}_{C_t,g-1}), \quad (2)$$

where $\bar{Y}_{g,t} = \mathbb{E}[Y_{it} \mid G_i = g]$ and $\bar{Y}_{C_t,t} = \mathbb{E}[Y_{it} \mid i \in C_t]$. I estimate and aggregate these effects using the Callaway and Sant’Anna (2021) DiD estimator (CSDiD). Event-study effects are obtained by re-indexing $ATT(g, t)$ in event time $e = t - g$ and aggregating across cohorts.

I also report average post-treatment effects over two- and four-year horizons. Formally,

$$\widehat{ATT}^H = \frac{1}{H} \sum_{h=0}^{H-1} \widehat{ATT}(h), \quad \widehat{ATT}(h) = \sum_g \omega_g(h) \widehat{ATT}(g, g+h), \quad (3)$$

where $\widehat{ATT}(g, g+h)$ is the group-time effect for cohort g at event time h and $\omega_g(h) = N_g(h) / \sum_{g'} N_{g'}(h)$, with $N_g(h)$ the number of counties first treated in g that contribute to horizon h . Event time h is indexed such that $h = 0$ denotes the year of the county’s first grant-assisted construction start. Accordingly, the two-year average aggregates $h = 0, 1$ and the four-year average aggregates $h = 0, \dots, 3$. Standard errors for the averages account for the covariance across horizons. The weights are normalized within each horizon, so cohorts that are not observed at h —because the panel ends or because the county enters a second construction episode—drop out of $\widehat{ATT}(h)$ rather than being

extrapolated.¹¹

Assumptions and threats to identification The staggered DiD design relies on three conditions: (i) no anticipation, (ii) parallel trends for treated cohorts relative to the chosen control group, and (iii) no spillovers across counties. In this setting, *no anticipation* means that outcomes do not change systematically *prior* to the first year of grant-assisted construction in ways attributable to the forthcoming project (e.g., pre-construction mobilization that occurs before the recorded construction start date). I assess this condition using pre-treatment event-time estimates.

The *parallel trends* condition requires that, absent treatment, outcomes for counties first treated in year g would have evolved similarly to outcomes for the comparison group over the same periods. A potential threat is endogenous targeting: because the allocation process incorporated population and documented treatment needs, treated counties may differ systematically from untreated counties in ways correlated with differential trends in outcomes. As discussed in Section 2, however, the CWA explicitly prohibited states from incorporating development objectives or projections of future population growth into priority lists, limiting the direct role of expected local economic growth in statutory grant prioritization. The empirical design further exploits variation in the timing of first construction starts rather than grant receipt alone. Appendix Section B.2 examines directly whether pre-program county characteristics predict treatment timing or grant amounts. Under the baseline control definition $C_t = \{i : G_i > t\}$, the identifying restriction can be stated as

$$\mathbb{E}[Y_{it}(\infty) - Y_{i,g-1}(\infty) \mid G_i = g] = \mathbb{E}[Y_{it}(\infty) - Y_{i,g-1}(\infty) \mid i \in C_t], \quad \forall t \geq g. \quad (4)$$

An alternative restriction excludes never-treated counties from the identifying comparison and requires parallel trends only between treated cohorts and not-yet-treated counties that are eventually treated:

$$\mathbb{E}[Y_{it}(\infty) - Y_{i,g-1}(\infty) \mid G_i = g] = \mathbb{E}[Y_{it}(\infty) - Y_{i,g-1}(\infty) \mid G_i > t, G_i < \infty], \quad \forall t \geq g. \quad (5)$$

¹¹Appendix Table A.7 reports summary statistics for grants associated with counties' first construction starts and the corresponding county-year observations. Appendix Section C.6 discusses the implications of dropping observations after a county enters a second construction episode.

This specification avoids requiring comparability between ever- and never-treated counties. I assess the plausibility of the parallel-trends assumption using event-study pre-trends and robustness to alternative comparison groups, including specifications that condition on pre-program county characteristics using the doubly robust estimator of Sant’Anna and Zhao (2020).

Finally, *no spillovers* requires that treatment in one county does not affect outcomes in other counties. Spillovers are plausibly limited for construction employment because projects are tied to specific wastewater treatment facilities and on-site work, although cross-county contractor activity could attenuate estimated effects if labor is reallocated across county lines. I investigate this possibility in Section 4.1, where I estimate the effect of grant-assisted construction on neighboring never-treated counties within the same commuting zone. Although the analysis is limited by the small number of eligible neighboring counties, the estimates provide suggestive evidence of limited spillovers in construction-sector labor markets. A separate issue for interpretation is whether grant-assisted projects displace other local construction activity. Evidence in Flynn and Smith (2022) indicates that CWA grants increased sewerage capital spending approximately dollar-for-dollar when compliance requirements were binding, suggesting limited crowd-out of local capital spending on the relevant margin. I return to the composition of construction-sector responses in Section 4.

4 Local Employment Effects

I examine employment effects in counties where federally assisted wastewater construction takes place, beginning with the direct response of the construction sector. Following Garin (2019), construction provides a natural first margin of adjustment because wastewater infrastructure projects directly require on-site construction labor and contracting services. Panel A of Table 2 reports baseline CSDiD estimates for construction employment, establishments, and average weekly wages. The coefficients summarize average effects over the first two and four years following a county’s first grant-assisted construction start. Figure 1 reports the corresponding event-study estimates, together with employment responses across construction subsectors.

The estimates indicate a short-run expansion in local construction activity. Construction employ-

ment increases by 19.885 workers per county-year on average over the first two years following construction start. Relative to the pre-treatment mean of 778.557 workers, this corresponds to an increase of approximately 2.55 percent. The four-year average remains positive at 12.989 workers per county-year but is less precisely estimated. The smaller average over the longer horizon is consistent with the duration of the underlying projects. As documented in Section 3, construction associated with the first treatment episode is progressively completed over the post-treatment period, so the four-year estimate averages together years in which most projects remain active with later years in which a substantial share have already been completed.

The other construction-sector outcomes move in the same direction. Over the first two post-treatment years, the number of construction establishments increases by 1.980, while average weekly construction wages increase by \$4.37 in 2017 dollars. The establishment response is precisely estimated, whereas the wage response is smaller and estimated less precisely. Over four years, both point estimates remain positive but are statistically indistinguishable from zero. The employment and establishment responses are therefore consistent with an increase in local construction activity following the onset of grant-assisted wastewater construction.

Employment effects across construction subsectors Panel B of Table 2 decomposes the construction-employment response across the three two-digit SIC construction industries: general building contractors (SIC 15), heavy construction contractors (SIC 16), and special trade contractors (SIC 17).¹² The response is concentrated in heavy construction, the category that contains water and sewer infrastructure work. Heavy-construction employment increases by 10.663 workers per county-year over the first two post-treatment years and by 16.105 workers over the first four years, with both estimates statistically significant at the 5 percent level. By comparison, the estimates for general building and special trade contractors are smaller and statistically indistinguishable from zero.

The concentration of the response in heavy construction is consistent with the sectoral composition of the projects themselves. It also provides some evidence against a large contemporaneous dis-

¹²SIC 15 covers general contractors engaged in residential, industrial, and commercial building construction; SIC 16 covers heavy construction other than buildings, including water and sewer main and pipeline construction classified under SIC 1623; and SIC 17 covers specialized trades such as plumbing, electrical work, painting, masonry, and roofing.

Table 2: Local construction sector effects

Panel A. Construction sector outcomes, levels			
	Construction emp.	# of establishments	Avg. weekly wage
2-yr post avg.	19.885*** (7.224)	1.980*** (0.450)	4.368* (2.340)
4-yr post avg.	12.989 (9.998)	1.022 (0.660)	4.212 (2.942)
N	38,992	38,994	38,992
Pre-treatment avg.	778.557	107.825	624.644
Panel B. Construction subsector employment effects, levels			
	General building	Heavy construction	Special trade
2-yr post avg.	2.070 (3.931)	10.663** (4.804)	6.030 (3.828)
4-yr post avg.	-2.662 (4.434)	16.105** (7.394)	-3.031 (5.342)
N	30,589	26,790	35,162
Pre-treatment avg.	265.838	198.177	421.312

Notes: This table reports CSDiD average treatment effects over the first 2 and 4 years after the start of grant-assisted construction. Panel A reports construction-sector employment, establishments, and average weekly wages. Panel B reports employment in general building contractors, heavy construction contractors, and special trade contractors. Standard errors are clustered at the county level and reported in parentheses. *Pre-treatment avg.* reports the average outcome in 1975, the first year with consistent county-level QCEW coverage. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

placement of employment in other construction industries: over the first two years, point estimates for both general building and special trade contractors are positive. At the four-year horizon, their point estimates turn negative but remain imprecisely estimated. The estimates therefore do not indicate substantial offsetting declines elsewhere in construction, although they do not rule out some reallocation across subsectors.

Job-years per dollar of federal assistance To provide a scale for the estimated employment response, I translate the average treatment effects into cumulative job-years. Because the two-year estimate averages event years $h = 0$ and $h = 1$, an average increase of 19.885 workers corresponds to 39.77 job-years per treated county over the first two years. Similarly, the four-year average of 12.989 workers corresponds to 51.96 job-years over event years $h = 0, \dots, 3$.

I normalize these cumulative employment effects by federal assistance associated with the county's

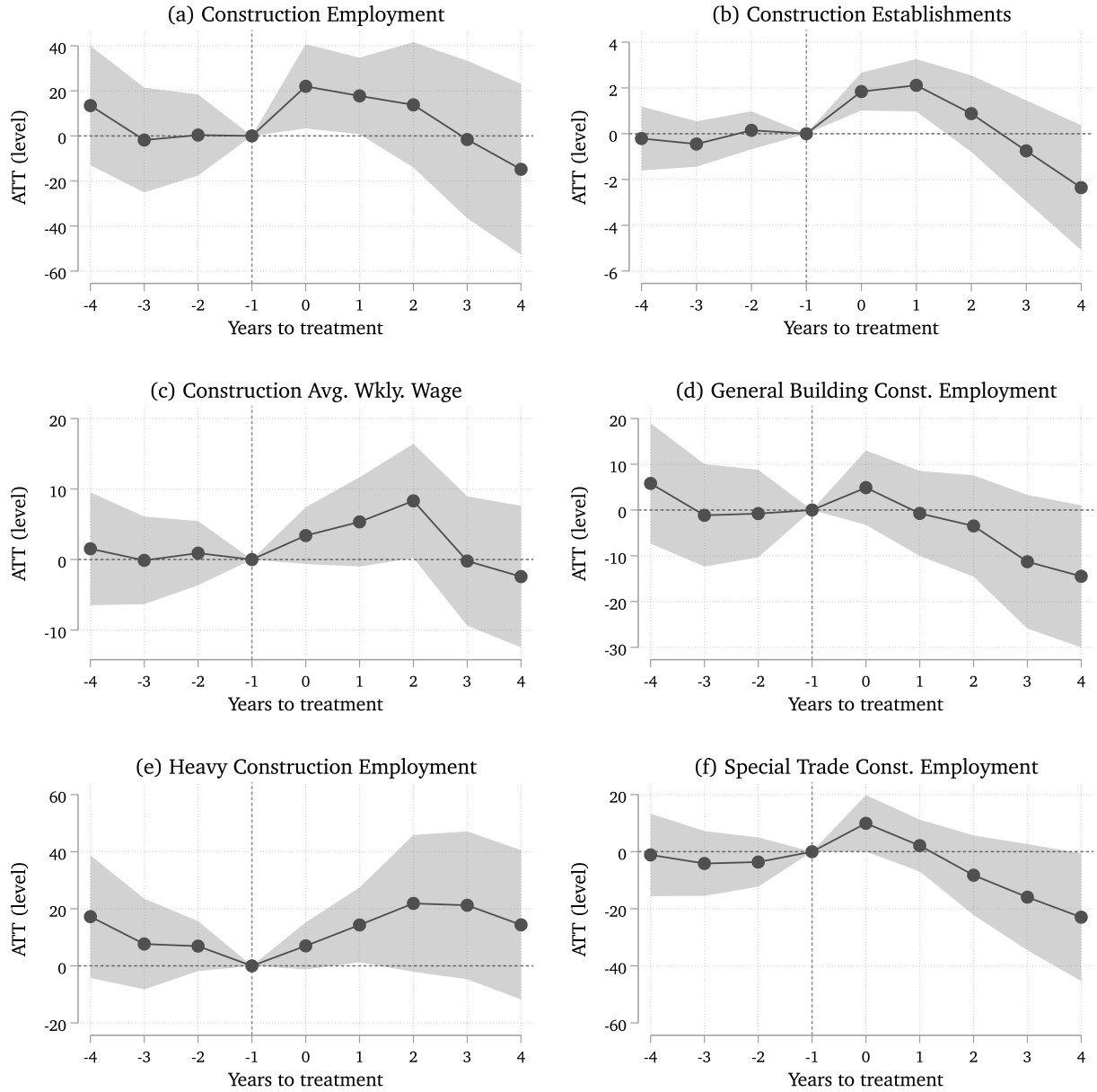


Figure 1: Event-study CSDiD estimates of construction-sector outcomes in levels. Outcomes include construction employment, establishments, average weekly wages, and employment in the three construction subsectors. Event time is measured relative to the start of grant-assisted construction. Shaded areas denote 95% confidence intervals.

first construction-start episode. Among first-treatment observations entering the QCEW estimation sample, the mean federal award is \$7.61 million in 2017 dollars (Appendix Table A.7). The resulting back-of-the-envelope estimates are approximately 0.52 job-years per \$100,000 of federal assistance

over two years and 0.68 job-years per \$100,000 over four years.^{13,14}

For comparison, Garin (2019) estimates approximately 0.5 job-years per \$100,000 of federal highway investment under the ARRA over a four-year horizon. The institutional settings and treatment definitions differ, so the comparison is only suggestive; the estimates here are of a similar order of magnitude.

4.1 Threats to internal validity and robustness

I examine several potential threats to the interpretation of the baseline construction-sector estimates. I first consider the choice between additive and proportional treatment effects and assess the level specification against estimates using logged outcomes. I then examine two identifying assumptions underlying the staggered DiD design: whether construction activity responds in anticipation of the recorded construction start and whether the timing of grant-assisted construction is systematically related to pre-existing local economic conditions. Finally, I test for employment spillovers to neighboring counties within the same commuting zone, which could contaminate the county-level comparison groups. Across these exercises, the construction-employment response is generally robust to alternative functional forms, comparison groups, and controls for pre-program characteristics, while the spatial-spillover estimates provide suggestive, though less precisely estimated, evidence against substantial cross-county effects.

Additive versus proportional specifications The baseline estimates are reported in levels, corresponding to an additive model of the construction-sector response. An alternative is a proportional specification in which outcomes are expressed in logs, so that treatment effects are interpreted as percentage changes relative to the size of the pre-existing local construction sector. The choice between these specifications is substantive. An additive model is appropriate if a grant-assisted project

¹³Mean eligible project cost at the first construction start is \$10.64 million in the QCEW estimation sample. Normalizing by total project cost, which includes the nonfederal matching share, yields approximately 0.37 and 0.49 job-years per \$100,000 over the two- and four-year windows, respectively.

¹⁴These calculations are descriptive normalizations of the reduced-form treatment effects rather than estimates of a continuous spending multiplier, since the baseline design identifies the effect of entering grant-assisted construction rather than the marginal effect of an additional dollar of federal assistance.

generates an increment to local construction demand that is primarily determined by the scale and engineering requirements of the project itself. A proportional model instead assumes that the absolute employment response increases with the size of the county's existing construction sector. Recent work emphasizes that functional-form choices in difference-in-differences settings should be guided by the economic structure of the treatment and the corresponding counterfactual model (Kahn-Lang & Lang, 2020; McConnell, 2024; Roth & Sant'Anna, 2023).

The institutional setting of the Construction Grants Program provides a basis for an additive specification. Wastewater treatment facilities are discrete engineering projects whose construction requirements depend on characteristics such as treatment capacity and facility design. Contemporary EPA cost-estimation guidance similarly modeled project costs as functions of treatment-plant capacity and engineering characteristics (Dames & Moore, 1978; U.S. EPA, 1976). Conditional on project scale, the resulting demand for on-site construction labor need not increase proportionally with the size of the host county's existing construction sector. In particular, a similarly sized wastewater project may require a comparable increment of construction labor in two counties with very different baseline construction workforces.

The grant data also show that the scale of the treatment used in the baseline design increases substantially less than proportionally with county size. Appendix Table B.10 reports elasticities of federal award amounts with respect to pre-program population and construction employment. Lifetime federal awards rise approximately proportionally with population: the controlled elasticity is 0.969, and unit elasticity cannot be rejected. This relationship is substantially weaker for the treatment-year award, which is the more relevant margin for a design based on the county's first construction start. The treatment-year award elasticity with respect to population is 0.540 in the bivariate specification and 0.341 after conditioning on pre-program county characteristics; unit elasticity is rejected in both cases.

The same pattern is more pronounced when county size is measured by the pre-treatment construction workforce. The elasticity of lifetime federal awards with respect to construction employment is 0.732 in the bivariate specification and 0.654 with controls. For treatment-year awards, the corresponding elasticities are only 0.347 and 0.241. Unit elasticity is rejected in all four specifications.

Table 3: Local construction sector effects, logs

Panel A. Construction sector outcomes, logs (percent)			
	Construction emp.	# of establishments	Avg. weekly wage
2-yr post avg.	0.972 (0.833)	0.163 (0.406)	0.751** (0.367)
4-yr post avg.	0.800 (1.178)	-1.013* (0.565)	0.664 (0.456)
N	38,992	38,994	38,992
Pre-treatment avg.	778.557	107.825	624.644
Panel B. Construction subsector employment effects, logs (percent)			
	General building	Heavy construction	Special trade
2-yr post avg.	1.754 (1.174)	2.820 (1.736)	-0.740 (0.888)
4-yr post avg.	0.156 (1.500)	2.855 (2.215)	-1.223 (1.210)
N	30,589	26,790	35,162
Pre-treatment avg.	265.838	198.177	421.312

Notes: This table reports CSDiD average treatment effects over the first 2 and 4 years after the start of grant-assisted construction. Outcomes are expressed in logs, and coefficients are reported in percent. Panel A reports construction-sector employment, establishments, and average weekly wages. Panel B reports employment in general building contractors, heavy construction contractors, and special trade contractors. Standard errors are clustered at the county level and reported in parentheses. *Pre-treatment avg.* reports the average outcome in levels in 1975, the first year with consistent county-level QCEW coverage. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Thus, larger construction labor markets receive larger awards, but the size of the initial grant-assisted construction episode rises much more slowly than the size of the existing construction sector. A proportional outcome model would therefore impose a substantially stronger relationship between baseline construction employment and the treatment response than is present in the treatment itself.

I nevertheless estimate the construction-sector effects using logged outcomes as a complementary functional-form check. Table 3 reports average effects over the first two and four post-treatment years, while Appendix Figure C.4 presents the corresponding event-study estimates. For construction employment, the two-year log estimate is 0.972 percent (s.e. 0.833), compared with an implied 2.55 percent increase obtained by scaling the level estimate by the pre-treatment mean. Although the log estimate is less precise, its 95 percent confidence interval contains the percentage change implied by the level specification. The same is true at the four-year horizon. Average weekly wages

display a similar correspondence: the two-year log estimate is 0.751 percent, compared with an implied increase of approximately 0.70 percent from the level estimate. Establishments are less stable under the log transformation, particularly at the four-year horizon, where the log estimate becomes negative.

The construction-subsector estimates provide a similar comparison. In levels, the employment response is concentrated in heavy construction, the category most directly related to water and sewer infrastructure. In logs, the heavy-construction coefficients remain positive at both horizons—2.820 percent over two years and 2.855 percent over four years—but are estimated less precisely. Thus, the proportional specification preserves the qualitative timing and sectoral concentration of the construction response, but provides weaker statistical evidence of post-treatment effects.

Additional specification checks provide no indication that the level results arise from systematic pre-treatment movements. Appendix Section E.1 reports in-time placebo tests in which fictitious construction-start dates are assigned before the true treatment date. For construction employment, the placebo estimates are small and statistically insignificant in both levels and logs at the two alternative placebo dates. Appendix Section E.2 further reports permutation tests that reassign treatment timing among ever-treated counties. The actual two-year construction-employment estimate in levels lies toward the tail of the resulting placebo distribution, with a randomization-inference p -value of 0.090; the corresponding log specification produces the same qualitative pattern, although with less separation from its permutation distribution.

The engineering structure of wastewater construction, the sub-proportional relationship between treatment-year grant size and county construction employment, and the similarity of the level and log event-study patterns support the additive specification as the preferred model for construction-sector outcomes. I therefore report effects in levels in the baseline analysis and use the proportional specification as a complementary robustness check.

Anticipation A key requirement of the staggered DiD design is that county labor-market outcomes do not respond to treatment before grant-assisted construction begins. In this setting, anticipation in construction employment could arise if the recorded construction start systematically lags the onset

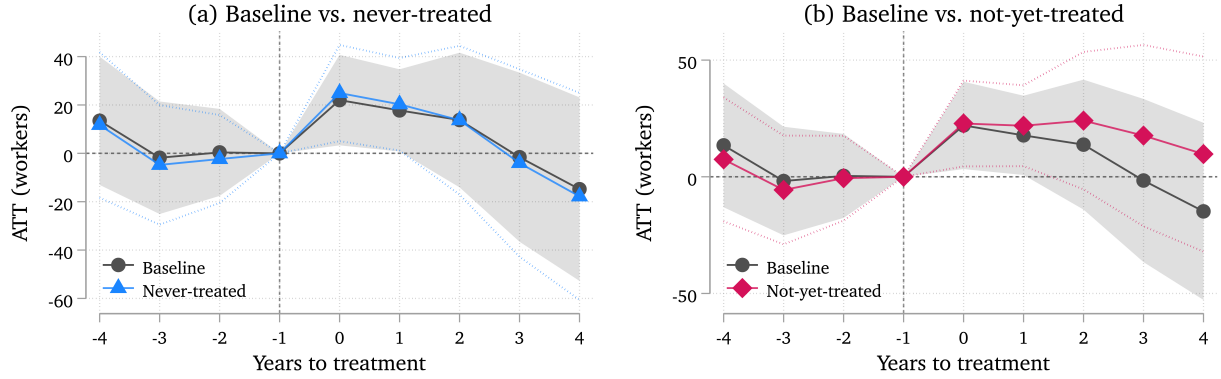


Figure 2: Event-study CSDiD estimates of construction employment under alternative control-group definitions. Panel (a) compares the baseline specification with estimates using only never-treated counties as controls. Panel (b) compares the baseline specification with estimates using only not-yet-treated counties as controls. The dark line denotes the baseline specification; colored lines denote the alternative control groups. Shaded areas and dotted lines denote 95% confidence intervals.

of on-site activity, for example because of early mobilization or site preparation, or if earlier stages of the grant process themselves generate substantial construction labor demand. The latter channel is less likely given the structure of the Construction Grants Program. The analysis defines treatment using Step 3 and combined Step 2+3 grants, which finance construction, whereas the preceding planning and design stages involve substantially smaller expenditures and do not entail the same demand for on-site construction labor.

The event-study estimates provide little evidence of anticipatory employment responses. Figure 1 shows that construction-employment coefficients prior to the recorded construction start are small and statistically indistinguishable from zero. The increase in employment emerges around the construction-start year rather than in the preceding years. The corresponding pre-treatment estimates for construction establishments and wages likewise show no systematic differential trends. These patterns are consistent with the recorded construction start capturing the onset of the economically relevant construction activity.

As an additional check, I re-estimate the construction-employment effects using only never-treated counties as the comparison group. The baseline estimator uses both never-treated and not-yet-treated counties as controls, so anticipatory adjustments among counties approaching their own construction starts could attenuate the estimated effect. Panel (a) of Figure 2 removes this possibil-

Table 4: Construction employment effects by control group

	(1) Baseline	(2) Never-treated	(3) Not-yet-treated
2-yr post avg.	19.885*** (7.224)	22.610*** (8.132)	22.362*** (7.222)
4-yr post avg.	12.989 (9.998)	13.730 (11.355)	21.611** (10.736)
N	38,992	38,653	22,211
Pre-treatment avg.	778.557	778.557	778.557

Notes: This table reports CSDiD average treatment effects on construction employment over the first 2 and 4 years after the start of grant-assisted construction. Column (1) uses never-treated and not-yet-treated counties as controls. Column (2) uses only never-treated counties. Column (3) uses only not-yet-treated counties. Standard errors are clustered at the county level and reported in parentheses. *Pre-treatment avg.* reports the average construction employment in 1975, the first year with consistent county-level QCEW coverage. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

ity by excluding not-yet-treated counties from the control group. The resulting event-study profile is similar to the baseline: pre-treatment estimates remain close to zero, and construction employment increases following the construction start. Table 4 reports the corresponding average effects over the first two and four post-treatment years. The estimates using only never-treated counties (Column (2)) are similar in sign and magnitude to the baseline estimates (Column (1)), providing little indication that anticipatory responses among not-yet-treated counties materially affect the main results.

Targeting and selection A second concern is endogenous targeting. Counties receiving grant-assisted projects differ systematically from counties that never receive assistance, raising the possibility that treatment status or timing is correlated with underlying economic conditions. As described in Section 2, municipalities applied for funding through state priority lists that were required to prioritize documented pollution-control needs and prohibited from incorporating geographic location, projections of future population growth, or broader development objectives unrelated to pollution abatement; these lists were subsequently reviewed and approved by the EPA (U.S. EPA, 1980). Nevertheless, allocation was explicitly related to factors such as population and measured treatment needs. Consistent with this institutional structure, Appendix Table A.1 documents substantial pre-program differences in levels between ever-treated and never-treated counties. These cross-sectional differences do not themselves violate the identifying assumption, but they would be

problematic if they were associated with differential counterfactual trends.

I assess this concern in three ways. First, I restrict the comparison group to counties that eventually receive grant-assisted construction, so that identification relies only on differences in treatment timing among ever-treated counties. Second, I condition directly on pre-program demographic, economic, and construction-sector characteristics using a doubly robust estimator. Third, I examine whether pre-program levels and growth rates predict treatment status and treatment timing. These exercises complement the event-study evidence in Figure 1, where construction employment exhibits no systematic differential trends prior to treatment.

The first exercise excludes never-treated counties and uses only not-yet-treated, eventually treated counties as controls. This comparison is useful because ever-treated and never-treated counties differ substantially in observable characteristics, whereas the restricted specification identifies the effect from counties exposed to the same program at different points in time. Panel (b) of Figure 2 presents the corresponding event-study estimates, and Column (3) of Table 4 reports the two- and four-year average effects. The estimates closely track the baseline specification in Column (1), and the pre-treatment coefficients remain centered around zero. The construction-employment response therefore does not depend on using never-treated counties as part of the comparison group.

The second exercise conditions directly on observed pre-program differences using the doubly robust IPW-OLS estimator of Sant’Anna and Zhao (2020). The first specification conditions on mean population and personal income per capita over 1969–1971 and their corresponding pre-program growth rates. The second additionally conditions on mean construction-sector personal income per capita and its growth rate over the same period. Table 5 reports the resulting average treatment effects. Columns (1) and (2) report effects over the first two post-treatment years, while Columns (3) and (4) report effects over the first four years. Within each horizon, the second specification adds the construction-sector controls. The estimated effects remain similar in magnitude to the unconditional baseline estimates, although they are less precisely estimated in some specifications.¹⁵

Finally, I examine directly whether pre-program economic conditions predict which counties receive

¹⁵Appendix Section C.2 reports the corresponding propensity-score diagnostics and shows adequate overlap in the covariate distributions used for estimation. These results indicate that conditioning on observed pre-program differences does not materially alter the estimated construction-employment response.

Table 5: Construction employment effects with covariate adjustment

	(1)	(2)	(3)	(4)
2-yr post avg.	20.923** (8.524)	19.456** (8.161)	20.540** (8.337)	19.462** (8.071)
4-yr post avg.	12.814 (9.951)	12.004 (9.850)	13.037 (9.945)	13.017 (9.877)
N	38,946	38,946	38,235	38,235
Pre-treatment avg.	778.557	778.557	778.557	778.557
Demographic controls (levels)	Yes	Yes	Yes	Yes
Demographic controls (growth)		Yes		Yes
Construction controls (levels)			Yes	Yes
Construction controls (growth)				Yes

Notes: This table reports CSDiD average treatment effects on construction employment over the first 2 and 4 years after the start of grant-assisted construction. Estimates use the doubly robust IPW-OLS (DRIPW) estimator of Sant’Anna and Zhao (2020). Columns vary the covariate set as indicated. *Demographic controls (levels)* include mean population and personal income per capita over 1969–1971. *Demographic controls (growth)* include population growth and personal income per capita growth from 1969 to 1971. *Construction controls (levels)* include mean construction-sector personal income per capita over 1969–1971. *Construction controls (growth)* include construction-sector personal income per capita growth from 1969 to 1971. Standard errors are clustered at the county level and reported in parentheses. *Pre-treatment avg.* reports average construction employment in 1975, the first year with consistent county-level QCEW coverage. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

grant-assisted construction and, among treated counties, when construction begins. I regress an indicator for ever receiving treatment and, separately, the construction-start year G_i among treated counties on pre-program population, personal income per capita, and construction-sector personal income per capita. For each characteristic, I distinguish its 1969–1971 level from its pre-program growth rate. This distinction is important for the identifying assumption: persistent differences in levels across counties are accommodated by the cohort-specific before-after comparisons underlying the CSDiD estimator, whereas systematic differences in pre-treatment trajectories would raise a more direct concern for parallel trends.

Appendix Table B.8 shows that treatment assignment is strongly related to county size but not to pre-program economic growth. Population significantly predicts whether a county is ever treated ($p < 0.001$), while none of the three pre-program growth rates does so (all $p > 0.6$). Among eventually treated counties, larger and higher-income counties enter construction earlier: pre-program population and income levels both significantly predict treatment timing ($p < 0.001$). In contrast, none of the three pre-program growth rates individually predicts treatment timing (all $p > 0.12$),

Table 6: Construction-sector spillover effects within commuting zones

	Construction emp.	# of establishments	Avg. weekly wage
2-yr post avg.	-4.127 (8.675)	-0.015 (0.305)	5.550 (7.699)
4-yr post avg.	-4.739 (10.723)	0.123 (0.458)	2.461 (7.916)
N	8,300	8,300	8,300
Pre-treatment avg.	176.68	27.15	555.21

Notes: This table reports CSDiD average treatment effects on construction-sector outcomes over the first 2 and 4 years after the first grant-assisted construction start in another county within the same commuting zone. The sample includes counties that are not directly treated during the relevant window. Outcomes are construction employment, construction establishments, and average weekly construction wages. Standard errors are clustered at the county level and reported in parentheses. *Pre-treatment avg.* reports the average outcome in 1975, the first year with consistent county-level QCEW coverage. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

and they are jointly insignificant ($F = 1.38$, $p = 0.246$). Thus, the program clearly did not allocate construction randomly across counties or over time, but the observed selection operates primarily through pre-existing levels rather than differential pre-program economic trajectories. Combined with the absence of event-study pre-trends and the stability of the estimates when restricting comparisons to eventually treated counties or conditioning on baseline characteristics, these patterns provide support for the parallel-trends assumption underlying the baseline design.

Spatial spillovers A final concern is that grant-assisted construction may affect labor markets beyond the county in which the project is located. Cross-county spillovers could arise if contractors and subcontractors hire workers or expand activity elsewhere in the surrounding labor market, or if construction activity is reallocated across county boundaries. In either case, interference between treated and comparison counties would complicate the interpretation of the county-level estimates.¹⁶ I therefore examine whether construction activity responds in neighboring counties that do not themselves receive CWA construction grants.

I define neighboring labor markets using commuting zones, which group counties according to cross-county commuting patterns (Tolbert & Sizer, 1996). Commuting zones are useful in this setting be-

¹⁶Auerbach et al. (2020), using Department of Defense procurement contracts, document positive geographic and industry spillovers from federal spending that decline with distance.

cause they capture economically integrated local labor markets while providing complete geographic coverage, including rural and nonmetropolitan counties that are common in the Construction Grants Program (Dupor & McCrory, 2018; Park et al., 2026). I use the USDA Economic Research Service crosswalk of the Tolbert and Sizer (1996) commuting-zone boundaries to county FIPS codes.

To estimate spillovers, I restrict the sample to counties that never directly receive a construction-stage grant. I define a never-treated county as exposed in the first year that another county in its commuting zone begins grant-assisted construction and apply the same staggered DiD framework to this commuting-zone exposure. The comparison group consists of never-treated counties that have not yet been exposed through another county in their commuting zone, including counties whose commuting zones are never exposed. Because own-grant recipients are excluded throughout, any estimated response captures changes associated with grant-assisted construction elsewhere in the same local labor market rather than direct construction activity in the county itself.

Table 6 reports the resulting estimates. I find no statistically detectable response in construction employment, establishments, or average weekly wages over either the two- or four-year horizon. The point estimates are small relative to the corresponding direct effects and provide no indication of substantial offsetting employment changes in neighboring counties. The estimates should nevertheless be interpreted cautiously because the spillover design has limited statistical power. As the program expands, an increasing share of commuting zones contains at least one directly treated county, substantially reducing the number of never-treated counties that remain unexposed. For example, the exposed and unexposed groups contain 318 and 300 counties, respectively, in 1976, compared with 548 and 70 counties near the end of the sample. Given the resulting precision, the design cannot reliably distinguish construction-employment spillovers smaller than approximately 16–18 percent of the pre-treatment mean from zero. The null estimates therefore rule out neither small nor moderate cross-county effects.

I address the declining comparison pool with a complementary snapshot design in Appendix Section C.3. Rather than allowing commuting-zone exposure to accumulate over the entire sample, this exercise fixes exposure status at a sequence of calendar years, preserving a more balanced comparison between exposed and unexposed counties at each snapshot. Appendix Table C.14 reports

the results. Across snapshot years, the construction-sector estimates again provide little evidence of systematic responses in employment, establishments, or wages in neighboring counties.

Overall, the commuting-zone analyses provide no evidence of large or systematic construction-sector spillovers into neighboring counties. This finding is consistent with the place-specific nature of wastewater infrastructure construction and provides some support for interpreting the baseline estimates as effects concentrated in the county where construction occurs. The spillover tests are not sufficiently precise, however, to rule out smaller cross-county responses. I therefore view this evidence as suggestive rather than definitive with respect to the no-spillovers assumption.

5 Treatment Effect Heterogeneity

5.1 Early versus late program eras

Federal assistance under the Construction Grants Program was heavily front-loaded. As shown in Panel (a) of Figure 3, both the number of grant-assisted construction starts and the volume of federal assistance were concentrated in the 1970s and declined thereafter. The institutional environment of the program also changed substantially over time. I therefore examine whether the construction-employment response differed across program eras.

I classify counties according to the era of their *first* grant-assisted construction start: 1973–1979, 1980–1984, and 1985 or later. These periods correspond approximately to distinct phases of the program surrounding the 1977 and 1981 amendments, allowing for the lag between legislative changes, grant approval, and realized construction.¹⁷ The 1977 amendments expanded program authorizations and modified eligibility and priority rules, while the 1981 amendments subsequently narrowed eligible project categories, restricted funding for excess reserve capacity, placed greater emphasis on cost effectiveness, and reduced the federal cost share to 55 percent beginning in fiscal year 1985 (Congressional Research Service, 2019b).¹⁸ These changes altered the fiscal and institu-

¹⁷Because treatment is defined by construction start rather than grant award, the era classification necessarily reflects realized construction timing rather than the dates on which statutory changes took effect.

¹⁸The 1977 amendments also included provisions affecting collection systems and allowed single grants combining design and construction for certain smaller projects.

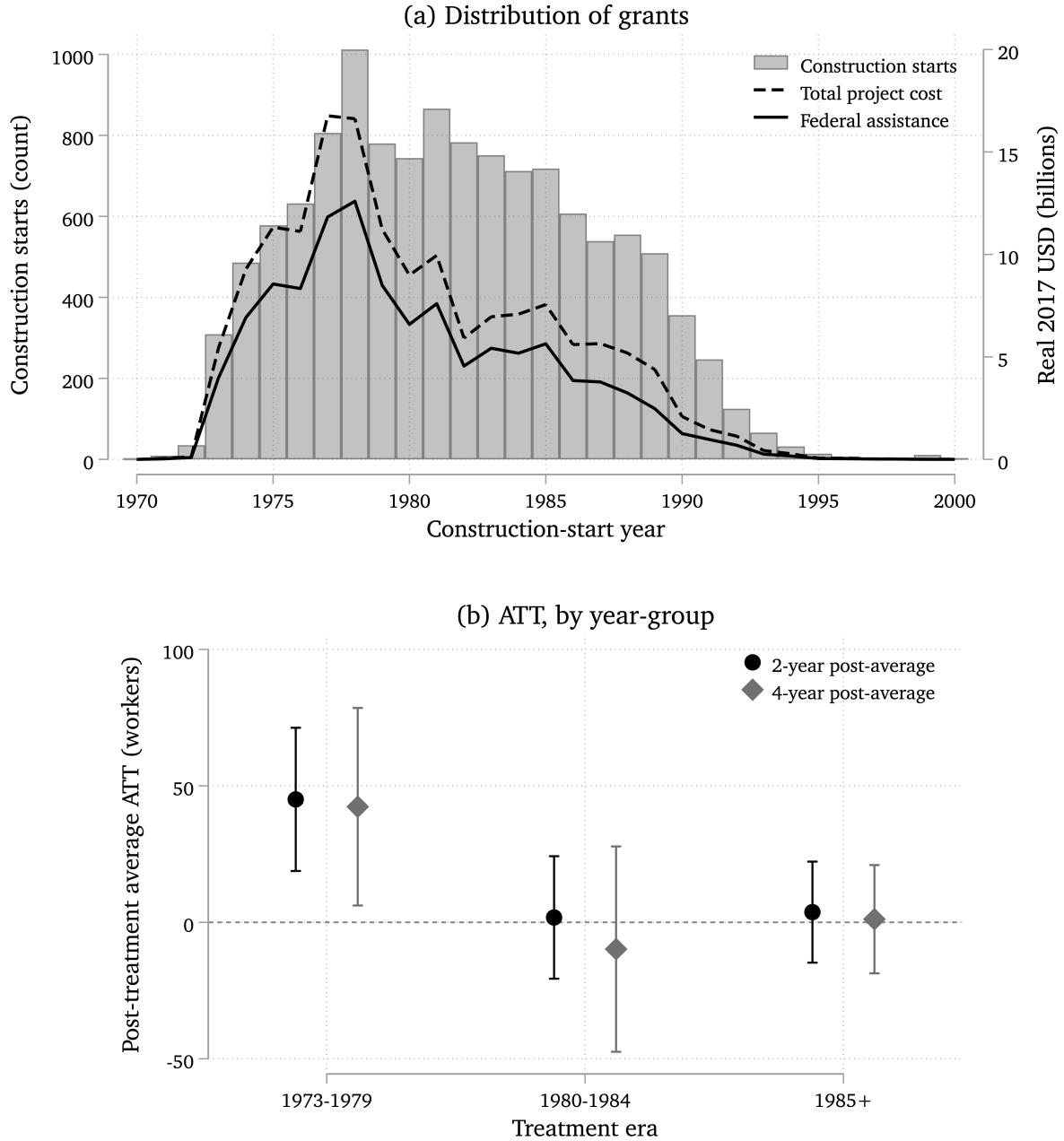


Figure 3: Construction activity and construction-employment effects by program era. Panel (a) plots the annual number of grant-assisted construction starts and total project costs and federal assistance, aggregated by construction-start year; dollar values are in billions of real 2017 USD. Panel (b) reports CSDiD average treatment effects on construction employment over the first 2 and 4 years, by era of first construction start (1973–1979, 1980–1984, and 1985 or later). Spikes denote 95% confidence intervals.

tional setting in which federally assisted wastewater projects were undertaken and provide a natural motivation for allowing their local employment effects to differ over the life of the program.

Panel (b) of Figure 3 reports the estimated construction-employment effects separately by era. For each specification, the treated sample consists of counties whose first construction start occurs within the indicated period, while counties untreated at time t form the comparison group. The employment response is concentrated in the early years of the program. Among counties first treated in 1973–1979, the estimated effects are positive and statistically significant over both the two-year and four-year horizons. The corresponding estimates for counties first treated in 1980–1984 and in 1985 or later are smaller and statistically indistinguishable from zero.

The decline in estimated employment effects over time coincides with substantial changes in the scale and institutional environment of the Construction Grants Program. The era estimates alone do not identify which of these changes accounts for the pattern, but they indicate that the local construction-employment response was concentrated in the earlier phase of the program.

5.2 Slack in the economy

A recurring hypothesis in the fiscal multiplier literature is that government spending generates larger real effects when there is greater economic slack (Auerbach & Gorodnichenko, 2012; Nakamura & Steinsson, 2014; Owyang et al., 2013; Ramey & Zubairy, 2018). When labor markets are slack, additional public demand can draw otherwise underutilized workers into employment; when labor markets are tight, the same demand may instead reallocate workers from existing activity or encounter capacity constraints. I examine whether the construction-employment effects of CWA projects exhibit this pattern using cross-county variation in local unemployment.

County-level unemployment rates come from the BLS Local Area Unemployment Statistics (LAUS). Because county-level LAUS data begin in 1976, I use each county’s 1976 unemployment rate—the earliest available measure—as a baseline proxy for local labor-market slack.¹⁹ Using a common calendar year also avoids comparing historical LAUS estimates across years, which is problematic because the BLS cautions that substate estimates from this period are not necessarily internally

¹⁹Appendix Section C.4 considers alternative classifications based on unemployment measured one and two years before each county’s construction start. Across these specifications, the two-year construction-employment response remains consistently larger in higher-unemployment counties, although the precision of the difference varies; the four-year estimates are considerably less precise.

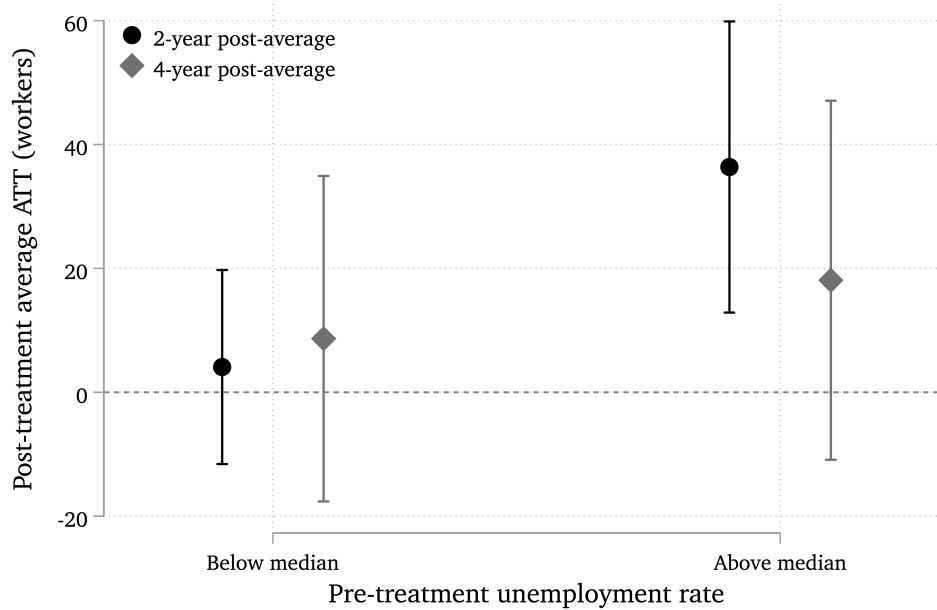


Figure 4: Construction-employment effects by local labor-market slack. Counties are classified using their 1976 unemployment rate from the BLS Local Area Unemployment Statistics (LAUS), split at the median of 6.25 percent. Points report CSDiD average treatment effects over the first 2 and 4 years after treatment. Spikes denote 95% confidence intervals.

consistent over time.²⁰ I divide counties according to whether their 1976 unemployment rate is above or below the median of 6.25 percent and estimate the baseline CSDiD specification separately for the two groups.

Figure 4 shows a pronounced difference in the construction-employment response across the two groups. In counties with above-median unemployment, the estimated employment effect is positive and statistically significant, particularly over the two-year horizon. In below-median-unemployment counties, the corresponding estimates are substantially smaller and statistically indistinguishable from zero. The pattern is therefore consistent with the state dependence emphasized in the fiscal multiplier literature: federally assisted construction generates larger net employment gains where local labor markets initially have greater unused capacity. In tighter labor markets, additional construction demand appears to translate less strongly into net local employment growth. This heterogeneity also suggests that the short-run employment consequences of infrastructure spending

²⁰County-level LAUS unemployment data prior to 1990 are not publicly posted and were obtained from the BLS by request. Appendix Section C.4 considers alternative classifications based on unemployment measured closer to each county's own construction start and obtains the same qualitative pattern.

depend importantly on the labor-market conditions under which projects are undertaken.

Are heterogeneity results driven by grant size? One potential concern is that the heterogeneity across program eras and local labor-market conditions reflects differences in the scale of grant-assisted construction rather than differences in employment responses. This concern is particularly relevant across program eras, since federal assistance was concentrated in the early years of the program, but could also arise if grant size differed systematically with local unemployment.

I examine this possibility in Appendix Section C.5. First, I estimate construction-employment effects separately by quartile of the federal award received in a county's first construction-start year. The estimates provide some evidence of a larger two-year response among counties receiving the largest awards, but there is no monotonic relationship between grant size and employment effects across quartiles, and the difference is not apparent over the four-year horizon. Second, I examine grant size, unemployment, and treatment timing jointly using county-level treatment-effect proxies. Conditional on grant size per capita and population, grant size is statistically insignificant at both horizons, while the treatment-timing terms remain statistically significant. Unemployment is imprecisely estimated in this exercise.

These results provide little evidence of a systematic dose-response relationship and indicate that differences in grant size alone do not account for the stronger employment effects among early program cohorts. The joint specification provides clearer evidence for the era result than for the slack heterogeneity. The treatment-timing terms remain statistically significant after conditioning on grant size, whereas the unemployment coefficient is imprecisely estimated and does not reproduce the positive slack relationship found in the subgroup estimates. I therefore interpret this exercise primarily as evidence that differences in initial grant size do not mechanically account for the stronger employment response among earlier program cohorts.

6 Total County Impact of the Construction Grants Program

A natural extension of the construction-sector analysis is to ask whether grant-assisted wastewater infrastructure projects generate broader effects within treated counties. Such effects could arise through demand for inputs and subcontracting services, increased spending by workers and firms, or changes in local labor supply and population. Total employment therefore captures a considerably broader response than construction employment: it combines any indirect labor-demand effects with changes in the number of workers and households located in the county.

Estimating these broader effects is less straightforward than estimating the direct construction-sector response for two reasons. First, there is no equally sharp event date. Construction start provides a natural treatment date for on-site construction labor, but activity elsewhere in the local economy may begin during planning, engineering, procurement, and other pre-construction stages (Leeper et al., 2010; Ramey, 2020). Second, the appropriate functional form is less clear. The direct construction response is naturally expressed in levels because the labor required to build a facility is tied to the scale and engineering requirements of the project rather than proportionally to the size of the county's existing construction sector. Broader local effects, by contrast, operate through the surrounding economy. Because treated counties differ substantially in economic size, a common proportional response is more plausible for these outcomes than it is for the direct construction response.

I therefore treat the estimates in this section as complementary evidence on broader county-level responses rather than as the paper's primary causal estimates. I use a treatment date two years before construction start as the preferred timing convention and report both level and proportional specifications. Appendix Section D reports a substantially broader set of specifications varying both treatment timing and functional form.

Treatment timing For the broader county outcomes, I define the preferred treatment date as $\tilde{G}_i = G_i - 2$, where G_i is the construction-start year used for the construction-sector analysis. The two-year adjustment is motivated by the timing of the grant process. Grant awards generally precede

physical construction, and activities such as planning, engineering, procurement, and contracting may generate local economic responses before workers arrive on the construction site. Appendix Table B.11 reports the distribution of time between grant award and construction start and provides the empirical basis for the two-year anticipation window.

This timing convention is necessarily less precise than the construction-start treatment used for the direct construction-sector estimates. Using G_i risks assigning pre-construction responses to the pre-treatment period, while using the Step 3 award date aligns treatment with the commitment of federal funds but not necessarily with the onset of local economic activity: the lag between award and construction start varies substantially across projects. Earlier Step 1 and Step 2 awards are still farther removed from physical construction and cannot be linked cleanly to each subsequent construction grant in all cases because the grant records do not provide a common project identifier across steps. A fixed shift before construction start avoids these matching problems but requires choosing an anticipation window.

For this reason, I do not interpret $G_i - 2$ as a uniquely identified date at which broader local effects begin. It is a preferred alignment that places treatment between the commitment of construction funding and the onset of physical construction for much of the sample.²¹ Appendix Section D therefore reports estimates using construction start, alternative grant-award dates, and shifts of one, two, and three years before construction start. In particular, Figures D.13 and D.14 overlay the event-study estimates under the one-, two-, and three-year adjustments, allowing the sensitivity of the estimated dynamics to the timing convention to be assessed directly.

Functional form The functional-form distinction is also more consequential for these outcomes than for construction employment. As discussed in Section 4.1, the direct construction response has a strong engineering basis for an additive specification: a wastewater project requires a bundle of construction labor determined primarily by project scale and design rather than by the size of the

²¹Appendix Table B.12 shows that, among all Step 3 (or Step 2+3) grants, construction begins within one year of award for 65.9% of projects, within two years for 85.3%, and within three years for 92.8%. The corresponding shares for the construction-defining grants in the estimation sample are 67.8%, 87.1%, and 94.1%, respectively. These distributions motivate using a two-year shift as the preferred anticipation adjustment while treating one- and three-year shifts as nearby robustness checks.

host county's pre-existing construction sector. No analogous restriction applies to indirect employment effects. If grant-funded activity raises demand through local suppliers, worker spending, or other multiplier channels, the resulting employment response may reasonably scale with the size of the local economy.

I therefore place greater weight on proportional specifications for sectoral and total employment in this section, while reporting levels for comparison and for translating estimates into economically interpretable magnitudes. This distinction matters empirically. Appendix Section D examines the sensitivity of the aggregate results to functional form and reports additional diagnostics on selection and treatment timing (Tables B.8, B.9, E.21, and E.22). Taken together, these exercises make the aggregate results less definitive than the direct construction-sector estimates. Accordingly, the discussion below emphasizes results that are reasonably stable across specifications rather than relying on statistical significance in a single level specification.

6.1 Sectoral and total employment

Table 7 reports employment effects for several non-construction sectors and for total county employment under the preferred two-year timing adjustment. Panel A reports effects in levels, while Panel B reports proportional effects. The level estimates are positive and precisely estimated for several sectors. Over the first two years, employment increases by an estimated 15.0 workers in transportation and utilities, 60.7 in manufacturing, 15.5 in wholesale trade, and 61.6 in retail trade. Total county employment increases by 216.9 workers per county-year. The corresponding four-year estimates remain positive and are generally larger.

The evidence is weaker when employment is measured in logs. The estimated two-year effect on total employment is 0.34 percent (s.e. 0.28), while the four-year effect is 0.44 percent (s.e. 0.39); both estimates are imprecise. Transportation and utilities, manufacturing, and wholesale trade likewise exhibit small and imprecise proportional effects despite their positive level estimates. Retail trade provides the clearest evidence of a proportional response: retail employment increases by an estimated 0.46 percent over two years and 0.93 percent over four years, with the latter statistically

Table 7: Sectoral and total employment effects

Panel A. Level outcomes							
	Transp. Util.	Manu.	Whole. trade	Retail trade	Agri.	Mining	Total emp.
2-yr post avg.	15.02*** (4.27)	60.66*** (17.16)	15.45*** (4.71)	61.55*** (11.53)	13.12* (7.34)	2.70 (8.00)	216.91*** (49.18)
4-yr post avg.	21.11*** (6.57)	97.13*** (26.21)	22.50*** (7.26)	83.02*** (16.33)	15.06* (9.08)	-1.95 (11.85)	311.94*** (76.84)
6-yr post avg.	22.41** (8.85)	102.56*** (34.54)	20.84** (9.59)	71.61*** (20.00)	10.25 (10.22)	-11.24 (15.55)	261.60*** (96.54)
N	35,457	35,079	34,721	38,477	21,986	13,800	38,746
Pre-treat. avg.	850.10	4,209.43	876.49	2,723.78	107.70	422.41	12,611.59
Panel B. Log outcomes (percent)							
	Transp. Util.	Manu.	Whole. trade	Retail trade	Agri.	Mining	Total emp.
2-yr post avg.	0.37 (0.58)	0.35 (0.56)	0.10 (0.54)	0.46* (0.26)	0.41 (1.90)	-0.28 (1.67)	0.34 (0.28)
4-yr post avg.	0.89 (0.76)	0.58 (0.77)	0.20 (0.69)	0.93** (0.36)	0.29 (2.29)	-1.88 (2.12)	0.44 (0.39)
6-yr post avg.	1.34 (0.94)	0.74 (0.98)	-0.55 (0.87)	0.78 (0.50)	1.45 (2.72)	-3.48 (2.71)	0.20 (0.52)
N	35,457	35,079	34,721	38,477	21,986	13,800	38,746
Pre-treat. avg.	850.10	4,209.43	876.49	2,723.78	107.70	422.41	12,611.59

Notes: This table reports CSDiD average treatment effects on sectoral and total county employment over the first 2, 4, and 6 years after the adjusted treatment date $\tilde{G}_i = G_i - 2$. Panel A reports effects in employment levels. Panel B reports effects in logs, with coefficients expressed in percent. Outcomes include transportation and utilities, manufacturing, wholesale trade, retail trade, agriculture, mining, and total employment. Standard errors are clustered at the county level and reported in parentheses. *Pre-treat. avg.* reports the average outcome in levels in the pre-treatment reference period. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

significant at the 5 percent level. Agriculture and mining show little systematic evidence of a post-treatment employment response.

In sum, the sectoral results provide suggestive, but not conclusive, evidence that employment gains extended beyond construction. Although the level estimates are positive across several non-construction sectors, most proportional estimates are small and imprecise, making the evidence sensitive to functional form. Retail trade provides the clearest evidence of an indirect response, consistent with increased local demand following grant-assisted construction. The total-employment estimates are

Table 8: Per-capita personal income effects

	Per-capita personal income
2-yr post avg.	0.025 (0.032)
4-yr post avg.	0.087** (0.038)
6-yr post avg.	0.100** (0.045)
N	63,898
Pre-treatment avg. (1973, in thousands of real 2017 dollars)	19.935

Notes: This table reports CSDiD average treatment effects on county personal income per capita over the first 2, 4, and 6 years after the adjusted treatment date $\tilde{G}_i = G_i - 2$. Personal income per capita is measured in thousands of real 2017 USD. Standard errors are clustered at the county level and reported in parentheses. *Pre-treatment avg.* reports average personal income per capita in 1973. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

similarly imprecise and therefore do not rule out modest aggregate employment effects, but they provide substantially weaker evidence than the direct construction-sector estimates.

6.2 Per-capita personal income

I next examine whether grant-assisted construction is associated with changes in county personal income per capita. This outcome provides a broader measure of local economic activity while avoiding the mechanical relationship between total employment and county size that complicates the interpretation of the employment results above.

Table 8 reports estimates under the preferred two-year timing adjustment. The estimated effect over the first two years is small and imprecise. The point estimates become larger at longer horizons: personal income per capita increases by an average of \$87 per person per year over the four-year window and \$100 over the six-year window, with both estimates statistically significant at the 5 percent level. As with total employment, however, the timing of a broader county-level response is less clearly defined than the timing of the direct construction response. Appendix Section D examines the sensitivity of these estimates to alternative treatment anchors and timing adjustments (Table D.20 and Figures D.12 and D.14).

The magnitude of the four-year estimate can be translated into a back-of-the-envelope income-based benchmark. An average annual increase of \$87 per person over four years implies approximately \$348 in cumulative per-capita personal income. Relative to mean federal assistance per capita associated with the county's first construction start (\$191.65 in 2017 dollars), this corresponds to approximately \$1.82 in cumulative local personal income per dollar of federal assistance. Normalizing instead by mean eligible project cost per capita (\$258.75), which includes the nonfederal matching share, yields approximately \$1.34 per dollar of total eligible project spending.²²

The personal-income estimates therefore provide suggestive evidence of broader local economic effects, particularly at longer horizons. At the same time, they are more sensitive than the construction-sector estimates to assumptions about when broader economic activity should be expected to respond. I therefore interpret these results as complementary evidence that the effects of grant-assisted construction may extend beyond the construction sector, rather than as a precise estimate of a local fiscal multiplier.

7 Conclusion

This paper studies the short-run local labor-market effects of the Clean Water Act (CWA) Construction Grants Program, one of the largest federal infrastructure investment programs in U.S. history. Using project-level construction start dates linked to county-level labor-market outcomes, I estimate dynamic treatment effects in a staggered difference-in-differences framework. The analysis focuses first on the construction sector, where the timing and expected direction of the labor-demand response are most closely tied to the physical implementation of grant-assisted projects, and then examines whether these effects extend to broader measures of local economic activity.

The clearest finding is a direct expansion of local construction activity when grant-assisted construction begins. Construction employment increases by approximately 20 workers per county-year over

²²These calculations are reduced-form accounting benchmarks rather than structural fiscal multipliers. They relate cumulative within-county personal-income changes to the public funds associated with the first construction start and do not account for financing, migration, inter-county spillovers, general-equilibrium effects, or the incidence of the underlying taxes and expenditures. The magnitude is nevertheless broadly consistent with the geographic cross-sectional fiscal-spending literature: Chodorow-Reich (2019) reports a preferred cross-study output multiplier of approximately 1.8, while individual studies vary substantially across settings (e.g., Corbi et al., 2019).

the first two years of a county's initial construction episode, accompanied by increases in construction establishments and wages. The employment response is concentrated in heavy construction, the subsector that includes water and sewer infrastructure work, with little systematic response in general building or specialized trade construction. The employment estimate corresponds to approximately 39.8 job-years per treated county over the first two years, or about 0.41 job-years per \$100,000 of federal assistance in 2017 dollars.

The magnitude of this response varies substantially across program eras and local labor-market conditions. Construction-employment gains are concentrated among counties entering the program during its earlier, more intensive years and are considerably larger in counties with greater pre-existing labor-market slack. In counties with above-median unemployment, grant-assisted construction generates clear employment gains, while estimates in tighter labor markets are substantially smaller. These patterns are robust to alternative measures of local slack and do not appear to be explained simply by differences in grant size. The results are therefore consistent with public construction demand generating larger net employment responses where underutilized local labor capacity is available.

Evidence that the construction response propagates through the broader county economy is considerably weaker. Level estimates indicate increases in several non-construction sectors and in total county employment, but most of these effects become small and imprecise under proportional specifications. Retail employment provides the clearest evidence of an employment response outside construction. Personal income per capita also increases at longer horizons under the preferred timing specification, with the four-year estimate implying a cumulative increase of approximately \$1.82 per dollar of federal assistance. Because the timing and functional form of these broader responses are less tightly linked to physical construction, I interpret this evidence as suggestive rather than as a precise estimate of a local fiscal multiplier.

In conclusion, the results distinguish between the direct local labor demand generated by infrastructure implementation and the broader stimulus effects sometimes attributed to public investment. The former is clearly visible in the construction sector and varies systematically with local labor-market conditions; evidence for the latter is more limited. This distinction is particularly relevant

for environmental infrastructure programs such as the CWA Construction Grants Program, whose primary objectives were regulatory compliance and pollution abatement rather than short-run economic stimulus. The results show that implementation nevertheless affected local labor markets, while cautioning against interpreting those employment effects themselves as measures of the program's net social benefits.

Several limitations are important for interpretation. The estimates are local and partial-equilibrium in nature and therefore do not capture general-equilibrium adjustments through commuting, migration, prices, or the financing of federal expenditures. The analysis of broader county outcomes also requires less sharply determined assumptions about when indirect economic responses begin, whether they operate additively or proportionally, and through which mechanisms they arise. Although the concentration of the short-run employment response in heavy construction is consistent with a direct labor-demand channel, the empirical design does not separately identify this mechanism from other channels through which wastewater infrastructure may affect local labor markets. In particular, improvements in downstream environmental quality and health could also affect labor supply or broader local economic activity. Distinguishing these mechanisms would require linking treatment plants to downstream populations through the hydrologic network. Future work integrating these implementation-stage labor-market responses with the environmental, public-health, and other benefits of wastewater infrastructure—as well as its financing and incidence—would provide a more complete assessment of the economic consequences of large-scale water infrastructure investment.

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Appendices

A Sample Construction

I construct the analysis sample by linking records from the Clean Water Act Construction Grants Program to county-level economic outcomes from the QCEW and BEA. The sample covers the 48 contiguous states and Washington, D.C. County definitions are harmonized across the three data sources using BEA geographic conventions, including the consolidation of several Virginia independent cities with adjacent counties.

Construction grants The primary source for Construction Grants Program awards is the EPA's Grant Information Control System (GICS), obtained through a Freedom of Information Act request. The data identify the recipient municipality, associated wastewater treatment facility, grant step, award and construction-start dates, federal award amount, and eligible project cost. Keiser and Shapiro (2019) use the same administrative data and provide additional validation of the records in their Appendix Section B.4.

I restrict the analysis to Step 3 grants, which finance construction, and Step 2+3 grants, which combine design and construction. Treatment timing is defined by the recorded construction-start date because the primary analysis focuses on the local labor-market response to the onset of physical construction. For records without a construction-start date, I use the award date. I restrict construction starts to 1973–1990, covering the principal years of the program represented in the analysis.

Grant records are assigned to counties by linking facilities to the EPA Clean Watershed Needs Survey (CWNS) facility registry and matching reported county names to county FIPS codes.²³ Records that cannot be assigned to a county are excluded. County definitions are then harmonized with the BEA and QCEW outcome data; in particular, Virginia independent cities are consolidated according to the BEA combined-area convention.

²³The CWNS facility registry is obtained from the replication files of Flynn and Marcus (2023).

Starting from 28,280 GICS grant records, I exclude records with zero federal awards, records that cannot be assigned to a county or have missing award amounts, grants other than Step 3 or Step 2+3, and grants outside the 1973–1990 construction window. Restricting these records to Step 3 and Step 2+3 grants yields 12,374 construction-related grants; imposing the 1973–1990 construction window leaves 11,921 records used to construct county treatment histories.

County sample The economic panel covers the 48 contiguous states and Washington, D.C. After harmonizing county definitions across the grant, QCEW, and BEA data, the panel contains 3,080 county-level geographic units. Of these, 2,437 counties receive at least one qualifying construction grant during the sample period and 643 are never treated.

A small number of counties do not have the 1969–1971 population information required for specifications that condition on pre-program characteristics. Five counties lack a usable pre-period population measure, primarily because they were created after the beginning of the pre-period or because of historical changes in geographic reporting. These counties are excluded only from specifications requiring the population covariate. Consequently, the corresponding sample contains 2,436 ever-treated and 639 never-treated counties.

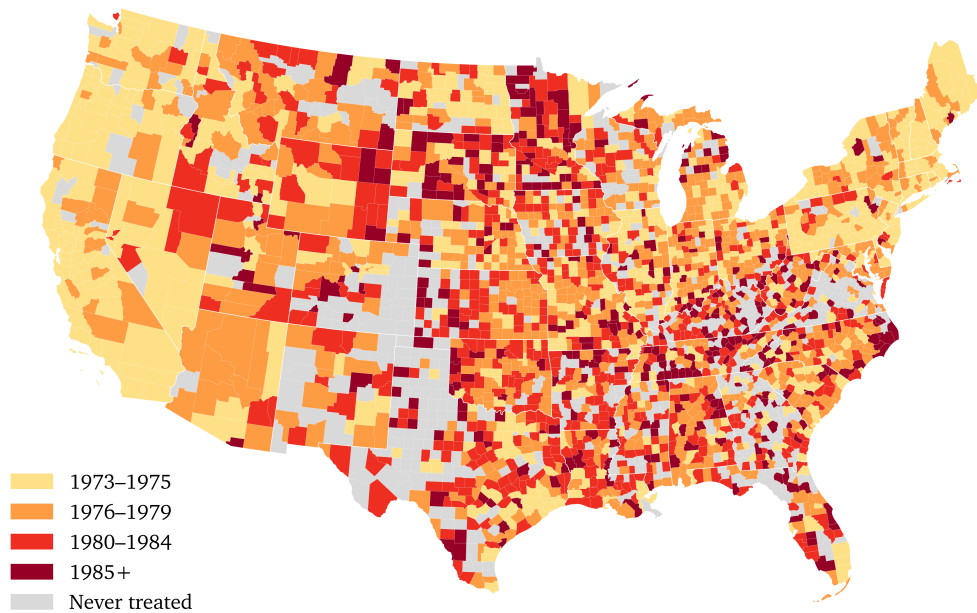
A.1 Estimation sample

In this section, I describe how the estimation sample is constructed and characterize its composition along four dimensions: the comparison between never-treated and ever-treated counties, the QCEW coverage restriction that excludes the 1973–74 cohorts from construction-employment specifications, the single-episode sample restriction and its scale, and the resulting final estimation sample by cohort and event-time horizon.

A.1.1 Comparing never-treated and ever-treated counties

The program’s reach was broad but uneven, both in the timing of construction and in the amount of federal assistance received. Figure [A.1](#) describes this variation geographically, while Table [A.1](#)

(a) First year of grant-assisted construction start



(b) Total real federal award per capita

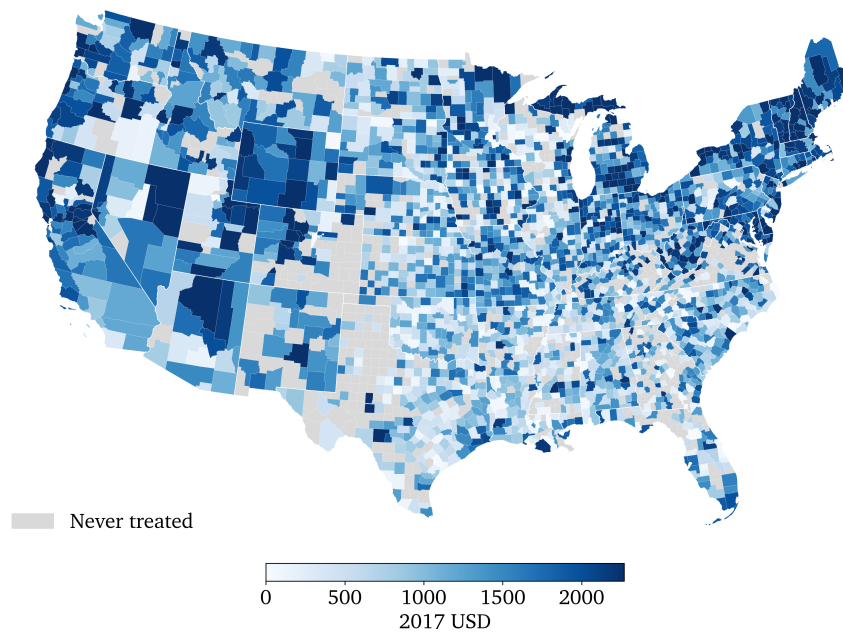


Figure A.1: Geographic distribution of grants under the CWA Construction Grants Program. Panel (a) maps the first year of grant-assisted construction start by county; gray denotes never-treated counties. Panel (b) maps cumulative federal grant awards per capita in real 2017 USD; darker shades indicate larger awards. The color scale in Panel (b) is winsorized at the 99th percentile.

compares counties that ever receive a construction grant with those that never do.

Panel (a) of Figure A.1 maps each county’s first grant-assisted construction start and therefore describes the geographic distribution of treatment cohorts used in estimation. Counties first enter treatment between 1973 and 1993, while counties shown in gray never receive a construction-stage grant. Panel (b) maps cumulative federal awards per resident in real 2017 dollars. The scale is winsorized at the 99th percentile (\$2,269 per capita) to prevent a small number of large per-capita awards in sparsely populated counties from compressing the remaining variation.

Table A.1 shows substantial differences in pre-program levels between ever-treated and never-treated counties. Over 1969–1971, ever-treated counties average 76,109 residents, compared with 27,329 among never-treated counties, and have higher total and per-capita personal income as well as higher construction-sector income per capita. The 1975 QCEW data show similar differences in county size, although these comparisons should be interpreted cautiously because the earliest construction cohorts had already entered treatment by 1975. These differences are consistent with the program’s allocation toward communities with larger populations and wastewater-treatment needs.

Differences in pre-program growth rates are considerably smaller. Ever-treated and never-treated counties are statistically indistinguishable in 1969–1971 growth in personal income, construction-sector income, and construction-sector income per capita. Population growth is higher among ever-treated counties (2.58 versus 1.72 percent, $p < 0.01$), while per-capita income growth is lower (5.17 versus 6.85 percent, $p = 0.02$). Thus, treatment status is clearly associated with pre-program economic levels, but the two groups exhibit fewer systematic differences in short-run pre-program growth. Because these summary statistics cover only a short pre-program window and do not account for treatment cohorts, I treat them primarily as descriptive evidence on sample composition. The event-study estimates and the additional selection tests reported in Section 4.1 provide the more direct evidence relevant to the parallel-trends assumption.

A.1.2 QCEW data restrictions

The QCEW SIC-based industry series used to measure construction employment, establishments, and average weekly wages begins in 1975. Ever-treated counties whose first grant-assisted con-

Table A.1: Summary statistics by Construction Grants Program treatment status

	Never treated	Ever treated	<i>p</i> -value
Panel A. BEA Personal Income (1969–1971; Real 2017 USD)			
Avg. population	27,329.23 (5,767.51)	76,108.59 (5,007.51)	0.00
Population growth (%)	1.72 (0.24)	2.58 (0.09)	0.00
Avg. personal income (thousands)	516,449.89 (129,479.40)	1,593,235.68 (123,838.13)	0.00
Personal income growth (%)	8.31 (0.56)	7.78 (0.15)	0.36
Avg. personal income per capita	15,518.94 (161.64)	16,784.04 (76.69)	0.00
Personal income per capita growth (%)	6.85 (0.70)	5.17 (0.15)	0.02
Avg. construction-sector personal income	20,212.69 (3,821.00)	85,978.31 (6,633.32)	0.00
Construction-sector personal income growth (%)	21.17 (4.30)	16.93 (1.81)	0.36
Avg. construction personal income / population	0.68 (0.02)	0.82 (0.01)	0.00
Construction personal income per capita growth (%)	19.14 (4.28)	13.93 (1.72)	0.26
Panel B. QCEW Employment Statistics (1975)			
Total employment	5,619.90 (1,061.51)	15,664.68 (920.48)	0.00
Total establishments	444.40 (82.85)	1,066.04 (54.19)	0.00
Avg. weekly wage, all sectors	523.13 (7.29)	535.02 (2.42)	0.12
Construction employment	351.62 (54.28)	950.69 (51.26)	0.00
Construction establishments	49.23 (6.55)	132.12 (6.18)	0.00
Construction avg. weekly wage	595.07 (9.23)	640.50 (3.95)	0.00
Observations	639	2,436	

Notes: This table reports summary statistics for never-treated and ever-treated counties under the CWA Construction Grants Program. Panel A reports averages over 1969–1971; growth rates are percentage changes from 1969 to 1971. Panel B reports QCEW outcomes in 1975, the first year with consistent county-level QCEW coverage. Standard errors are reported in parentheses. *p*-value reports the *p*-value from a test of equality between the never-treated and ever-treated means. Dollar values are in real 2017 USD.

struction start occurred in 1973 or 1974 therefore have no pre-treatment QCEW observations. The Callaway–Sant’Anna estimator cannot estimate treatment effects for these cohorts, so they do not contribute to the construction-sector estimates. This restriction affects 430 of 2,437 ever-treated

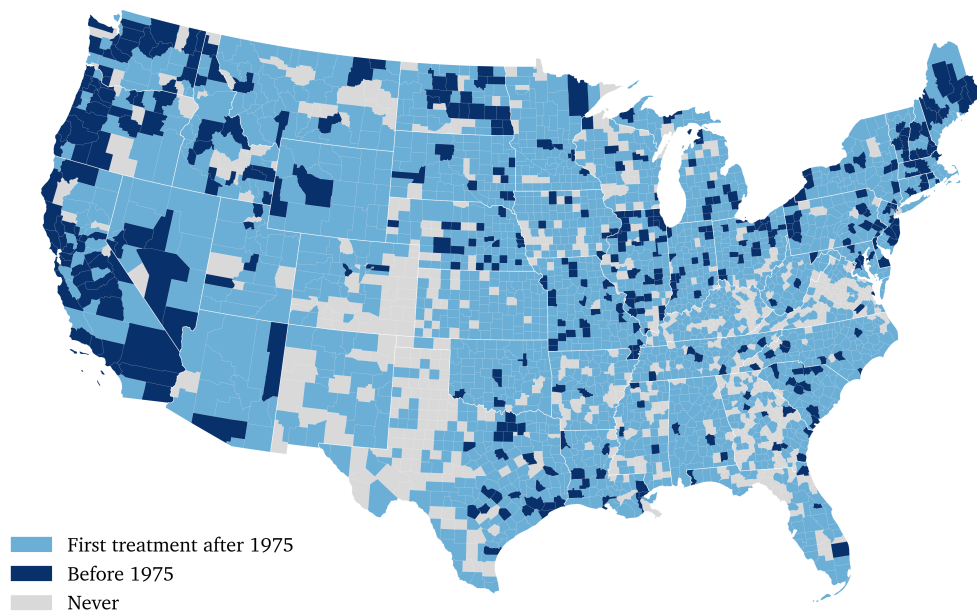


Figure A.2: Counties by construction-employment estimation status. Light blue denotes ever-treated counties first treated in 1975 or later. Dark blue denotes counties first treated in 1973–1974, before consistent QCEW construction-employment coverage begins. Gray denotes never-treated counties. Of 2,437 ever-treated counties, 430 (17.6%) were first treated in 1973–1974.

counties (17.6%), shown in Figure A.2. The remaining 2,007 ever-treated counties were first treated in 1975 or later, while all 643 never-treated counties remain available as potential comparisons.

Figure A.3 shows the corresponding distribution of first treatment years. Construction starts are concentrated in the second half of the 1970s and decline thereafter. Because QCEW coverage begins in 1975, the available pre-treatment window also varies across retained cohorts: counties first treated in 1976 have one year of pre-treatment QCEW data, those first treated in 1977 have two years, and later cohorts contribute progressively longer pre-treatment histories. Consequently, estimates at longer pre-treatment event times are identified by later-treated cohorts. Table A.6 reports the resulting sample composition by cohort and event-time horizon.

Whether the QCEW restriction affects the population represented by the construction-sector estimates depends on how the excluded early cohorts compare with the retained cohorts. Table A.2 compares their pre-program characteristics.

The excluded 1973–1974 cohorts are substantially larger than the retained cohorts. They average

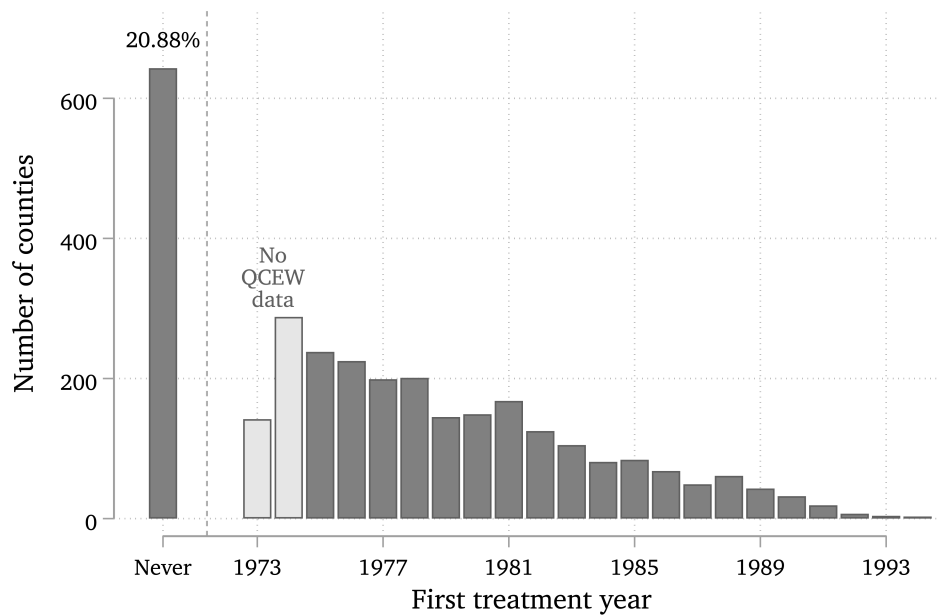


Figure A.3: Distribution of counties by first treatment year. *Never* denotes counties that never receive a construction-stage grant. Light gray bars denote the 1973–1974 treatment cohorts, which precede consistent QCEW construction-employment coverage.

189,849 residents, compared with 51,865 among counties first treated in 1975 or later, and \$4.30 billion in personal income, compared with \$1.02 billion. Other pre-program level characteristics likewise differ significantly between the two groups. The 1975 QCEW differences should be interpreted cautiously because these outcomes are already post-treatment for the excluded cohorts.

In contrast, pre-program growth rates measured over 1969–1971 are similar across the two groups. I find no statistically significant differences in population growth (2.86 versus 2.51 percent, $p = 0.16$), personal income growth (7.71 versus 7.79 percent, $p = 0.82$), per-capita income growth (4.79 versus 5.25 percent, $p = 0.21$), construction-sector income growth (17.70 versus 16.78 percent, $p = 0.89$), or construction income per capita growth (14.07 versus 13.93 percent, $p = 0.98$). Thus, the QCEW start date disproportionately excludes some of the program’s earliest and largest recipient counties, but the excluded and retained cohorts do not exhibit detectable differences in pre-program growth over the available 1969–1971 window. The construction-sector estimates should therefore be interpreted as applying to counties first entering grant-assisted construction from 1975 onward rather than to the earliest 1973–1974 cohorts.

Table A.2: Summary statistics for pre-1975 and 1975+ treatment cohorts

	Pre-1975	1975+	<i>p</i> -value
Panel A. BEA Personal Income (1969–1971; Real 2017 USD)			
Avg. population	189,849.12 (24,597.44)	51,864.70 (2,776.11)	0.00
Population growth (%)	2.86 (0.22)	2.51 (0.10)	0.16
Avg. personal income (thousands)	4,298,603.29 (614,407.07)	1,016,506.60 (66,701.67)	0.00
Personal income growth (%)	7.71 (0.36)	7.79 (0.17)	0.82
Avg. personal income per capita	18,763.02 (199.27)	16,364.07 (79.79)	0.00
Personal income per capita growth (%)	4.79 (0.33)	5.25 (0.17)	0.21
Avg. construction-sector personal income	235,128.95 (32,333.53)	54,174.67 (3,780.12)	0.00
Construction-sector personal income growth (%)	17.70 (6.61)	16.78 (1.69)	0.89
Avg. construction personal income / population	0.98 (0.04)	0.78 (0.01)	0.00
Construction personal income per capita growth (%)	14.07 (6.09)	13.93 (1.63)	0.98
Panel B. QCEW Employment Statistics (1975)			
Total employment	21,629.70 (2,716.40)	14,912.11 (979.20)	0.02
Total establishments	1,433.40 (156.98)	1,019.58 (57.78)	0.01
Avg. weekly wage, all sectors	554.07 (7.04)	532.62 (2.57)	0.00
Construction employment	1,261.41 (152.62)	911.75 (54.48)	0.03
Construction establishments	179.96 (18.18)	126.11 (6.57)	0.01
Construction avg. weekly wage	692.34 (11.51)	633.87 (4.18)	0.00
Observations	429	2,003	

Notes: This table reports summary statistics for ever-treated counties, split between counties first treated in 1973–1974 and counties first treated in 1975 or later. Panel A reports averages over 1969–1971; growth rates are percentage changes from 1969 to 1971. Panel B reports QCEW outcomes in 1975, the first year with consistent county-level QCEW coverage. Standard errors are reported in parentheses. *p*-value reports the *p*-value from a test of equality between the two group means. Dollar values are in real 2017 USD.

A.1.3 Single-episode restriction

The baseline specification defines treatment as the start of a county’s first grant-assisted construction episode. Because many counties subsequently receive additional construction grants, later episodes

Table A.3: Scope of the single-episode sample restriction

	Count	Share (%)
Panel A. Ever-treated counties, by episode count (share of ever-treated)		
One construction episode only	679	27.9
Two or more construction episodes	1,758	72.1
Panel B. County-years (share of the full panel)		
Full panel	98,560	100.0
Ever-treated counties	77,984	79.1
Never-treated counties	20,576	20.9
Excluded by the restriction	34,563	35.1
as a share of ever-treated county-years		44.3
Panel C. Federal awards, millions of 2017 dollars (share of all awards)		
All awards	42,932.2	100.0
Retained: first-episode awards	7,368.3	17.2
Excluded: later-episode awards	35,563.8	82.8

Notes: This table reports the scope of the single-episode sample restriction, which drops county-year observations from the start of a county's second grant-assisted construction episode onward. Panel A reports ever-treated counties by the number of construction episodes. Panel B reports county-years in the full panel and the number excluded by the restriction. Shares are relative to the full panel except where otherwise indicated. Panel C reports federal awards retained and excluded by the restriction. Dollar values are in millions of real 2017 USD.

can overlap with the post-treatment path associated with the first episode. To isolate the response to the initial construction episode, I therefore retain observations through the year preceding a county's second construction start and drop subsequent county-year observations. The resulting estimand describes the employment response following a county's first grant-assisted construction episode before another episode begins.

Repeat construction episodes are common. Panel A of Table A.3 shows that 1,758 of 2,437 ever-treated counties (72.1 percent) begin a second episode during the sample period, while 679 counties receive only one episode. As shown in Panel B, the restriction removes 34,563 county-year observations, or 44.3 percent of ever-treated county-years. Panel C shows that the restriction is more consequential in terms of program expenditures: \$7.37 billion in federal awards are associated with observations retained in the estimation sample, compared with \$35.56 billion associated with excluded observations. Thus, 82.8 percent of observed federal awards occur in episodes outside the baseline estimation sample. The baseline estimates should accordingly be interpreted as effects asso-

Table A.4: Ever-treated counties remaining by event-time horizon

Horizon (yr)	0	1	2	3	4	5	6	7	8
Counties	2,437	1,883	1,552	1,341	1,164	1,059	966	906	847
Share (%)	100.0	77.3	63.7	55.0	47.8	43.5	39.6	37.2	34.8

Notes: This table reports the number and share of ever-treated counties remaining in the estimation sample at each event-time horizon. Horizon 0 is the year of the county's first grant-assisted construction start. Counties exit the sample once a second grant-assisted construction episode begins.

ciated with the initial construction episode rather than effects of a county's cumulative participation in the Construction Grants Program.

The restriction also changes the composition of the treated sample as the event-time horizon increases. Table A.4 reports the number of ever-treated counties that remain available at each horizon before a subsequent construction episode begins. Of the ever-treated counties, 77.3 percent remain in the sample at $h = 1$, while 55.0 percent remain at $h = 3$. Longer-horizon treatment effects therefore place progressively greater weight on counties with more widely spaced construction episodes.

Table A.5 further characterizes repeat participation. The median ever-treated county experiences two construction episodes and receives \$3.44 million in federal awards over the sample period, of which \$1.01 million is associated with its first episode. The corresponding means are 3.46 episodes and \$17.62 million in total awards, compared with \$3.02 million associated with the first episode. Among the 1,758 counties experiencing a second episode, the median interval between the first and second construction starts is two years, while the mean is 3.66 years, reflecting a right-skewed distribution of intervals.

These patterns clarify both the purpose and the cost of the baseline restriction. Excluding observations after subsequent construction starts produces a cleaner event-time interpretation around the first construction episode, but it substantially reduces the contribution of repeatedly treated counties, particularly at longer horizons and in terms of program dollars represented. For this reason, I treat the single-episode specification as the baseline design for isolating the initial construction response rather than as evidence on the cumulative effect of all Construction Grants Program investment received by a county. Appendix Section C.6 relaxes this restriction and assesses whether the main conclusions are sensitive to retaining observations following subsequent construction episodes.

Table A.5: Distribution of grants and single-episode sample restrictions across counties

	Mean	SD	p5	Median	p95
Panel A. Grants and federal awards per ever-treated county ($N = 2,437$)					
Construction episodes	3.46	2.92	1.00	2.00	10.00
All federal awards (millions)	17.62	65.45	0.28	3.44	75.03
First-episode award (millions)	3.02	8.70	0.09	1.01	11.04
Later-episode awards (millions)	14.59	61.20	0.00	1.42	69.09
Panel B. Years from first to second construction start ($N = 1,758$)					
Years to second construction start	3.66	3.23	1.00	2.00	11.00
Panel C. Share of a county's sample excluded, all ever-treated counties ($N = 2,437$)					
County-yrs, all panel yrs	44.32	30.20	0.00	53.12	81.25
County-yrs, post-treatment yrs only	60.68	39.41	0.00	81.25	96.30
Federal awards	50.27	38.92	0.00	58.46	98.54
Panel D. Share of a county's sample excluded, counties with a second episode ($N = 1,758$)					
County-yrs, all panel yrs	61.44	14.56	37.50	62.50	81.25
County-yrs, post-treatment yrs only	84.12	13.43	56.00	88.46	96.30
Federal awards	69.69	27.31	14.03	78.01	98.95

Notes: This table reports the county-level distribution of grants and observations excluded by the single-episode sample restriction. Panel A reports the number of construction episodes and federal awards received by ever-treated counties. Federal awards are in millions of real 2017 dollars. *First-episode award* refers to awards associated with a county's first construction episode, and *later-episode awards* refers to awards associated with subsequent episodes. Panel B reports the number of years between the first and second construction starts among counties with a second episode. Panels C and D report the shares of county-years, post-treatment county-years, and federal awards excluded by the restriction. Panel C includes all ever-treated counties; Panel D includes only counties with a second episode.

A.1.4 Final baseline estimation sample

The final baseline sample combines the restrictions described above. Treatment is defined by a county's first grant-assisted construction episode, observations from a second construction start onward are excluded, and construction-sector specifications additionally require first treatment in 1975 or later because the QCEW industry series begins in 1975. Table A.6 reports the resulting sample composition by treatment cohort and event-time horizon. Table A.7 describes how the distribution of grant characteristics changes as the treatment definition and QCEW restriction are applied.

Table A.6 makes explicit how these restrictions affect identification across cohorts and event time. The absence of observations at longer pre-treatment horizons for the earliest retained cohorts follows

Table A.6: Estimation sample by cohort and event-time horizon

Cohort	Counties	Horizon (years since the first construction start)							
		-4	-3	-2	-1	0	1	2	3
1976	Treated	0	0	0	214	213	154	107	86
	Not yet treated	0	0	0	1,456	1,445	1,258	1,066	925
	Never treated	0	0	0	585	565	569	568	566
1977	Treated	0	0	190	195	195	147	120	95
	Not yet treated	0	0	1,255	1,280	1,272	1,079	936	787
	Never treated	0	0	565	584	573	567	566	562
1978	Treated	0	192	196	200	199	134	110	88
	Not yet treated	0	1,066	1,076	1,125	1,117	971	819	658
	Never treated	0	569	573	598	582	582	576	574
1979	Treated	140	140	143	145	144	111	91	77
	Not yet treated	926	939	974	987	980	827	665	535
	Never treated	568	567	582	602	590	581	578	577
1980	Treated	143	145	145	146	146	122	102	90
	Not yet treated	793	826	835	839	827	663	532	429
	Never treated	566	582	590	597	584	579	578	567
1981	Treated	163	164	164	165	164	132	122	99
	Not yet treated	656	663	663	666	659	527	425	330
	Never treated	576	581	584	593	580	578	566	545
1982	Treated	125	125	125	125	124	98	80	74
	Not yet treated	540	538	534	543	532	427	330	262
	Never treated	578	579	580	595	580	567	549	546
1983	Treated	102	101	102	103	103	87	76	74
	Not yet treated	430	426	430	436	425	328	260	192
	Never treated	578	578	580	596	574	554	549	547
1984	Treated	78	79	80	80	77	70	59	46
	Not yet treated	347	348	345	355	329	259	190	150
	Never treated	566	567	574	585	550	540	538	537
1985	Treated	78	76	76	79	78	65	58	55
	Not yet treated	252	252	253	259	254	184	145	92
	Never treated	549	554	550	569	542	535	532	532
1986	Treated	64	64	63	66	65	56	48	42
	Not yet treated	196	195	191	203	190	150	95	57
	Never treated	549	540	542	567	541	534	533	524
1987	Treated	40	39	40	41	39	31	24	20
	Not yet treated	150	145	150	158	151	97	58	31
	Never treated	538	535	541	565	541	534	528	517
1988	Treated	53	53	54	56	54	48	47	46
	Not yet treated	92	97	97	102	98	57	30	13
	Never treated	532	534	541	569	546	532	529	521
1989	Treated	38	38	39	39	36	34	34	31
	Not yet treated	57	59	59	61	59	32	13	8
	Never treated	533	534	546	566	539	533	530	524
1990	Treated	25	26	26	27	27	26	24	25
	Not yet treated	33	31	33	33	31	13	8	4
	Never treated	528	532	539	561	533	526	520	528

Notes: This table reports the number of counties available at each event-time horizon by treatment cohort. Within each cohort, rows report treated counties, not-yet-treated counties, and never-treated counties used in estimation. Event time is measured relative to the county's first grant-assisted construction start.

mechanically from the 1975 start of QCEW coverage. For example, a cohort first treated in 1976 can contribute at $h = -1$ but not at earlier event times; consequently, $h = -2$ is identified beginning with the 1977 cohort, $h = -3$ with the 1978 cohort, and $h = -4$ with the 1979 cohort. At positive horizons, treated counts decline within cohorts as counties begin subsequent construction episodes and leave the single-episode sample. For example, the 1976 cohort contributes 214 counties at $h = -1$ but 86 at $h = 3$.

The composition of the comparison group also changes over the program's life. The number of not-yet-treated counties declines for later cohorts as progressively fewer counties remain untreated, whereas never-treated counties continue to provide a comparison group throughout the sample period. Thus, both the treated units and the relative importance of the available comparison groups vary across cohort-by-event-time comparisons, as is explicit in Table A.6.

Table A.7 characterizes the grants represented at successive stages of sample construction. Panel A contains all 8,436 county-years in which grant-assisted construction begins. Panel B retains each county's first construction start, yielding the treatment episodes used in the baseline design, while Panel C further restricts to the 2,007 counties first treated in 1975 or later and therefore eligible for the construction-sector analysis.

The restrictions alter the distribution of project size. Mean federal assistance falls from \$13.07 million across all grant-receiving county-years in Panel A to \$9.66 million for first construction episodes in Panel B and \$7.61 million in the 1975+ construction sample in Panel C. The latter decline is consistent with the earlier evidence that the excluded 1973–1974 cohorts contained some of the program's largest recipients. In contrast, federal assistance relative to county population increases across the three samples: mean assistance per capita rises from \$131 to \$192 and then \$200, while the corresponding median rises from \$61 to \$105 and \$115. The baseline construction sample therefore contains smaller projects in absolute dollar terms than the unrestricted set of construction starts, but projects that are somewhat larger relative to the populations of the counties receiving them.

In sum, the restrictions listed in this section define the population and treatment episode represented by the baseline construction-sector estimates. They describe the response to a county's first grant-assisted construction episode among counties entering treatment from 1975 onward, prior to the

Table A.7: Grant characteristics across sample restrictions

	Mean	SD	p5	Median	p95
Panel A. All county-year observations with grants					
<i>County-year observations (N = 8,436)</i>					
Number of grants	1.41	1.07	1.00	1.00	3.00
Project cost (million)	17.93	57.58	0.37	4.65	72.13
Federal grant award (million)	13.07	42.34	0.26	3.32	52.70
Per capita project cost	178.92	307.80	5.25	85.93	656.24
Per capita federal grant award	131.32	229.88	3.72	61.34	485.22
Panel B. First-treatment-year observations					
<i>County-year observations (N = 2,437)</i>					
Number of grants	1.17	0.54	1.00	1.00	2.00
Project cost (million)	13.28	47.20	0.35	3.70	46.47
Federal grant award (million)	9.66	31.99	0.26	2.74	35.12
Per capita project cost	258.75	344.10	11.19	143.80	869.89
Per capita federal grant award	191.65	251.28	8.54	104.54	651.12
Panel C. First-treatment-year observations, 1975 onward					
<i>County-year observations (N = 2,007)</i>					
Number of grants	1.16	0.51	1.00	1.00	2.00
Project cost (million)	10.64	38.99	0.36	3.56	38.45
Federal grant award (million)	7.61	24.08	0.27	2.62	28.84
Per capita project cost	270.35	344.28	13.38	158.90	881.25
Per capita federal grant award	199.83	250.09	9.99	114.75	669.60

Notes: This table reports grant characteristics across successive sample restrictions. Panel A includes all county-year observations in which grant-assisted construction begins. Panel B restricts to each county's first construction-start year. Panel C further restricts to counties first treated in 1975 or later. Dollar values are in real 2017 USD. *Project cost* refers to the total eligible project cost for federal assistance, including the federal grant award and the local cost share.

onset of any subsequent construction episode. The preceding tables document how this design changes both the composition of treated counties and the grant expenditures represented in the estimation sample.

B Additional Institutional Details related to Program Administration

This appendix provides additional detail on the administration and timing of the CWA Construction Grants Program. I first describe the legislative history, state priority-list procedures, program funding rules, and the three-step grant process, drawing primarily on the EPA's *Handbook of Procedures* (U.S. EPA, 1980). I then use the administrative grant records to examine the timing of awards and construction starts and provide additional evidence on how grant receipt and treatment timing relate to pre-program county characteristics. These analyses complement the institutional and identification discussions in Sections 2 and 4.1.

B.1 Additional institutional details

Legislative background Federal support for municipal wastewater treatment construction predates the Clean Water Act. A loan program was first authorized in 1948 but never implemented for lack of appropriated funds. The Federal Water Pollution Control Act of 1956 (P.L. 84-660) introduced the first federal construction grants, allocated to states by a formula based on relative population and per capita income, with individual projects selected by the states and grants capped at 30% of eligible cost, not to exceed \$250,000 (U.S. EPA, 1980, p. I-3). The 1966 amendments removed the dollar cap and raised the federal cost share to 55%. P.L. 92-500 in 1972 raised the federal share to 75%, broadened eligible project types, and introduced the three-step grant process (Step 1 planning, Step 2 design, Step 3 construction) that structures the data used in this paper (U.S. EPA, 1980, pp. I-3–I-4). The Clean Water Act of 1977 (P.L. 95-217) subsequently introduced mid-course revisions, including mandatory evaluation of innovative and alternative treatment technologies and an increased 85% federal share for qualifying Step 2 and Step 3 projects (U.S. EPA, 1980, p. I-4).

State planning and the project priority list Each state was required to maintain a continuing planning process under Section 303(e) of the Act, encompassing effluent limitations, areawide and basin-level water quality management plans, and, of direct relevance here, an inventory and ranking of needed treatment works (U.S. EPA, 1980, pp. II-1–II-5). This ranking took the form of an annually

submitted *project priority list*, from which projects were certified by the state for construction grant funding.

The state priority system used to construct this list was required to rest on a specific set of criteria: the severity of the pollution problem, the existing population affected, the need for preservation of high-quality waters, the applicable category of treatment need (e.g., secondary treatment, infiltration/inflow correction, sewer system rehabilitation, new collectors and interceptors, or combined sewer overflow correction), and, at the margin, treatment-works and step sequencing, allotment deadlines, and total funds available (U.S. EPA, 1980, pp. II-8–II-9). The regulations governing this system explicitly excluded local economic conditions from consideration: “the State may not consider: the project area’s development needs not related to pollution abatement, the geographical region within the State, or future population growth projections” (U.S. EPA, 1980, p. II-9).

Once approved, priority lists could not be revised informally. The annual list required a public hearing before acceptance by the Regional Administrator, and only amendments judged immaterial by the Regional Administrator could bypass this requirement (U.S. EPA, 1980, p. II-10). Advancement through each grant step required the state to certify, on a standard EPA form, that the project remained entitled to priority ahead of every lower-ranked project on the list; reviewers were instructed to confirm that funding a given project would not “jeopardize the funding of any projects of higher priority” (U.S. EPA, 1980, p. IV-9). This certification requirement was applied independently at the Step 1, Step 2, and Step 3 stages.

Program funding Congress authorized construction grants funding in multi-year blocks, but funds were only available as annually appropriated and were then allotted to states by a statutory formula based on need and population (U.S. EPA, 1980, p. II-11). State allotments remained available for two fiscal years, after which unobligated funds were pooled and reallocated to states that had fully obligated their own allotments. States were required to reserve a minimum share of their annual allotment for grant cost overruns (at least 5%) and were permitted, at their option, to reserve up to an additional 10% for Step 1 and Step 2 projects not yet on the current priority list, along with smaller mandatory or optional reserves for innovative/alternative technology and small-community

alternatives (U.S. EPA, 1980, pp. II-11–II-13). Projects that could not be funded within available reserves due to cost overruns were deferred to the following year’s list rather than funded through discretionary reallocation. Separately, states could enter into delegation agreements with EPA regional offices under which the state assumed responsibility for technical or administrative review of specified documents, intended to reduce duplication of effort and shorten grant processing time (U.S. EPA, 1980, p. II-13).

The three-step process and the timing of construction Step 1 grants funded facility planning, including evaluation of treatment alternatives and selection of a preferred project; Step 2 grants funded preparation of construction plans and specifications; and Step 3 grants funded construction itself. The construction start date used to define treatment timing in this paper is not fixed until relatively late in this sequence. A preliminary Plan of Operation, submitted as part of the Step 3 application, was required to identify implementation actions and time frames, but the Handbook specifically notes that “specific dates may, of necessity, be omitted until a construction start date is known” (U.S. EPA, 1980, p. VI-36). The construction start date itself was set only after the Step 3 grant was accepted, through a formal procurement sequence: EPA authorization to advertise for competitive bids, review of submitted bids, resolution of any bid protests, and authorization to award a construction contract (U.S. EPA, 1980, p. VI-19), culminating in a preconstruction conference among the grantee, contractor, state agency, and EPA before construction activity began.

B.2 Local economic conditions and grant receipt, timing, and amount

The priority system described above excluded local development needs, geographic region within the state, and projected population growth from the criteria states could use to rank projects. This does not imply that grant receipt was unrelated to local characteristics: larger municipalities served more residents, generated greater treatment needs, and were more likely to require federally assisted wastewater infrastructure. This subsection therefore examines how local economic conditions relate to three distinct margins of program participation observed in the data: whether a county ever receives a construction grant, when its first grant award and construction start occur, and the amount

of federal assistance associated with treatment. I first relate these outcomes to pre-program county characteristics measured before the enactment of P.L. 92-500. I then examine whether changes in local economic conditions during the program predict the timing of first award or construction start using a discrete-time hazard model. Finally, I examine how grant amounts scale with county characteristics and population, which also informs the choice between additive and proportional outcome specifications.

Covariates in the first exercise are measured over 1969–1971, before P.L. 92-500 and before any county in the sample is treated: log population, population growth, personal income per capita, income per capita growth, the construction sector’s share of personal income, and the growth of that share. All but log population are standardized, so their coefficients correspond to a one-standard-deviation change in the relevant characteristic. Population is included throughout because larger counties mechanically operate more treatment works, generate more eligible projects, and are more likely to appear somewhere on a state’s priority list. The lower panel of Table B.8 therefore reports joint tests that hold the population terms aside: all remaining covariates, all growth rates, growth rates excluding population growth, and the two construction-sector covariates.

Determinants of grant receipt Column (1) of Table B.8 examines whether pre-program characteristics predict whether a county ever receives a construction grant. Population is the only covariate that significantly predicts receipt: a doubling of county population is associated with a 6.7 percentage-point increase in the probability of ever being treated ($0.096 \times \ln 2$). None of the remaining five covariates is individually significant, and they are jointly insignificant ($p = 0.484$). The growth-rate covariates are also jointly insignificant ($p = 0.989$), as are the two construction-sector covariates ($p = 0.286$).

This result helps interpret the unconditional differences between ever- and never-treated counties reported in Appendix Table A.1. Recipient counties are substantially larger and differ from never-treated counties along several pre-program dimensions. Conditional on population, however, pre-program income, income growth, and the size and growth of the local construction sector provide little additional predictive information about which counties eventually enter the program. Grant

Table B.8: Pre-program correlates of grant receipt, timing, and amount

Covariates	Receipt	Timing		Federal grant amount	
	(1)	(2) Award	(3) Const.	(4) Lifetime	(5) Treat-yr.
Population (log)	0.096*** (0.006)	-1.175*** (0.069)	-1.215*** (0.076)	-0.011 (0.018)	-0.484*** (0.024)
Population growth (%)	0.001 (0.008)	-0.033 (0.108)	-0.063 (0.115)	0.177*** (0.023)	0.166*** (0.030)
Income per capita	0.007 (0.008)	-0.661*** (0.096)	-0.929*** (0.106)	0.212*** (0.024)	-0.004 (0.030)
Income p.c. growth (%)	-0.001 (0.008)	-0.198 (0.137)	-0.250* (0.151)	0.052* (0.031)	-0.026 (0.037)
Const. share of income	0.011 (0.007)	0.051 (0.103)	0.043 (0.105)	0.053** (0.024)	0.044 (0.028)
Const. share growth (%)	-0.001 (0.008)	-0.108 (0.092)	-0.089 (0.104)	-0.015 (0.033)	-0.013 (0.034)
N	3,032	2,417	2,413	2,439	2,413
All excluding pop (<i>p</i>)	0.484	0.000	0.000	0.000	0.556
Growth rates (<i>p</i>)	0.989	0.214	0.242	0.000	0.000
Exc pop growth (<i>p</i>)	0.968	0.108	0.124	0.241	0.682
Const. block (<i>p</i>)	0.286	0.500	0.688	0.081	0.286

Notes: This table reports OLS estimates relating pre-program (1969–1971) local conditions to measures of Clean Water Act Construction Grants Program treatment. Column (1) is a linear probability model for whether a county ever receives a Step 3/4 grant (full sample). Column (2) restricts to counties that ever receive a grant award and uses the year of first award as the dependent variable; column (3) restricts to counties that ever begin construction and uses the year of first construction start. Columns (4)–(5) restrict to counties with a construction start and use the log of federal award dollars per capita (based on 1969–1971 population) as the dependent variable: column (4) uses each county’s lifetime cumulative federal award received across the program, and column (5) uses only federal award dollars tied to grants dated in the county’s own treatment year. Population is included in log levels; population growth, income per capita, income per capita growth, the construction sector’s share of personal income, and its growth are standardized (mean zero, unit standard deviation). Robust standard errors are reported in parentheses. The lower panel reports *p*-values from *F*-tests of joint significance for the indicated groups of covariates. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

receipt is therefore clearly selected on county size, but there is considerably less evidence that counties experiencing different pre-program economic trajectories were systematically more likely to participate.

Determinants of award and construction-start timing Columns (2) and (3) restrict attention to treated counties and examine whether pre-program characteristics predict when treatment occurs. Negative coefficients indicate earlier treatment. Larger counties enter the program earlier: a doubling of population is associated with first award occurring approximately 0.81 years earlier and first

construction start approximately 0.84 years earlier. Counties with higher pre-program income per capita are also treated earlier, by approximately 0.66 years per standard deviation for first award and 0.93 years for construction start. Accordingly, the joint test of all non-population covariates rejects in both columns ($p < 0.001$), driven primarily by the income level.

The growth-rate variables tell a different story. They are jointly insignificant for both first-award timing ($p = 0.214$) and construction-start timing ($p = 0.242$). The same is true when population growth is excluded from the block ($p = 0.108$ and $p = 0.124$, respectively), and the construction-sector variables are also jointly insignificant ($p = 0.500$ and $p = 0.688$). Thus, counties that were larger and richer before the program tended to enter earlier, but counties experiencing faster pre-program economic growth were not systematically treated earlier or later.

These cross-sectional regressions establish that treatment timing is related to some pre-program county levels, but they do not address the more immediate identification concern: whether changes in a county's economic conditions during the program predict when treatment occurs. Table B.9 addresses this question directly using time-varying county characteristics.

Table B.9 estimates a discrete-time hazard of first treatment using the panel of not-yet-treated county-years between 1973 and 1990. Local economic conditions are now allowed to vary over time: levels are measured at $t - 1$ and growth rates over $t - 3$ to $t - 1$, with each covariate standardized within year. State and year fixed effects absorb persistent differences across states and the common national evolution of the program. Column (1) defines treatment as Step 3 award receipt, while column (2) defines treatment as construction start.

County size remains the dominant predictor in both specifications. A one-standard-deviation increase in log population is associated with approximately a doubling of the hazard of both award receipt and construction start. Beyond population, however, the two treatment definitions produce notably different results. For award receipt, recent local economic conditions have predictive content: a one-standard-deviation increase in population growth raises the award hazard by 7.7 percent, income per capita growth raises it by 5.5 percent, and the construction-sector share raises it by 4.6 percent. The joint tests reject for all non-population covariates ($p = 0.004$), for the growth-rate block ($p < 0.001$), and for growth rates excluding population growth ($p = 0.002$); the construction-sector

Table B.9: Local economic conditions and treatment timing

Covariates	Timing	
	(1) Award	(2) Const. start
Population (log)	0.742*** (0.039)	0.756*** (0.039)
Population growth (%)	0.074*** (0.025)	0.036 (0.026)
Income per capita	-0.043 (0.034)	-0.028 (0.035)
Income p.c. growth (%)	0.054*** (0.019)	0.025 (0.020)
Const. share of income	0.045* (0.023)	0.021 (0.025)
Const. share growth (%)	-0.058* (0.032)	-0.004 (0.026)
N	25,468	28,111
All excluding pop (<i>p</i>)	0.004	0.641
Growth rates (<i>p</i>)	0.000	0.344
Exc pop growth (<i>p</i>)	0.002	0.427
Const. block (<i>p</i>)	0.069	0.688

Notes: This table reports discrete-time hazard estimates of first treatment among not-yet-treated county-years from 1973–1990, using a complementary log-log specification. Column (1) defines treatment as first Step 3/4 award receipt; column (2) defines treatment as first construction start. Counties never treated during the sample period are right-censored. Covariate levels are measured at $t - 1$, and growth rates are measured from $t - 3$ to $t - 1$. All covariates are standardized to mean zero and unit standard deviation within year. Both specifications include state and year fixed effects. Standard errors are clustered at the county level and reported in parentheses. *All excluding pop* reports the *p*-value from a joint test of all covariates other than population and population growth; *Growth rates* jointly tests the three growth-rate covariates; *Exc pop growth* jointly tests income per capita growth and construction-share growth; and *Const. block* jointly tests the construction-share level and growth covariates. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

block is marginally significant ($p = 0.069$).

These relationships are not present when treatment is defined by construction start. In column (2), every non-population coefficient lies within 1.5 standard errors of zero, and none of the corresponding joint tests rejects: the *p*-values are 0.641 for all non-population covariates, 0.344 for the growth-rate block, 0.427 for growth rates excluding population growth, and 0.688 for the construction-sector block. Thus, while recent changes in local economic conditions contain some information about when a Step 3 award is received, they do not predict when physical construction actually begins.

This distinction is important because the baseline treatment in this paper is dated by construction start rather than award receipt. The institutional process described above provides a plausible explanation for why the two dates need not respond similarly to contemporaneous local conditions. Award timing reflects a project's position on the state priority list and the availability of the state's annual federal allotment. After an award is made, construction can begin only after a separate procurement process involving authorization to advertise for bids, bid review, resolution of protests, contract award, and a preconstruction conference. The resulting interval between award and construction start is documented in Table B.11: the median is approximately 0.75 years, with a fifth-to-ninety-fifth percentile range spanning roughly three years. Consistent with this institutional separation between award and implementation, the relationship between recent local economic conditions and award timing is no longer evident when treatment is dated by the start of construction.

Determinants of grant amount The final exercise examines whether the amount of federal assistance associated with treatment is systematically related to pre-program county characteristics. Columns (4) and (5) of Table B.8 regress log federal award dollars per capita on the same pre-program covariates. Column (4) uses each county's cumulative federal assistance over the life of the program, while column (5) uses only awards associated with grants dated in the county's treatment year. The distinction is important because lifetime assistance reflects repeated participation over many years, whereas treatment-year assistance more closely measures the scale of the initial construction episode studied in the baseline analysis.

Lifetime federal assistance is related to several pre-program characteristics. Population growth (0.177 per standard deviation), income per capita (0.212), income per capita growth (0.052), and the construction-sector share (0.053) all enter positively, and the joint tests reject both for all non-population covariates and for the growth-rate block ($p < 0.001$ in each case). This pattern is consistent with the cumulative nature of program participation: counties whose population and economic activity grow over time may face greater wastewater-treatment needs and subsequently return to the program for additional projects.

The corresponding relationships are substantially weaker for the federal assistance associated with a

Table B.10: Elasticity of federal grant awards with respect to county size

Outcome	Population		Construction employment	
	(i) Bivariate	(ii) Controls	(i) Bivariate	(ii) Controls
Lifetime	1.077*** (0.017)	0.969*** (0.030)	0.732*** (0.019)	0.654*** (0.028)
Treat. yr.	0.540*** (0.023)	0.341*** (0.028)	0.347*** (0.021)	0.241*** (0.025)
Lifetime – H0: elasticity = 1 (p)	<0.001	0.312	<0.001	<0.001
Treat. yr. – H0: elasticity = 1 (p)	<0.001	<0.001	<0.001	<0.001

Notes: This table reports OLS estimates of the elasticity of federal grant awards with respect to county size. County size is measured by pre-treatment population in the first two columns and pre-treatment construction employment in the last two columns. Both federal grant awards and the corresponding county-size measure enter in logs. Each coefficient is estimated in a separate regression. *Lifetime* refers to cumulative federal grant awards received by a county over the program period, and *Treat. yr.* refers to federal grant awards received in the county's first treatment year. *Bivariate* includes only the corresponding county-size measure, while *Controls* additionally includes the pre-program covariates used in Table B.8. The bottom two rows report p -values from tests of the null hypothesis that the corresponding elasticity equals one. Robust standard errors are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

county's initial treatment episode. Once the population terms are set aside, the remaining covariates do not jointly predict treatment-year assistance ($p = 0.556$); neither do growth rates excluding population growth ($p = 0.682$) or the construction-sector variables ($p = 0.286$). Pre-program income per capita, which predicts earlier treatment in columns (2) and (3), has essentially no relationship with treatment-year federal assistance (-0.004 , s.e. 0.030). Thus, while cumulative program spending is systematically related to local characteristics, there is considerably less evidence that the scale of the initial construction episode is determined by pre-program economic conditions beyond county population.

The relationship between grant size and population is itself informative about how the treatment should be represented in the employment specifications. Because the dependent variable in column (5) is federal assistance per capita, its coefficient on log population of -0.484 implies that total treatment-year federal assistance increases less than proportionally with county population. Table B.10 examines this relationship directly by regressing log federal award dollars on log population.

Lifetime federal assistance scales approximately proportionally with county population. The esti-

mated population elasticity is 1.077 in the bivariate specification and 0.969 after conditioning on the remaining pre-program covariates and state fixed effects; in the latter specification, proportionality cannot be rejected ($p = 0.312$). This is consistent with repeated program participation eventually generating greater cumulative investment in larger counties.

Treatment-year assistance behaves differently. Its estimated population elasticity is 0.540 in the bivariate specification and 0.341 after conditioning on the remaining covariates and state fixed effects. Proportionality is strongly rejected in both specifications ($p < 0.001$). The latter estimate implies that doubling county population is associated with only about a 27 percent increase in federal assistance during the treatment year, rather than a doubling of the transfer.

This subproportional scaling is consistent with the institutional unit at which the program operates. Construction grants finance individual treatment works serving defined service areas rather than county governments as a whole, so the scale of a county's first project need not increase proportionally with the county's total population. The estimates in Table [B.10](#) are consistent with this feature: treatment-year federal assistance increases with county population, but substantially less than proportionally. This provides support for reporting employment effects in levels for outcomes closely tied to the physical construction project, where the treatment is a discrete local construction episode rather than a population-proportional transfer. At the same time, it does not imply that proportional specifications are uninformative, particularly for broader county-level outcomes whose responses may scale with the size of the local economy. I therefore report both additive and proportional specifications and place greater weight on the additive specification for the direct construction-sector outcomes.

B.3 Timing of the planning-to-construction sequence

The Construction Grants Program involved a sequence of planning, design, grant award, procurement, construction, and completion rather than a single implementation date. The timing of this sequence matters for two choices in the empirical analysis. First, the construction-sector analysis is anchored on the observed construction start and focuses on relatively short post-treatment horizons;

Table B.11: Timing between CWA grant and construction milestones

	N	Mean	SD	p5	Median	p95
Step 1 receipt → Step 3 receipt	1,139	4.40	3.07	0.90	3.54	10.93
Step 1 receipt → Const. start	1,137	5.64	3.37	1.47	4.96	12.06
Step 3 receipt → Const. start (all)	11,921	1.17	1.36	0.25	0.75	3.50
Step 3 receipt → Const. start (estim. sample)	2,437	1.05	1.09	0.00	0.75	3.25
Const. start → Completion (all)	11,801	2.62	2.02	0.50	2.00	6.75
Const. start → Completion (estim. sample)	2,417	2.84	2.18	0.75	2.25	7.50

Notes: This table reports the time elapsed, in years, between milestones in the CWA Construction Grants process. *Receipt* refers to the grant award date. Step 1-to-Step 3 and Step 1-to-construction-start intervals use facility-level matches between construction-defining Step 3 (or Step 2+3) grants and preceding Step 1 grants. *All* includes all Step 3 (or Step 2+3) grants awarded between 1973 and 1990. *Estimation sample* includes the grant defining each county's first construction start. Construction-duration statistics exclude records with missing, non-positive, or greater-than-15-year durations.

Section 3 notes that construction lasts approximately 2.6 years on average and that most projects are completed within four years. Second, the broader county-level outcomes in Section 6 allow economic activity to begin before physical construction by shifting the treatment date earlier than the observed construction start. This section documents the empirical timing underlying these choices.

Table B.11 reports intervals between four milestones: Step 1 receipt, Step 3 receipt, construction start, and construction completion. For the first two intervals, I link each county's construction-defining Step 3 (or combined Step 2+3) grant to the closest preceding Step 1 grant at the same facility. Because the administrative data contain no common project identifier linking Step 1 and Step 3 grants, these matches should be interpreted as approximate facility-level links rather than verified grant sequences. The procedure identifies 1,139 matches for the Step 1-to-Step 3 interval and 1,137 for the Step 1-to-construction-start interval, out of the 2,437 construction-defining grants used to assign county treatment dates. The Step 3-to-construction-start and construction-duration intervals do not require this heuristic linkage and are therefore reported both for the full set of construction grants and for the subset defining county treatment.

The pre-construction process is lengthy and heterogeneous. Among the facility-level matches, the median interval from Step 1 receipt to Step 3 receipt is 3.54 years, with a 5th-to-95th percentile range of 0.90 to 10.93 years. The median interval from Step 1 receipt to construction start is 4.96 years, with a corresponding range of 1.47 to 12.06 years. These long and dispersed intervals make

Table B.12: Cumulative share of grants reaching construction by years since award

	N	1 yr	2 yrs	3 yrs
Step 3 receipt → Const. start (all)	11,921	65.9%	85.3%	92.8%
Step 3 receipt → Const. start (estim. sample)	2,437	67.8%	87.1%	94.1%

Notes: This table reports the cumulative share of Step 3 (or Step 2+3) grants for which construction begins within one, two, and three years of the grant award date. *All* includes all Step 3 (or Step 2+3) grants awarded between 1973 and 1990. *Estimation sample* includes the grant defining each county's first construction start.

the earlier planning stages poor anchors for the onset of local construction activity.

The interval between the construction-stage award and physical construction is considerably shorter, but it is not uniform across projects. Among all Step 3 and Step 2+3 grants, the median interval from award to construction start is 0.75 years, with the 5th and 95th percentiles at 0.25 and 3.50 years. The corresponding distribution is similar for the grants defining county treatment, with a median of 0.75 years and a 5th-to-95th percentile range of 0.00 to 3.25 years. Thus, counties receiving construction-stage awards at the same point in calendar time need not begin physical construction at the same point thereafter. This dispersion is one reason that the construction start date, rather than the grant award date, provides the cleaner event-time anchor for outcomes directly tied to on-site construction.

Although the upper tail is long, most construction-stage grants proceed to construction relatively quickly. Table B.12 shows that construction begins within one year of Step 3 receipt for 65.9 percent of all grants and 67.8 percent of the construction-defining grants in the estimation sample. Within two years, these shares rise to 85.3 and 87.1 percent, respectively, and within three years to 92.8 and 94.1 percent. The two-year adjustment used for the broader county-level outcomes in Section 6 therefore spans the award-to-construction interval for the large majority of treatment-defining grants. It should not, however, be interpreted as identifying a unique date at which indirect economic effects begin: the underlying interval varies across projects, which is why Appendix Section D also reports results shifting the construction-start date by one and three years.

Finally, once physical construction begins, projects remain active for several years. Construction duration averages 2.62 years among all grants with valid start and completion dates, with a median of 2.00 years. Among the grants defining county treatment, the corresponding mean is 2.84 years

and the median is 2.25 years. These durations provide the empirical basis for emphasizing the first several years following construction start in the construction-sector event study: they encompass the period during which the direct labor demand generated by the project is most likely to be active.

C Additional Analyses on Construction Sector Effects

C.1 Proportional estimation specifications

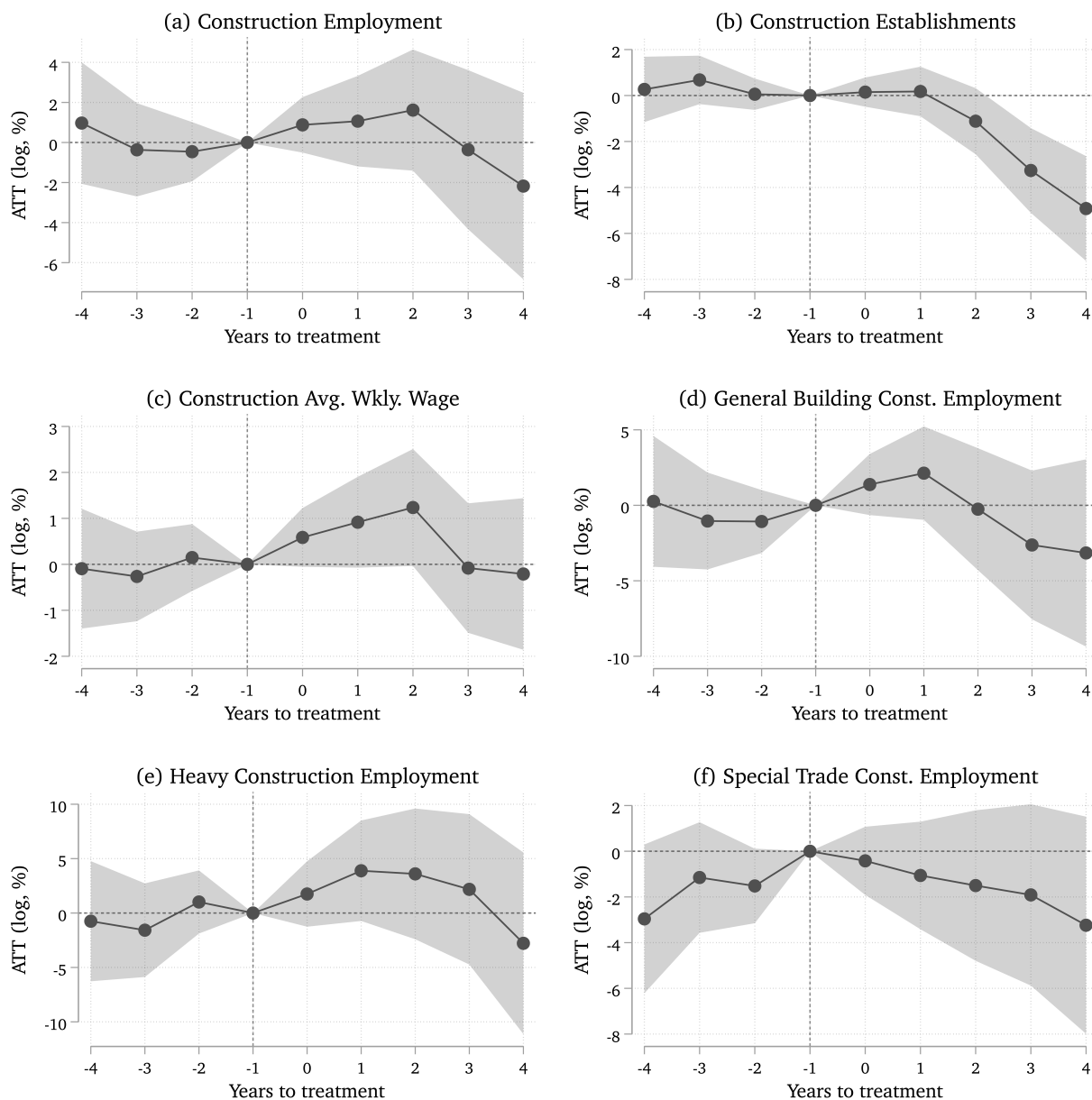


Figure C.4: Event-study CSDiD estimates of local construction-sector effects, log outcomes. Panels report construction employment, establishments, average weekly wages, and employment in the three construction subsectors. Event time is measured relative to the construction start year, with the year before treatment omitted. Coefficients are expressed in percent. Shaded areas denote 95% confidence intervals.

Figure C.4 presents the event-study counterparts to the proportional specifications reported in Table 3. As discussed in the main text, the additive specification is preferred for the direct construction response because the scale of the initial grant-assisted construction episode rises substantially less than proportionally with the size of the existing county construction sector. The log estimates provide a complementary functional-form check.

Across the aggregate construction outcomes, the pre-treatment coefficients are small and statistically indistinguishable from zero. Following construction start, the employment trajectory broadly resembles that obtained in levels, although the proportional estimates are less precise. The subsector decomposition also preserves the main sectoral pattern: the clearest positive post-treatment response occurs in heavy construction, which contains water and sewer infrastructure construction, while the other construction subsectors show less systematic responses. Overall, the event studies therefore show similar timing and sectoral concentration under the proportional specification, while confirming the weaker statistical precision of the log estimates reported in Table 3.

C.2 Propensity-score overlap and weight diagnostics

This section reports propensity-score overlap and weight diagnostics for the doubly robust IPW-OLS (DRIPW) estimates in Table 5. The DRIPW estimator of Sant’Anna and Zhao (2020) combines an outcome regression for the untreated potential outcome with inverse-probability weights constructed from cohort-specific propensity scores. It is doubly robust in the sense that consistency requires correct specification of either the outcome regression or the propensity-score model, but not necessarily both.

For each treatment cohort g , I estimate the propensity score using a logit of membership in cohort g , relative to counties not yet treated as of g , on the covariates in the full specification of Table 5: pre-program population, personal income per capita, and construction-sector income per capita, together with their growth rates over 1969–1971. Population enters in logs.

Figure C.5 shows substantial overlap between the propensity-score distributions of treated and comparison counties in every estimated cohort. Table C.13 provides two additional diagnostics.

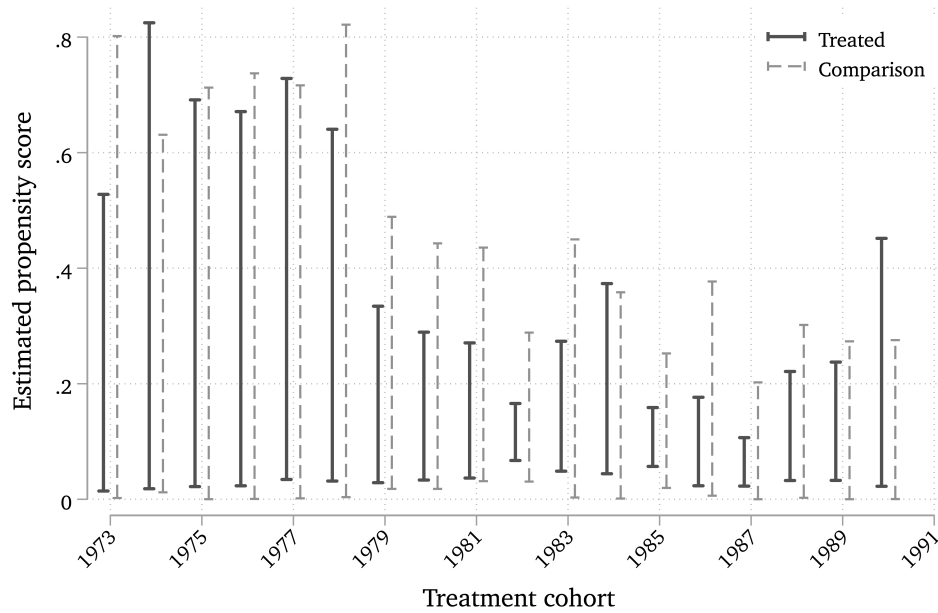


Figure C.5: Propensity-score overlap by treatment cohort under the full control specification in Table 5. For each cohort, propensity scores are estimated among treated counties and counties not yet treated as of that cohort’s treatment year. Solid spikes show the range of estimated propensity scores among treated counties, and dashed spikes show the corresponding range among comparison counties. Cohorts with fewer than ten treated or ten comparison counties are omitted.

The diagnostics indicate little evidence of a common-support or weight-concentration problem. The share of treated counties outside the comparison group’s propensity-score support ranges from zero to 3.2 percent across cohorts and has a median of zero; only 4 of the 18 estimated cohorts contain any off-support treated counties. Weight concentration is similarly limited: the single most heavily weighted comparison county never accounts for more than 2.80 percent of a cohort’s total inverse-probability weight.

These diagnostics indicate that the covariate-adjusted estimates in Table 5 are supported by comparable observations and are not driven by a small number of comparison counties receiving extreme weights. They do not establish the conditional parallel-trends assumption or determine whether including pre-program growth rates improves identification. The latter concern is addressed in Table 5 by comparing otherwise identical specifications with and without the growth-rate controls.

Table C.13: Propensity-score support and weight concentration by treatment cohort

Cohort	N (treated)	N (comparison)	Off-support (%)	Max weight share (%)
1973	140	2,892	0.0	2.80
1974	285	2,607	0.4	0.59
1975	237	2,370	0.0	1.03
1976	224	2,146	0.0	1.22
1977	198	1,948	0.5	1.25
1978	200	1,748	0.0	2.18
1979	144	1,604	0.0	0.66
1980	147	1,457	0.0	0.54
1981	166	1,291	0.0	0.46
1982	124	1,167	0.0	0.33
1983	102	1,065	0.0	0.80
1984	80	985	1.2	0.70
1985	84	901	0.0	0.40
1986	68	833	0.0	0.88
1987	48	785	0.0	0.53
1988	60	725	0.0	0.72
1989	41	684	0.0	0.92
1990	31	653	3.2	1.24

Notes: This table reports propensity-score support and inverse-probability-weight diagnostics by treatment cohort for the full covariate specification in Table 5. For each cohort, the propensity score is estimated using treated counties in that cohort and counties not yet treated as of g . *Off-support* refers to the share of treated counties whose estimated propensity score falls outside the range of estimated propensity scores among comparison counties. *Max weight share* refers to the share of total inverse-probability weight, $\hat{p}/(1 - \hat{p})$, assigned to the single most-weighted comparison county. Cohorts with fewer than 10 treated or 10 comparison counties are omitted.

C.3 Additional analyses on spatial spillovers across counties

Section 4.1 examines whether grant-assisted construction generates employment effects in neighboring counties within the same commuting zone (CZ). One limitation of that exercise is that the pool of CZ-unexposed counties becomes increasingly small as the program expands geographically. This section implements a complementary snapshot-year design that fixes CZ exposure at particular points in the program and thereby preserves a more balanced comparison between exposed and unexposed counties.

For each snapshot year $Y \in \{1975, \dots, 1979\}$ and post-window length $h \in \{1, 3\}$, I restrict the sample to counties whose own treatment status remains unchanged throughout the relevant window: counties that are never treated or whose first construction start occurs strictly after $Y + h$. Among

Table C.14: Snapshot-year construction-sector spillover estimates

	1975	1976	1977	1978	1979
Panel A: Construction employment					
2-yr post avg.	-9.15 (5.68)	8.00 (10.78)	-10.93 (11.98)	-3.25 (12.43)	-1.15 (10.22)
4-yr post avg.	-5.00 (10.28)	4.58 (14.69)	3.88 (16.97)	11.76 (15.36)	2.48 (14.59)
Panel B: Construction sector establishments					
2-yr post avg.	-0.75** (0.35)	0.28 (0.48)	0.81 (0.66)	1.47** (0.70)	0.32 (0.72)
4-yr post avg.	0.28 (0.54)	0.95 (0.73)	1.63 (1.01)	2.18** (1.03)	-0.47 (1.27)
Panel C: Construction sector average weekly wages					
2-yr post avg.	1.16 (2.17)	2.85 (4.83)	-4.20 (4.86)	-0.08 (5.71)	9.74 (6.58)
4-yr post avg.	-0.14 (4.44)	-0.54 (6.39)	1.22 (6.74)	-3.54 (7.04)	5.54 (9.31)
N exposed (2 yrs)	903	974	968	995	970
N unexposed (2 yrs)	882	682	537	446	339
N exposed (4 yrs)	668	746	770	763	742
N unexposed (4 yrs)	524	410	309	225	161

Notes: This table reports snapshot-year spillover estimates for construction-sector outcomes. For each snapshot year from 1975–1979, a county is *exposed* if another county in the same commuting zone had initiated grant-assisted construction by that year. Counties are included only if they are never directly treated or their own first construction start occurs after the relevant post-treatment window, and are excluded if commuting-zone exposure changes during the window. The pre-period is year $Y - 1$, and the post-period is the average outcome over the following 2 or 4 years. Panel A reports construction employment, Panel B construction establishments, and Panel C average weekly construction wages. *N exposed* and *N unexposed* report sample sizes for Panel A. Standard errors are clustered at the county level and reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

these counties, I classify a commuting zone as exposed if another county in the CZ had initiated grant-assisted construction by year Y . I further exclude counties whose CZ exposure status changes during the post-treatment window. I then compare the change from the pre-period outcome in year $Y - 1$ to the average outcome from Y through $Y + h$ between counties in exposed and unexposed commuting zones. Because counties experiencing their own construction start during the window are excluded, any differential change reflects exposure to construction elsewhere in the commuting zone rather than contemporaneous direct treatment in the county itself.

This specification complements rather than replaces the CSDiD analysis in Section 4.1. Fixing expo-

sure at a sequence of snapshot years avoids relying on the progressively shrinking pool of counties that remain CZ-unexposed throughout the full panel, but comes at the cost of estimating separate two-period comparisons rather than exploiting the full staggered timing of exposure. The eligible sample also becomes smaller at later snapshots and longer horizons as construction spreads across commuting zones.

Table C.14 reports the estimates across the five snapshot years and two horizons. None of the construction-employment or wage coefficients is statistically distinguishable from zero at conventional levels. Three of the ten establishment coefficients are significant at the 5 percent level, but their signs vary and the estimates do not exhibit a consistent pattern across snapshot years or horizons. The snapshot estimates therefore do not reveal a systematic construction-sector response in neighboring counties.

These results are consistent with the direct commuting-zone analysis in Section 4.1, but should not be interpreted as establishing the absence of spatial spillovers. The snapshot design addresses one particular limitation of the main spillover exercise—the erosion of the CZ-unexposed comparison pool—while introducing its own restrictions and retaining limited statistical power. I therefore view the two exercises together as providing no clear evidence of economically systematic cross-county spillovers, rather than as ruling out smaller effects that the available designs may not detect.

C.4 Additional measures of local slack

The heterogeneity analysis in Section 5.2 classifies counties using their 1976 unemployment rate, the earliest year for which county-level BLS LAUS data are available. Using a common calendar year avoids comparing unemployment estimates across different years in the early LAUS data, but for counties treated substantially later, the 1976 reading may not closely reflect labor-market conditions immediately preceding construction. This section examines whether the main heterogeneity pattern is robust to measuring unemployment closer to each county's own construction start.

I consider three alternatives to the main specification. First, I classify counties using unemployment one year before their construction start ($g - 1$). Second, I use unemployment two years before con-

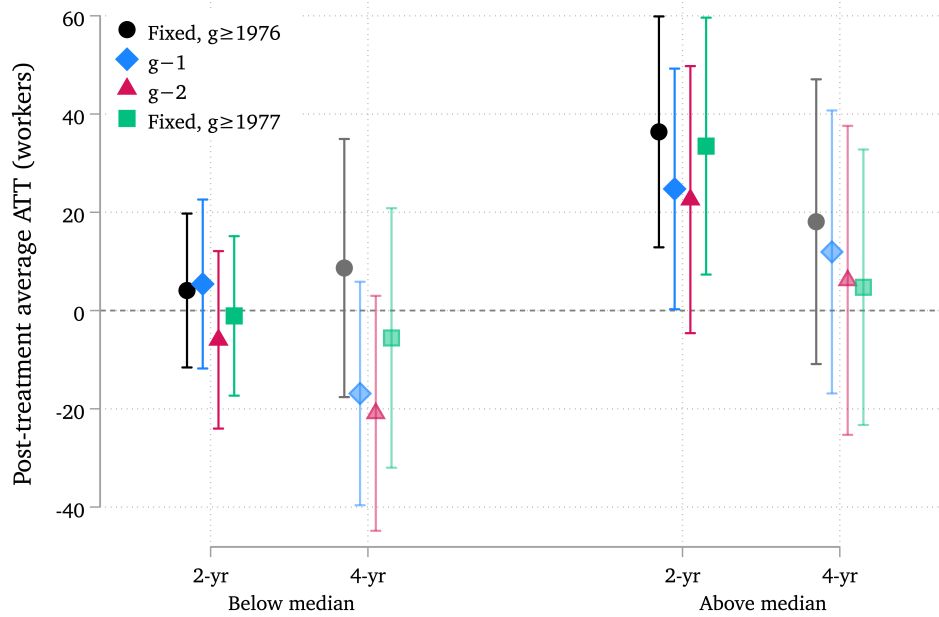


Figure C.6: Construction-employment effects under alternative measures of local labor-market slack. *Fixed*, $g \geq 1976$ classifies counties using their 1976 unemployment rate, while *Fixed*, $g \geq 1977$ additionally excludes the 1976 treatment cohort. $g-1$ and $g-2$ classify counties using unemployment one and two years before construction start, respectively, split at the median within each reference year. Markers report CSDiD average treatment effects over the first 2 and 4 years after treatment. Spikes denote 95% confidence intervals.

struction ($g-2$). For these two specifications, the high- and low-unemployment groups are defined relative to the median within each reference year, so counties are compared with others observed at the same point in the national business cycle. Third, I retain the fixed 1976 classification but exclude the cohort whose construction begins in 1976, ensuring that the classifying unemployment rate strictly precedes treatment.

These alternatives involve a tradeoff. The $g-1$ and $g-2$ measures better approximate labor-market conditions near construction, but they also move the classification closer to treatment and reduce the set of cohorts that can be classified because county LAUS data begin in 1976. The common 1976 measure is therefore preferable as a baseline measure, while the treatment-relative measures provide a useful robustness check on whether the results depend on conditions observed in that particular year.

Figure C.6 and Table C.15 show a consistent qualitative pattern across the alternative classifications.

Table C.15: Construction employment effects by alternative measures of local unemployment

	Fixed, $g \geq 1976$		$g-1$		$g-2$		Fixed, $g \geq 1977$	
	(1) Lo	(2) Hi	(3) Lo	(4) Hi	(5) Lo	(6) Hi	(7) Lo	(8) Hi
2-yr post-avg.	4.08 (7.99)	36.37*** (11.99)	5.40 (8.77)	24.75** (12.50)	-5.96 (9.22)	22.58 (13.87)	-1.09 (8.28)	33.47** (13.34)
4-yr post-avg.	8.65 (13.40)	18.09 (14.79)	-16.89 (11.61)	11.93 (14.70)	-20.92* (12.20)	6.15 (16.05)	-5.58 (13.48)	4.74 (14.31)
N	20,127	18,732	25,955	26,554	25,179	25,814	19,446	17,871
2-yr diff. (p)	0.025		0.205		0.087		0.028	
4-yr diff. (p)	0.636		0.124		0.179		0.600	

Notes: This table reports CSDiD average treatment effects on construction employment over the first 2 and 4 years after treatment, estimated separately for counties with low and high local unemployment. Columns (1)–(2) classify counties using the 1976 unemployment rate, split at the pooled median of 6.25 percent. Columns (3)–(4) and (5)–(6) classify counties using unemployment one and two years before construction start, respectively, split at the median within each reference year. Columns (7)–(8) use the 1976 classification and exclude counties first treated in 1976. *Lo* and *Hi* refer to counties with below- and above-median unemployment under the corresponding classification. Standard errors are clustered at the county level and reported in parentheses. *2-yr diff. (p)* and *4-yr diff. (p)* report p -values from tests of equality between the *Lo* and *Hi* estimates at each horizon. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

At the 2-year horizon, estimated construction-employment gains are larger in higher-unemployment counties under all four specifications: 36.37 versus 4.08 workers using the fixed 1976 measure, 24.75 versus 5.40 using $g-1$, 22.58 versus -5.96 using $g-2$, and 33.47 versus -1.09 when the 1976 cohort is excluded from the fixed-year specification. The difference is statistically significant at the 5 percent level under the two fixed-year specifications, marginally significant under $g-2$, and not statistically significant under $g-1$. The 4-year estimates are substantially less precise, and the difference between high- and low-unemployment counties is not statistically significant under any classification. Overall, the larger near-term construction-employment response in higher-unemployment counties is not specific to measuring local slack in 1976, although the precision of the subgroup difference varies across specifications.

C.5 Heterogeneous effects based on grant size

The baseline CSDiD estimates treat the start of grant-assisted construction as a binary treatment, although the scale of the initial construction episode varies substantially across counties. If larger projects require more construction labor, counties receiving larger awards may experience larger

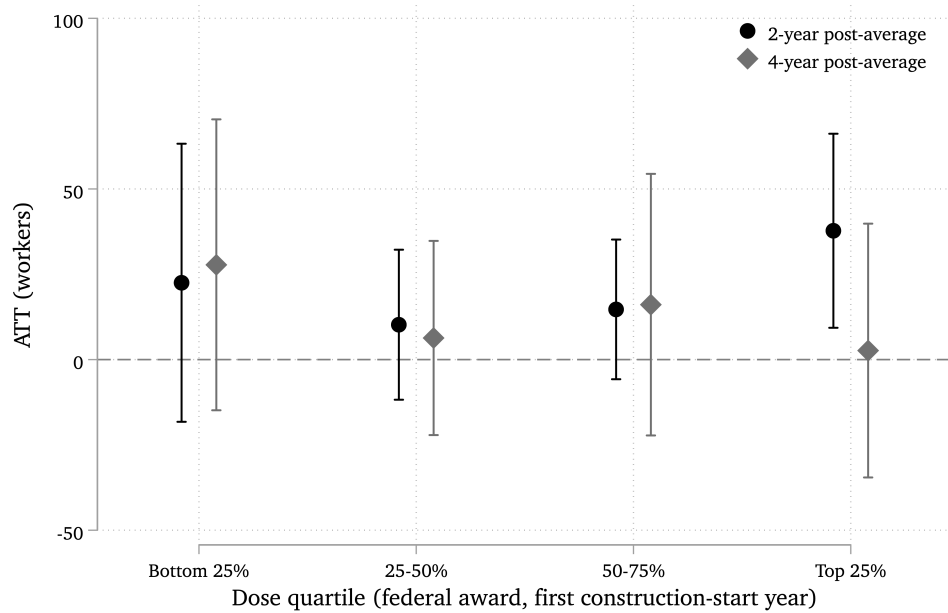


Figure C.7: Construction-employment effects by grant-size quartile. Grant size is measured by the federal award received in a county’s first construction-start year, with treated counties divided into equal-count quartiles. Markers report CSDiD average treatment effects over the first 2 and 4 years after treatment, estimated separately by quartile using not-yet-treated counties as controls. Spikes denote 95% confidence intervals.

employment responses. This section examines whether construction-employment effects vary systematically with the size of the federal award associated with a county’s first construction episode. I then ask whether differences in grant size can account for the heterogeneity across program eras and local labor-market conditions documented in Section 5.

Employment effects by grant size I first divide treated counties into quartiles based on the federal award received in their first construction-start year and estimate the CSDiD specification separately within each quartile, using not-yet-treated counties as the comparison group. Grant size is measured using the treatment-year award rather than cumulative program funding because the former corresponds directly to the construction episode defining treatment, whereas cumulative awards incorporate subsequent funding decisions made after treatment has begun.

Figure C.7 provides some evidence that the largest initial projects generate larger near-term employment responses, but not a clear monotonic dose-response relationship. At the 2-year horizon, the

largest estimate occurs in the top grant-size quartile, whose 95 percent confidence interval ranges from 9.3 to 66.2 workers and excludes zero. The estimates across the remaining quartiles do not increase systematically with grant size. At the 4-year horizon, estimates are less precise and the top-quartile effect falls close to zero. The evidence therefore suggests that particularly large projects may generate larger employment gains in the near term, but does not support a stable relationship in which construction employment rises proportionally with the size of the initial federal award.

Comparisons across grant-size quartiles should be interpreted cautiously. Each quartile-specific CS-DiD estimate is constructed using its own treated sample and associated not-yet-treated comparisons, so differences across quartiles need not reflect grant size alone if the corresponding treated groups would otherwise have followed different employment paths. Dividing the sample into four groups also reduces precision, particularly at longer horizons as the available comparison group contracts. I therefore use these estimates as descriptive evidence on treatment-intensity heterogeneity rather than as a separately identified causal dose-response function.

Can grant size explain the era and slack heterogeneity? Grant size could also confound the heterogeneity documented in Section 5. Federal assistance was concentrated in the early years of the program, so the stronger construction-employment response among early treatment cohorts could partly reflect larger initial projects. Grant size may likewise differ between counties with high and low unemployment. To examine whether these relationships account for the main heterogeneity results, I consider grant size, treatment timing, and local unemployment jointly.

For each eligible treated county, I construct a simple treatment-effect proxy as the change in construction employment from the year before first treatment to two or four years afterward, net of the average change over the same calendar window among not-yet-treated counties. I regress this measure jointly on treatment-year federal awards per capita, pre-program population, the 1976 unemployment rate, and a quadratic in first-treatment year. This exercise is not an alternative treatment-effect estimator; rather, it provides a way to examine whether the heterogeneity dimensions emphasized in the main text remain associated with the treatment-effect proxy after conditioning jointly on grant size and the other included characteristics.

Table C.16: Treatment-effect proxies and grant size, timing, and unemployment

	2-yr	4-yr
log(Fed. award per capita)	-13.716 (11.390)	-21.902 (23.184)
log(Population)	28.730 (38.972)	29.721 (29.018)
Unemployment rate	-1.448 (2.242)	-0.703 (3.425)
First-treat year (centered 1973)	-49.525** (19.046)	-67.630*** (19.092)
First-treat year ²	2.354** (0.847)	3.385*** (0.859)
N	1,130	849

Notes: This table reports OLS regressions of county-level treatment-effect proxies on grant size, population, unemployment, and treatment timing. The treatment-effect proxy is the change in construction employment from $g - 1$ to the indicated post-treatment horizon, net of the average change over the same calendar window among not-yet-treated counties. *Fed. award per capita* refers to federal grant dollars received in the treatment year divided by county population. *First-treat year* is centered at 1973. The 2-year and 4-year specifications require no second construction episode before the corresponding horizon. Standard errors are clustered by first-treatment year and reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C.16 reports the results. Grant size per capita is not statistically significant at either horizon, nor is population. The unemployment coefficient is also imprecisely estimated and, unlike the subgroup results in Section 5.2, does not indicate a larger response in higher-unemployment counties once grant size, population, and treatment timing are included jointly. In contrast, the treatment-year terms remain statistically significant at both horizons. Thus, although early program cohorts received larger awards on average, the relationship between treatment timing and subsequent construction-employment gains is not eliminated after conditioning on grant size.

These exercises provide little evidence that differences in initial grant size alone account for the era heterogeneity reported in the main text. The quartile estimates do not display a stable monotonic dose-response relationship, and controlling directly for treatment-year grant size does not eliminate the relationship between treatment timing and the employment response. The evidence is less informative for the slack heterogeneity: although the separate subgroup specifications consistently produce larger near-term estimates in higher-unemployment counties, that relationship does not survive in the joint horse-race regression. I therefore use this exercise primarily to assess whether grant size mechanically explains the era heterogeneity and do not interpret it as providing additional

support for the slack result.

C.6 Relaxing the single-episode sample restriction

The baseline specification isolates a county's first grant-assisted construction episode by dropping observations from the start of a second episode onward. As documented in Appendix Section A.1, this restriction becomes increasingly consequential at longer horizons, with 77.3 percent of ever-treated counties still contributing at $h = 1$ and 55.0 percent at $h = 3$. This section examines how the estimates change when subsequent construction episodes are retained.

Relaxing the restriction does not simply recover observations lost to attrition. The restored county-years are periods in which additional grant-assisted construction is underway. Among ever-treated counties, 72.1 percent begin a second episode during the sample period, and the median repeat county does so two years after its first construction start. Subsequent episodes also account for substantially more federal funding than the first episode alone (Table A.3). The unrestricted specification therefore incorporates additional treatment as the event-time horizon increases.

The two specifications consequently correspond to different economic objects. The baseline estimates the employment response associated with a county's first construction episode while observations remain uncontaminated by subsequent episodes. The unrestricted specification instead traces employment after a county first enters the Construction Grants Program while allowing additional grant-assisted construction to accumulate thereafter. The latter is informative about the broader employment footprint of program participation, but treatment intensity then varies across counties and event-time horizons in a way that is not captured by the initial treatment indicator.

Figure C.8 compares the resulting event-study paths. The estimates coincide through the treatment year by construction, since the single-episode restriction removes observations only after a county begins a subsequent construction episode. Both specifications therefore produce the same pre-treatment coefficients and the same treatment-year estimate of 22.0 workers (s.e. 9.5). The paths diverge beginning at $h = 1$. In the baseline sample, the estimated effect declines from 17.8 workers at $h = 1$ to 13.8 at $h = 2$ and -1.6 at $h = 3$. In the unrestricted panel, the corresponding es-

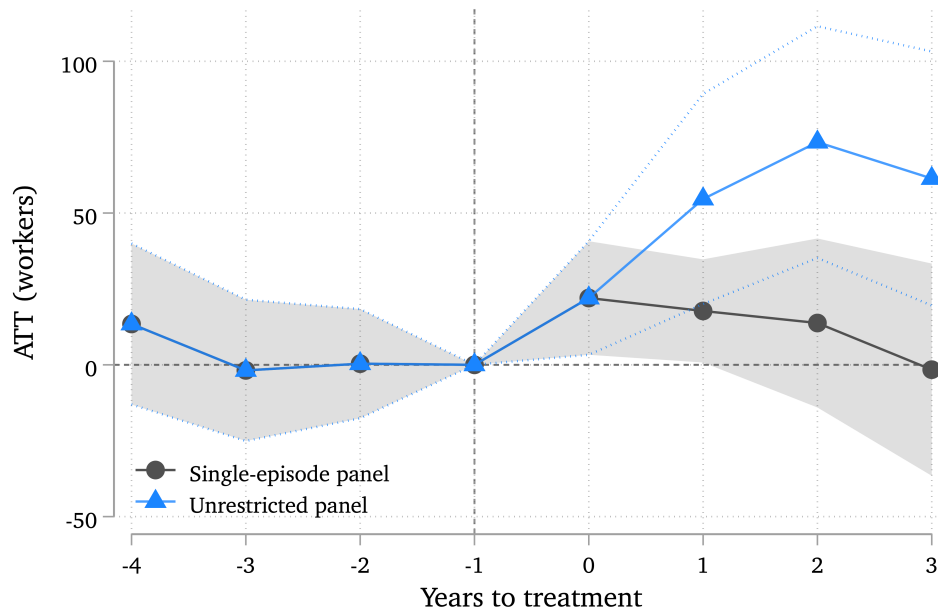


Figure C.8: Event-study estimates of construction employment under alternative sample restrictions. The dark line reports the baseline single-episode specification, which drops county-year observations from the start of a county's second grant-assisted construction episode onward. The blue line reports the unrestricted specification, which retains subsequent construction episodes. Shaded areas and dotted lines denote 95% confidence intervals.

estimates are 54.6, 73.4, and 61.4 workers, each statistically significant at the one percent level. This divergence emerges at the horizons when subsequent construction episodes increasingly enter the unrestricted sample, consistent with the descriptive evidence that repeat treatment occurs relatively soon after a county's initial construction start.

The comparison should not be interpreted as a direct estimate of attrition bias. Two forces operate simultaneously. First, the restored observations contain additional grant-assisted construction and therefore additional treatment. Second, counties that receive subsequent grants relatively quickly may differ from counties that do not, creating endogenous sample composition in the baseline as the horizon increases. Because the same repeat-treatment counties generate both mechanisms, the unrestricted comparison cannot separately identify them.

I retain the single-episode specification as the baseline because it preserves a clearly defined treatment: a county's first grant-assisted construction episode. This is particularly important for analyses relating the employment response to the size of the initial grant. The unrestricted estimates

nevertheless provide useful complementary evidence about cumulative program participation. Importantly, they are substantially larger than the baseline estimates at positive horizons, indicating that the single-episode restriction does not generate the paper's positive construction-employment response by retaining unusually high-employment observations. The tradeoff is one of interpretation: the baseline isolates the initial construction episode, whereas the unrestricted specification incorporates the additional construction activity generated by subsequent grants.

D Additional Analyses on Total County Effects

D.1 Robustness to the choice of treatment anchor

Alternative treatment-timing definitions The timing comparisons also change the policy event to which the estimated effects are anchored. The baseline construction-sector analysis defines treatment by the start of physical construction. For the broader employment and income outcomes, shifting treatment two years earlier instead captures a broader pre-construction and construction window, allowing economic responses associated with planning, engineering, procurement, and other activities preceding the recorded construction start to enter the post-treatment period. Anchoring treatment at Step 3 or Step 1 receipt moves the event still earlier in the administrative process. The resulting estimates therefore should not be interpreted as alternative estimates of an identical parameter: each timing definition assigns a different portion of the planning-to-construction sequence to the post-treatment period.

This distinction is particularly important because the administrative milestones are separated by substantial and heterogeneous intervals. As documented in Appendix Section B.3, the median interval from Step 1 receipt to construction start is 4.96 years, while the median interval from Step 3 receipt to construction start is 0.75 years, with considerable dispersion around both. Moreover, Appendix Section B.2 shows that grant-award timing is related to contemporaneous local economic conditions, whereas these relationships largely disappear when treatment is dated by construction start. Thus, moving the treatment anchor earlier does more than relabel event time: it both changes the economic activity included in the post-treatment estimand and moves treatment toward administrative dates whose timing is more closely related to local conditions.

Tables D.17 and D.18 report sectoral employment estimates under five specifications: construction start minus two years, the same timing estimated using DRIPW controls for pre-program population and per-capita income, the unshifted construction-start date, Step 3 award receipt, and Step 1 award receipt. Figures D.9 and D.10 additionally report average pre-treatment coefficients. The level estimates for transportation and utilities, manufacturing, wholesale trade, and retail trade are generally positive across treatment definitions. The construction-start-minus-two-year estimates are also sim-

ilar when pre-program controls are added through DRIPW. The log estimates are less precise, as elsewhere in the paper.

The more informative differences appear in the pre-treatment coefficients. Under construction start minus two years, with or without the DRIPW controls, the pre-treatment estimates are generally close to zero. The pre-treatment comparisons become less stable under some of the other treatment anchors, most clearly when treatment is dated by Step 1 receipt. Step 1 produces sizeable pre-treatment coefficients for several sectoral outcomes, including wholesale and retail employment in levels and wholesale, retail, and mining employment in logs, and its estimates are considerably less precise because they use the smaller facility-matched sample. The unshifted construction-start and Step 3 specifications produce larger pre-treatment differences for some outcomes as well. These patterns are consistent with the administrative evidence above that earlier grant milestones are imperfect treatment anchors for isolating the subsequent local economic response.

The same distinction appears for total employment. Panel A of Table [D.19](#) shows positive level effects under all five timing definitions: the two-year estimates range from approximately 217 to 286 additional workers, and the estimates remain positive through the six-year horizon. Figure [D.11](#), however, shows that similar post-treatment estimates do not imply equally clean identifying comparisons. The construction-start-minus-two-year specifications have comparatively small pre-treatment coefficients, whereas the unshifted construction-start, Step 3, and particularly Step 1 definitions produce larger negative pre-treatment estimates. For Step 1 receipt, the average pre-treatment coefficient in levels is approximately –240 workers, with a sizeable negative estimate also appearing in logs. The positive post-treatment estimates under these administrative anchors therefore need to be interpreted together with their less favorable pre-treatment behavior.

Per-capita personal income is more sensitive to the treatment anchor. Table [D.20](#) shows relatively modest effects when treatment is dated two years before construction start, at construction start, or at Step 3 receipt. In contrast, the Step 1 specification produces substantially larger estimates—approximately \$320, \$475, and \$491 per capita over the two-, four-, and six-year horizons. Figure [D.12](#) again qualifies this comparison: Step 1 also differs from the other specifications in the pre-treatment period. Combined with the smaller matched sample and the long and heterogeneous

interval between Step 1 receipt and construction documented above, these estimates should not be read as evidence that the causal effect simply begins several years before construction.

These exercises illustrate two consequences of changing the treatment date. First, the estimand changes as progressively earlier portions of the planning-to-construction sequence are included in the post-treatment period. Second, the identifying comparisons themselves become less favorable under some of the earlier administrative anchors, particularly Step 1 receipt. The construction-start-minus-two-year specification used for the broader local outcomes is therefore best interpreted as capturing economic activity surrounding the transition from grant funding into construction, rather than as the effect of construction initiation itself. The Step 1, Step 3, and unshifted construction-start results provide alternative views of where that response is placed along the project timeline, but they do not identify an identical policy parameter.

Table D.17: Sectoral employment effects under alternative treatment-timing definitions, levels

	Transportation and Utilities					Manufacturing				
	(i)	(ii)	(iii)	(iv)	(v)	(i)	(ii)	(iii)	(iv)	(v)
2-yr post avg.	15.02*** (4.27)	14.88** (6.18)	12.67*** (3.68)	15.40*** (4.43)	5.80 (10.67)	60.66*** (17.16)	56.93*** (16.16)	87.48*** (17.97)	68.21*** (17.18)	96.82*** (29.14)
4-yr post avg.	21.11*** (6.57)	20.55*** (7.63)	14.88*** (5.34)	19.14*** (6.08)	19.04* (10.42)	97.13*** (26.21)	85.45*** (22.62)	94.92*** (24.93)	99.37*** (26.68)	118.24*** (37.31)
6-yr post avg.	22.41** (8.85)	21.98** (8.64)	15.86** (7.11)	20.97** (8.27)	21.35* (12.83)	102.56*** (34.54)	79.84*** (24.82)	108.54*** (33.38)	118.04*** (34.74)	111.57** (43.43)
N	35,457	35,421	38,547	36,120	22,807	35,079	35,032	38,117	35,857	22,712
Pre-treat. avg.	552.73	552.73	850.10	737.74	569.60	2,714.67	2,714.67	4,209.43	3,669.25	2,766.88
Wholesale trade										
Retail trade										
2-yr post avg.	15.45*** (4.71)	13.90*** (4.87)	18.03*** (3.99)	17.25*** (4.77)	11.59* (5.92)	61.55*** (11.53)	62.05*** (16.98)	60.90*** (9.27)	60.53*** (10.12)	65.70*** (14.86)
4-yr post avg.	22.50*** (7.26)	20.02*** (7.20)	15.06*** (5.77)	19.43*** (6.86)	15.65* (8.42)	83.02*** (16.33)	81.20*** (21.30)	52.34*** (13.38)	67.62*** (15.52)	89.29*** (23.97)
6-yr post avg.	20.84** (9.59)	19.18** (8.56)	10.60 (7.90)	17.92* (9.67)	16.90 (11.42)	71.61*** (20.00)	72.88*** (21.58)	43.19** (17.74)	60.65*** (20.03)	87.10*** (30.24)
N	34,721	34,687	37,804	35,528	22,341	38,477	38,424	41,675	39,329	25,349
Pre-treat. avg.	547.70	547.70	876.49	761.09	562.85	1,728.62	1,728.62	2,723.78	2,371.66	1,752.71

Notes: This table reports GSDiD average treatment effects on sectoral employment over the first 2, 4, and 6 years under alternative treatment-timing definitions. Columns (i)–(v) use: (i) construction start minus 2 years; (ii) construction start minus 2 years with DRIPW covariate adjustment; (iii) construction start; (iv) Step 3 award receipt; and (v) Step 1 award receipt using the matched sample. DRIPW controls include pre-program population and personal income per capita. Standard errors are clustered at the county level and reported in parentheses. *Pre-treat. avg.* reports the average outcome in the pre-treatment reference period for the estimation sample corresponding to each specification. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table D.18: Sectoral employment effects under alternative treatment-timing definitions, logs

	Transportation and Utilities					Manufacturing				
	(i)	(ii)	(iii)	(iv)	(v)	(i)	(ii)	(iii)	(iv)	(v)
2-yr post avg.	0.37 (0.58)	0.40 (0.59)	-0.17 (0.48)	0.43 (0.56)	0.70 (0.87)	0.35 (0.56)	0.42 (0.57)	0.40 (0.51)	-0.08 (0.55)	1.71* (0.94)
4-yr post avg.	0.89 (0.76)	0.92 (0.76)	0.02 (0.69)	0.28 (0.76)	0.72 (1.19)	0.58 (0.77)	0.63 (0.79)	0.59 (0.71)	0.62 (0.81)	1.67 (1.26)
6-yr post avg.	1.34 (0.94)	1.29 (0.94)	0.40 (0.91)	0.64 (0.94)	0.46 (1.46)	0.74 (0.98)	0.72 (1.00)	0.68 (0.92)	0.88 (1.03)	0.99 (1.58)
N	35,457	35,421	38,547	36,120	22,807	35,079	35,032	38,117	35,857	22,712
Pre-treat. avg.	552.73	552.73	850.10	737.74	569.60	2,714.67	2,714.67	4,209.43	3,669.25	2,766.88
Wholesale trade										
Retail trade										
2-yr post avg.	0.10 (0.54)	0.10 (0.54)	1.09** (0.49)	0.35 (0.58)	-0.37 (1.04)	0.46* (0.26)	0.46* (0.26)	0.47* (0.26)	0.29 (0.26)	0.79* (0.47)
4-yr post avg.	0.20 (0.69)	0.18 (0.69)	0.35 (0.69)	0.23 (0.76)	-0.00 (1.23)	0.93** (0.36)	0.93** (0.37)	0.18 (0.38)	0.41 (0.39)	0.79 (0.63)
6-yr post avg.	-0.55 (0.87)	-0.57 (0.87)	-0.44 (0.88)	-0.26 (0.93)	0.11 (1.43)	0.78 (0.50)	0.73 (0.50)	-0.11 (0.50)	0.33 (0.51)	0.23 (0.78)
N	34,721	34,687	37,804	35,528	22,341	38,477	38,424	41,675	39,329	25,349
Pre-treat. avg.	547.70	547.70	876.49	761.09	562.85	1,728.62	1,728.62	2,723.78	2,371.66	1,752.71

Notes: This table reports CSDiD average treatment effects on log sectoral employment over the first 2, 4, and 6 years under alternative treatment-timing definitions. Columns (i)–(v) use: (i) construction start minus 2 years; (ii) construction start minus 2 years with DRIPW covariate adjustment; (iii) construction start; (iv) Step 3 award receipt; and (v) Step 1 award receipt using the matched sample. DRIPW controls include pre-program population and personal income per capita. Standard errors are clustered at the county level and reported in parentheses. *Pre-treat. avg.* reports the average outcome in the pre-treatment reference period for the estimation sample corresponding to each specification. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

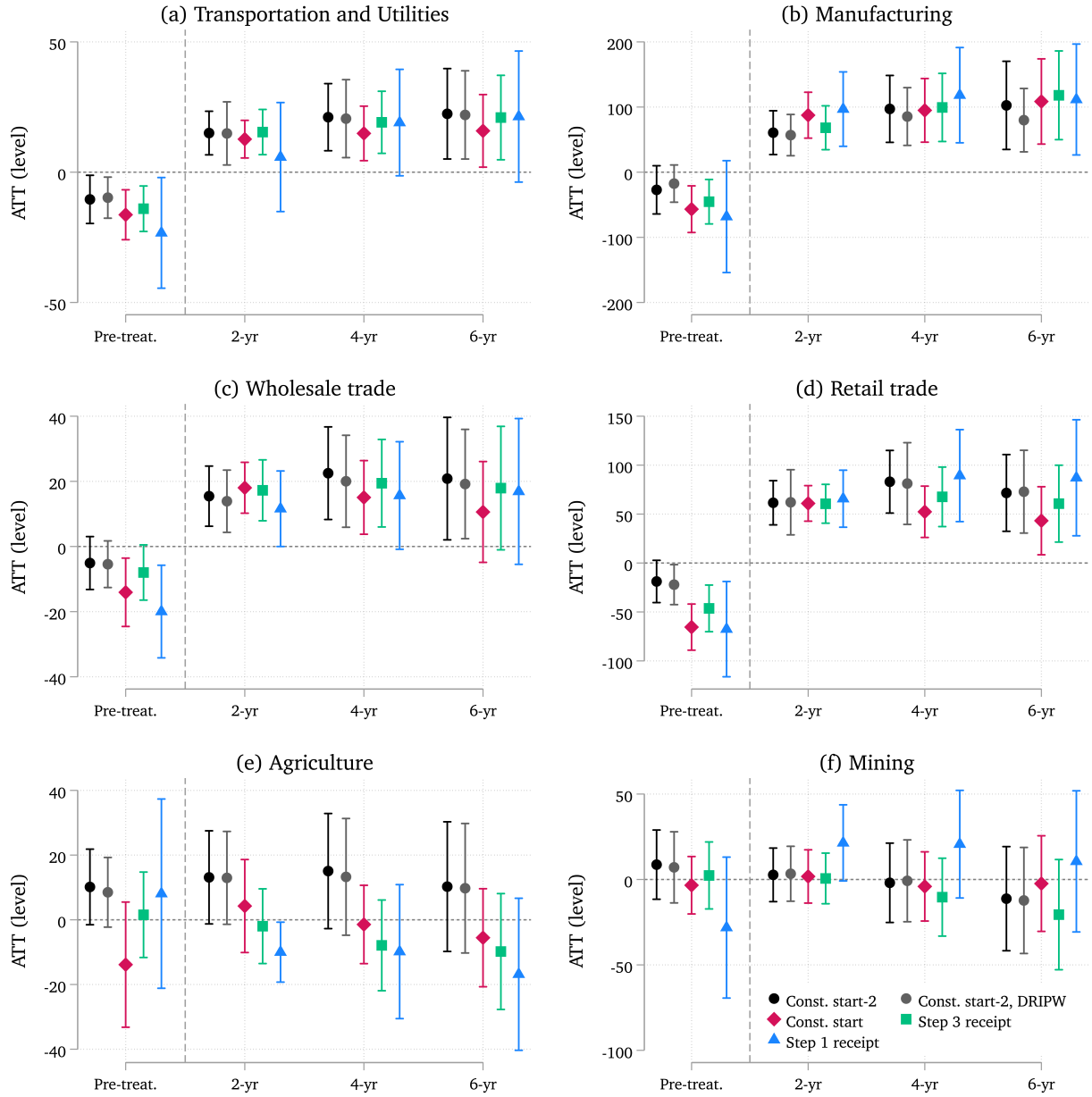


Figure D.9: Sectoral employment effects under alternative treatment-timing definitions, levels. Markers report CSDiD average treatment effects over the pre-treatment period and the first 2, 4, and 6 years after treatment. Specifications use construction start minus 2 years, construction start minus 2 years with DRIPW covariate adjustment, construction start, Step 3 award receipt, and Step 1 award receipt using the matched sample. Spikes denote 95% confidence intervals.

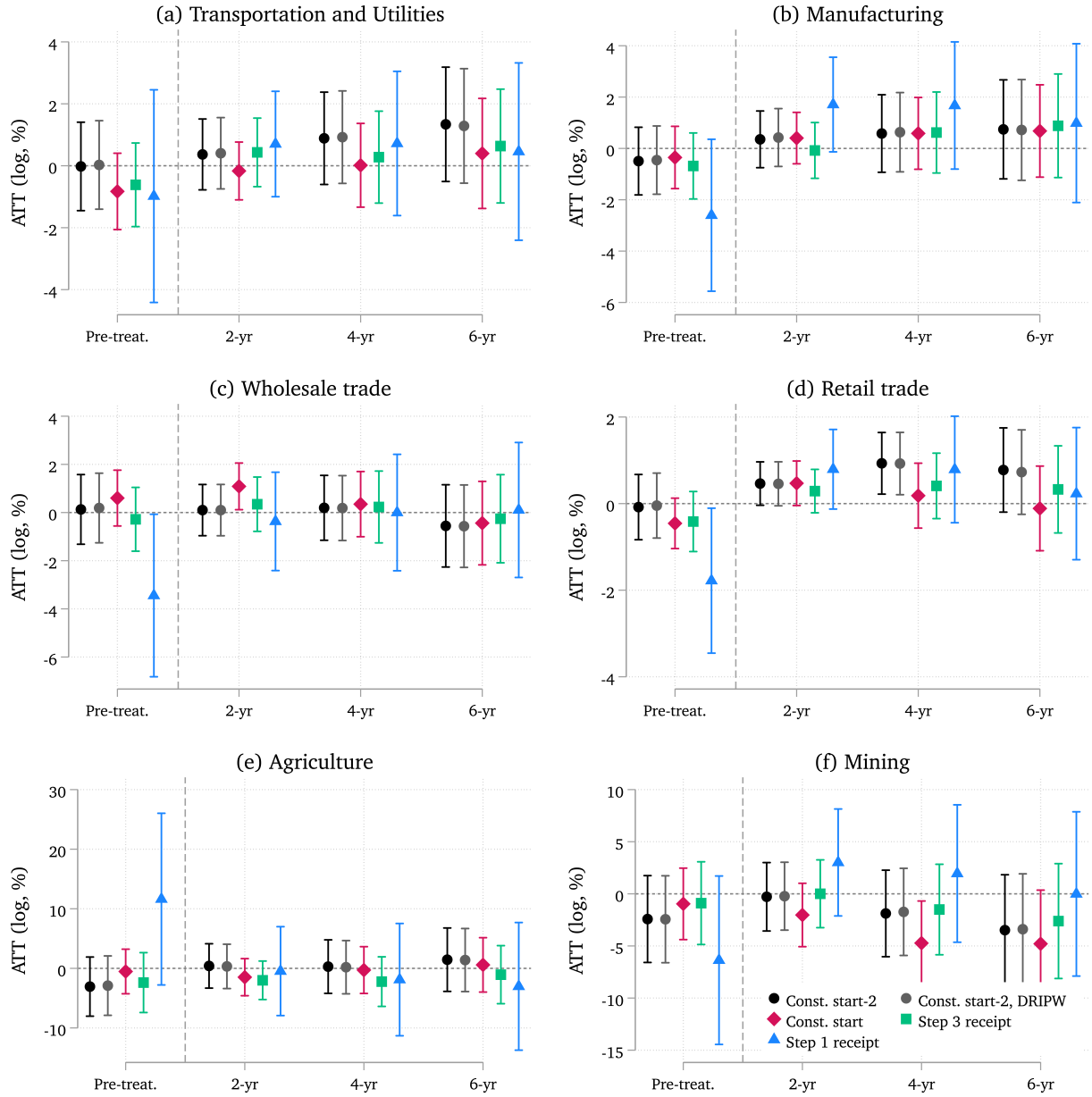


Figure D.10: Sectoral employment effects under alternative treatment-timing definitions, logs. Markers report CSDiD average treatment effects over the pre-treatment period and the first 2, 4, and 6 years after treatment. Specifications use construction start minus 2 years, construction start minus 2 years with DRIPW covariate adjustment, construction start, Step 3 award receipt, and Step 1 award receipt using the matched sample. Coefficients are expressed in percent. Spikes denote 95% confidence intervals.

Table D.19: Total employment effects under alternative treatment-timing definitions

Panel A. Level outcome					
	Total employment				
	(i)	(ii)	(iii)	(iv)	(v)
2-yr post avg.	216.91*** (49.18)	215.25*** (63.89)	286.14*** (43.56)	251.21*** (48.65)	285.27*** (65.61)
4-yr post avg.	311.94*** (76.84)	295.90*** (89.75)	251.34*** (68.31)	280.13*** (78.20)	355.26*** (99.98)
6-yr post avg.	261.60*** (96.54)	259.81*** (94.31)	194.88** (89.93)	230.37** (96.01)	323.95** (130.04)
N	38,746	38,690	41,934	39,595	25,582
Pre-treatment avg.	7,945.94	7,945.94	12,611.59	10,912.44	8,025.77
Panel B. Log outcome (percent)					
	Total employment				
	(i)	(ii)	(iii)	(iv)	(v)
2-yr post avg.	0.34 (0.28)	0.35 (0.28)	0.02 (0.26)	0.02 (0.26)	1.31*** (0.47)
4-yr post avg.	0.44 (0.39)	0.39 (0.39)	-0.28 (0.38)	-0.13 (0.38)	0.71 (0.66)
6-yr post avg.	0.20 (0.52)	0.11 (0.52)	-0.67 (0.49)	-0.35 (0.50)	-0.22 (0.82)
N	38,746	38,690	41,934	39,595	25,582
Pre-treatment avg.	7,945.94	7,945.94	12,611.59	10,912.44	8,025.77

Notes: This table reports CSDiD average treatment effects on total employment over the first 2, 4, and 6 years under alternative treatment-timing definitions. Panel A reports effects in employment levels. Panel B reports effects in logs, with coefficients expressed in percent. Columns (i)–(v) use: (i) construction start minus 2 years; (ii) construction start minus 2 years with DRIPW covariate adjustment; (iii) construction start; (iv) Step 3 award receipt; and (v) Step 1 award receipt using the matched sample. DRIPW controls include pre-program population and personal income per capita. Standard errors are clustered at the county level and reported in parentheses. *Pre-treatment avg.* reports the average total employment level in the pre-treatment reference period for the estimation sample corresponding to each specification. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

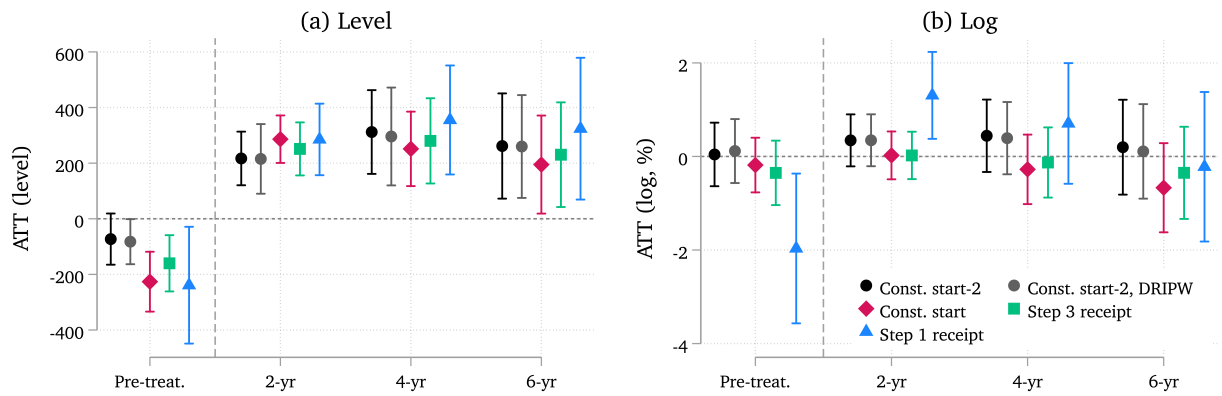


Figure D.11: Total employment effects under alternative treatment-timing definitions. Panel (a) reports effects in levels, and Panel (b) reports effects in logs. Markers report CSDiD average treatment effects over the pre-treatment period and the first 2, 4, and 6 years after treatment. Specifications use construction start minus 2 years, construction start minus 2 years with DRIPW covariate adjustment, construction start, Step 3 award receipt, and Step 1 award receipt using the matched sample. Log coefficients are expressed in percent. Spikes denote 95% confidence intervals.

Table D.20: Per-capita personal income effects under alternative treatment-timing definitions

	Per-capita personal income			
	(i)	(ii)	(iii)	(iv)
2-yr post avg.	0.025 (0.032)	0.117*** (0.036)	0.031 (0.033)	0.320*** (0.060)
4-yr post avg.	0.087** (0.038)	0.107** (0.045)	0.079* (0.043)	0.475*** (0.079)
6-yr post avg.	0.100** (0.045)	0.077 (0.055)	0.071 (0.054)	0.491*** (0.092)
N	63,898	63,898	62,276	45,901
Pre-treat. avg.	19.632	19.935	19.935	19.406

Notes: This table reports CSDiD average treatment effects on per-capita personal income over the first 2, 4, and 6 years under alternative treatment-timing definitions. Columns (i)–(iv) use: (i) construction start minus 2 years; (ii) construction start; (iii) Step 3 award receipt; and (iv) Step 1 award receipt using the matched sample. Per-capita personal income is measured in thousands of real 2017 USD. Standard errors are clustered at the county level and reported in parentheses. *Pre-treat. avg.* reports the average per-capita personal income level in the pre-treatment reference period for the estimation sample corresponding to each specification. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

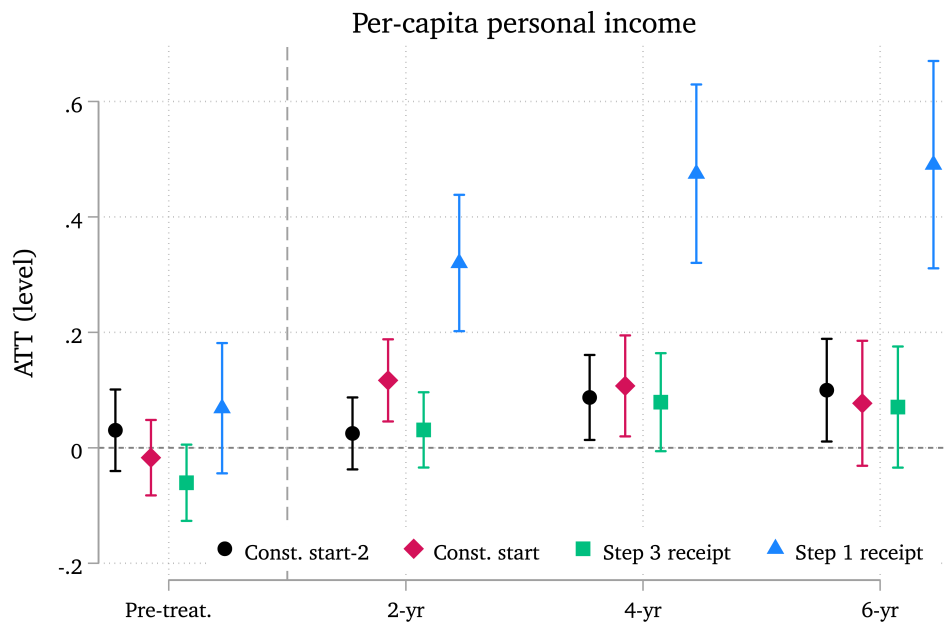


Figure D.12: Per-capita personal income effects under alternative treatment-timing definitions. Markers report CSDiD average treatment effects over the pre-treatment period and the first 2, 4, and 6 years after treatment. Specifications use construction start minus 2 years, construction start, Step 3 award receipt, and Step 1 award receipt using the matched sample. Per-capita personal income is measured in thousands of real 2017 USD. Spikes denote 95% confidence intervals.

Alternative shifts relative to construction start Section 6 dates treatment two years before construction start when examining total employment and per-capita personal income. The purpose of this adjustment is to allow broader local economic responses to begin during activities preceding physical construction, rather than to interpret construction start as the unique date at which these outcomes can respond. Among the grants defining counties' first construction starts, 88% were awarded within two years of construction start. I assess sensitivity to the particular two-year adjustment by alternatively defining treatment as one, two, or three years before construction start, $\tilde{G}_i = G_i - t$ for $t \in \{1, 2, 3\}$.

Figures D.13 and D.14 report the corresponding event-study estimates. For total employment, changing the adjustment primarily shifts the estimated response along the event-time axis. In levels, the response peaks around $h = 1$ when treatment is dated one year before construction start, around $h = 2$ under the preferred two-year adjustment, and around $h = 3$ when treatment is dated three years earlier. The magnitudes around these respective peaks are also similar. This displacement is what would be expected if the three specifications are capturing largely the same calendar-time employment response but assigning a different date as event time zero. The log specifications show the same general pattern, although the estimates are smaller and less precise.

Per-capita personal income displays a similar, though less sharply defined, timing pattern. Moving the treatment date further before construction generally shifts the positive portion of the event-study path toward later event times. The one-year adjustment rises earlier, while the three-year adjustment develops more gradually and reaches its largest estimates later in the event window. Across the three definitions, the pre-treatment estimates are imprecise and not statistically distinguishable from zero.

These comparisons therefore do not identify the precise date at which broader local effects begin. Instead, they show that the post-treatment patterns reported in Section 6 are not specific to choosing exactly two years before construction start as the adjusted treatment date. Shifting that date by one year in either direction mainly changes where the response appears in event time, as expected when the underlying economic response is tied to the same project sequence.

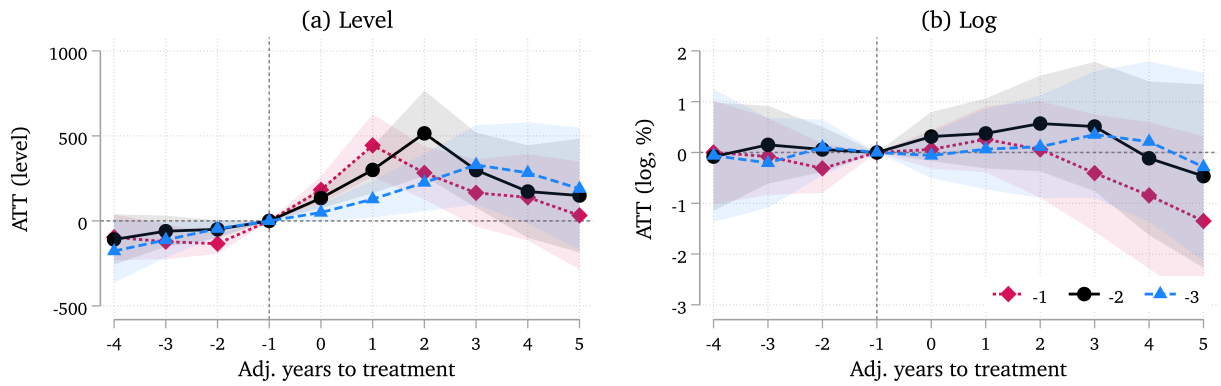


Figure D.13: Event-study estimates of total employment under alternative treatment-timing adjustments. Panel (a) reports effects in levels, and Panel (b) reports effects in logs. The three series shift treatment to 1, 2, or 3 years before construction start. Log coefficients are expressed in percent. Shaded areas denote 95% confidence intervals.

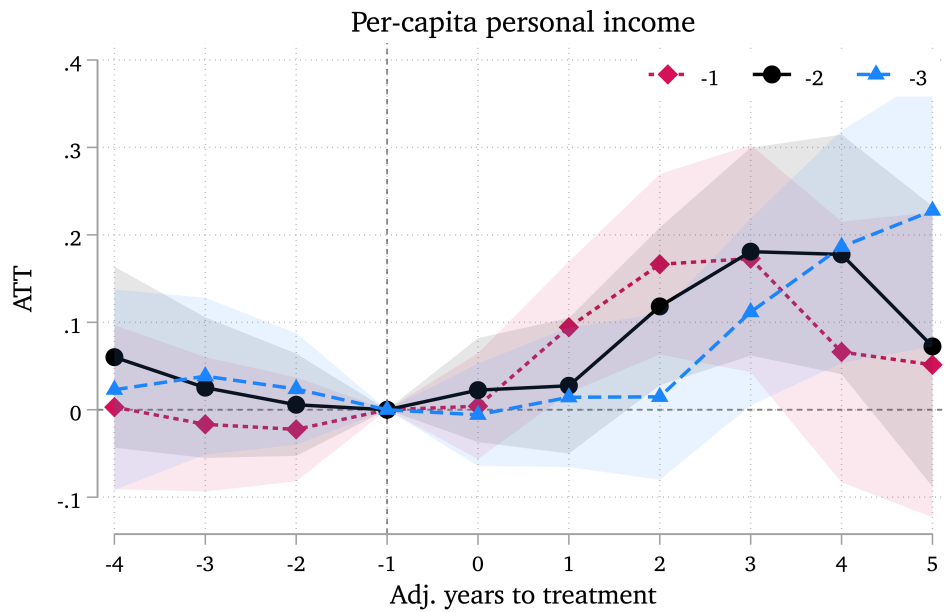


Figure D.14: Event-study estimates of per-capita personal income under alternative treatment-timing adjustments. The three series shift treatment to 1, 2, or 3 years before construction start. Per-capita personal income is measured in thousands of real 2017 USD. Shaded areas denote 95% confidence intervals.

E Additional Robustness Exercises

E.1 In-time placebo tests

Table E.21: In-time placebo tests

Panel A. Construction employment				
	Level		Log	
	$k = 2$	$k = 3$	$k = 2$	$k = 3$
2-yr post avg.	-1.289 (8.599) [0.881]	-8.362 (7.832) [0.286]	0.002 (0.010) [0.845]	-0.013 (0.011) [0.226]
N	22,835	22,213	22,835	22,213
Panel B. Total employment				
	Level		Log	
	$k = 2$	$k = 3$	$k = 2$	$k = 3$
2-yr post avg.	58.204 (36.502) [0.111]	106.853 (43.380) [0.014]	-0.001 (0.003) [0.712]	0.004 (0.004) [0.325]
N	20,123	19,681	20,123	19,681

Notes: This table reports CSDiD average placebo effects over the first 2 years following fictitious treatment dates assigned before the treatment date used in the corresponding baseline specification. k denotes the number of years by which the fictitious treatment date is moved earlier. For construction employment, the baseline treatment date is construction start. For total employment, the baseline treatment date is construction start minus 2 years. The sample includes only county-years strictly before the corresponding baseline treatment date. Panels report level and log outcomes. Standard errors are clustered at the county level and reported in parentheses. p -values are reported in brackets.

The event-study estimates provide direct evidence on outcomes before treatment, but I also conduct in-time placebo tests that assign treatment before the treatment date used in the corresponding baseline specification. For each outcome, I restrict the estimation sample to observations strictly preceding the true treatment date, move treatment backward by an additional two or three years, and re-estimate the CSDiD specification. The resulting “post-treatment” period therefore occurs entirely before the treatment event used in the main analysis. Table E.21 reports the average placebo effect over the first two years following each fictitious treatment date.

Construction employment placebo tests Panel A reports the results for construction employment, for which the baseline treatment date is construction start. The placebo estimates are small and statistically insignificant under both fictitious dates. In levels, the estimates are -1.289 workers ($p = 0.881$) when treatment is moved two years earlier and -8.362 workers ($p = 0.286$) when it is moved three years earlier. The corresponding log estimates are also close to zero and statistically insignificant ($p = 0.845$ and $p = 0.226$, respectively). These results are consistent with the event-study evidence for construction employment: I find no systematic change in construction employment during periods preceding the recorded construction start.

Total employment placebo tests Panel B repeats the exercise for total employment. Here the reference treatment date is already shifted two years before construction start, as in Section 6. Moving this date back another two years produces a level placebo estimate of 58.2 workers, which is not statistically significant ($p = 0.111$). Moving it back three years produces a larger estimate of 106.9 workers that is statistically significant ($p = 0.014$). Thus, unlike construction employment, the level specification does not pass every in-time placebo test. The corresponding log estimates are much smaller and statistically insignificant under both fictitious dates ($p = 0.712$ and $p = 0.325$).

These results provide mixed evidence for total employment in levels and cleaner placebo evidence for the log specification. The significant $k = 3$ estimate should also be interpreted in light of the treatment-timing exercise above. Because the aggregate specification already dates treatment two years before construction start, this placebo assigns treatment five years before construction start. It therefore tests for differences substantially earlier in the project sequence than the window the aggregate analysis is intended to capture. I nevertheless report it as evidence that the level estimates for broader employment outcomes are more sensitive to treatment timing than the construction-sector results.

E.2 Permutation tests on ever-treated counties

As a separate inference exercise, I conduct permutation tests that compare the estimated effects under the observed treatment timing with effects obtained after randomly reassigning treatment

Table E.22: Permutation tests

Panel A. Construction employment		
	Level	Log
2-yr post avg.	22.930 (8.643) [0.090]	0.023 (0.012) [0.143]
Panel B. Total employment		
	Level	Log
2-yr post avg.	284.991 (71.883) [0.006]	0.005 (0.004) [0.356]

Notes: This table reports randomization-inference tests for the CSDiD average treatment effect over the first 2 post-treatment years. Panel A reports construction employment; Panel B reports total employment. Outcomes are shown in levels and logs. In each of 1,000 draws, first-treatment dates are permuted without replacement across ever-treated counties, preserving the empirical distribution of treatment years. Never-treated counties remain untreated. Construction employment uses construction start as the treatment date; total employment uses construction start minus 2 years. Standard errors are clustered at the county level and reported in parentheses. Bracketed values report randomization-inference p -values, defined as the share of permuted estimates at least as large in absolute value as the observed estimate.

timing across ever-treated counties. Whereas the conventional CSDiD p -value relies on the estimator's sampling distribution, the permutation test asks how unusual the observed estimate is relative to estimates generated under alternative assignments of treatment timing.

In each of 1,000 draws, I permute first-treatment dates across ever-treated counties without replacement, preserving the empirical distribution of treatment years and therefore the number of counties assigned to each cohort. Never-treated counties retain their status as comparison units. I then re-estimate the same two-year average CSDiD effect under each assignment. For construction employment, treatment is dated by construction start. For total employment, the reassigned treatment date is shifted two years earlier, maintaining the same timing convention as the main aggregate specification. The randomization-inference p -value is the share of the 1,000 permuted estimates at least as large in absolute value as the estimate obtained under the observed assignment.

Table E.22 reports the results. The strongest permutation evidence is for total employment in levels: only 0.6 percent of the reassigned treatment schedules generate an estimate at least as large in absolute value as the observed effect. The corresponding share is 9.0 percent for construction

employment in levels. Thus, the observed level effect for total employment lies well into the tail of its permutation distribution, while the construction-employment effect is unusual at the 10 percent level under this alternative inference procedure.

The log specifications provide weaker evidence. A permuted estimate at least as large as the observed effect occurs in 14.3 percent of draws for construction employment and 30.0 percent for total employment. The corresponding conventional p -values are also relatively large, so the permutation exercise does not materially change the inference for these specifications. Neither log specification rejects the null at conventional levels under randomization inference.

In sum, the placebo and permutation exercises do not provide uniformly stronger inference than the baseline estimates. The construction-employment specification passes both in-time placebo tests, while its observed level effect is moderately unusual relative to randomly reassigned treatment timing. For total employment, the observed level effect is highly unusual under permutation, but one of the earlier in-time placebo assignments also produces a significant level estimate. The diagnostics therefore reinforce the comparatively clean timing evidence for the construction-sector result, while supporting a more cautious interpretation of the broader total-employment response.